High-Performance Computer Architecture

A Companion to ECE 4100/6100

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Contents

Before We Begin **v**

1 Introduction **1**
1.1 What Computer Architecture Is . . . 1
1.2 Core Methods: Parallelism, Speculation, and Prediction . . . 1
1.3 Parallelism by Grain Size . . . 2
1.4 A Budget of Transistors . . . 3
1.5 Flynn's Taxonomy . . . 4
1.6 Vectors: Packed-Word SIMD vs. Pipelined Vector Units . . . 5
1.7 Common Chapter 1 Pitfalls . . . 6
1.8 What to Carry Forward . . . 6
1.9 Practice Questions . . . 6

2 Performance Evaluation **7**
2.1 Why Performance Evaluation Matters . . . 7
2.2 Throughput vs. Runtime . . . 7
2.3 CPU Runtime Equation . . . 8
2.4 RISC vs. CISC Through the Performance Equation . . . 9
2.5 Amdahl's Law . . . 9
2.6 Gustafson's Law and Why It Does Not Contradict Amdahl . . . 10
2.7 Combining Performance Numbers Correctly . . . 11
2.8 Workload Dependence and Benchmarking . . . 12
2.9 Cost-Aware Performance Comparison . . . 12
2.10 What to Carry Forward . . . 12
2.11 Practice Questions . . . 13

3 Cache Memories **14**
3.1 Memory Technology and the Hierarchy . . . 14
3.2 Typical Hierarchy Levels . . . 14
3.3 Locality: Why Caches Work . . . 15
3.4 Cache Geometry and Address Partitioning . . . 15
3.5 Direct-Mapped, Set-Associative, and Fully Associative . . . 17
3.6 Hit, Miss, and Average Access Time . . . 18
3.7 The Three C's of Misses . . . 18
3.8 Replacement Policies and Their Tradeoffs . . . 19
3.9 Write Policies . . . 21
3.10 Reducing Miss Ratio . . . 22
3.11 Reducing Miss Penalty . . . 23
3.12 Reducing Hit Time . . . 25
3.13 Virtual Memory Technology, Terminology, and Analogy to Caching . . . 26

3.14 Compiler and Data-Layout Techniques . . . 26
3.15 What to Carry Forward . . . 28
3.16 Practice Questions . . . 28

4 Pipelining **30**
4.1 Pipeline Motivation . . . 30
4.2 Five-Stage Example Pipeline . . . 30
4.3 Pipeline Speedup Calculation . . . 30
4.4 CPI Model and Stall Accounting . . . 31
4.5 Program Characteristics Plus Hazards Produce Stalls . . . 32
4.6 Structural Hazards . . . 32
4.7 Data Hazards . . . 32
4.8 Control Hazards . . . 33
4.9 Branch Target Buffer (BTB) . . . 34
4.10 Whether-To Branch: Direction Prediction . . . 36
4.11 Simple Dynamic Direction Predictors . . . 36
4.12 Global and Local History Predictors . . . 37
4.13 Perceptron Direction Prediction . . . 39
4.14 Hybrid and Tournament Predictors . . . 40
4.15 Timing-Diagram Interpretation . . . 40
4.16 What to Carry Forward . . . 41
4.17 Practice Questions . . . 42

5 Integrated Synthesis and Cache Simulation Concepts **43**
5.1 From an Event to an Explanation . . . 43
5.2 Integrated Themes from Chapters 1 and 2 . . . 43
5.3 Integrated Themes from Chapter 3 . . . 44
5.4 Integrated Themes from Chapter 4 . . . 44
5.5 Cache Simulation Conceptual Model . . . 46
5.6 Hierarchy Semantics in the Model . . . 46
5.7 Prefetching Concepts . . . 46
5.8 Statistics and AAT Interpretation . . . 47
5.9 Conceptual Tradeoffs Across Parameters . . . 47
5.10 What to Carry Forward . . . 47
5.11 Practice Questions . . . 47

6 Dynamic ILP and Out-of-Order **49**
6.1 Why Cycles per Instruction Can Go Below One . . . 49
6.2 Historical Motivation . . . 49
6.3 Superscalar Pipeline View . . . 50
6.4 Dependency Types and Why They Matter . . . 50
6.5 Fire/Complete Taxonomy . . . 51
6.6 Scoreboarding Intuition (FICO Style) . . . 52
6.7 Function Units and Latency . . . 52
6.8 Worked FICO Timing Example . . . 53
6.9 Register Renaming as the Next Step . . . 54

6.10 Tomasulo's Algorithm . . . 55
6.11 Result Buses and Physical Register Files . . . 57
6.12 RATs, Ready Bits, and Dispatch in Renamed Machines . . . 57
6.13 Precise State and Recovery . . . 58
6.14 Reorder Buffers and Retirement . . . 58
6.15 Future Files and Checkpoint/Repair . . . 62
6.16 Speculation Without and With Prediction . . . 63
6.17 Security Implications of Speculative Execution . . . 63
6.18 Implementation Pressure . . . 63
6.19 Interpreting Out-of-Order Processor Statistics . . . 63
6.20 What to Carry Forward . . . 64
6.21 Practice Questions . . . 64

7 Compiler-Scheduled ILP 66
7.1 VLIW as a Compiler-Scheduled Alternative . . . 66
7.2 Dependence Graphs and List Scheduling . . . 67
7.3 Multi-Op Encoding, Tail Bits, and Pause Fields . . . 69
7.4 Software Pipelining and Cyclic Scheduling . . . 69
7.5 Iterative Modulo Scheduling . . . 71
7.6 Worked Software-Pipelining Example . . . 72
7.7 Memory Disambiguation . . . 74
7.8 What to Carry Forward . . . 74
7.9 Practice Questions . . . 74

8 Multiprocessors and Coherence 76
8.1 From Instruction-Level to Thread-Level Parallelism . . . 76
8.2 Memory Organization and System Structure . . . 77
8.3 Interconnects: Shared Buses and Point-to-Point Links . . . 78
8.4 The Cache-Coherence Problem . . . 78
8.5 Coherence Solution Families . . . 79
8.6 Snooping Coherence and the MSI Protocol . . . 80
8.7 Transient States and a Worked Two-Cache Trace . . . 81
8.8 MESI, MOESI, and MOESIF Refinements . . . 82
8.9 Directory-Based Coherence . . . 85
8.10 Directory Scaling and Distributed Organizations . . . 88
8.11 Sharing Patterns and False Sharing . . . 90
8.12 Programming Models: Shared Memory and Message Passing . . . 91
8.13 Synchronization Primitives and Coherence Traffic . . . 92
8.14 Scaling and Architectural Limits . . . 93
8.15 What to Carry Forward . . . 93
8.16 Practice Questions . . . 94

9 Synchronization and Memory Consistency 97
9.1 Protocol States Versus Protocol Behavior . . . 97
9.2 Atomic Primitives for Shared-Memory Synchronization . . . 97
9.3 Reducing Synchronization Contention . . . 99

9.4 Transactional Memory . . . 101
9.5 Memory Consistency Beyond Coherence . . . 102
9.6 What to Carry Forward . . . 105
9.7 Practice Questions . . . 106

10 Parallel Interconnection Networks 108
10.1 Why Parallel Interconnects Are Their Own Topic . . . 108
10.2 Direct and Indirect Topologies . . . 108
10.3 Direct Networks: Hypercubes, Meshes, and Tori . . . 109
10.4 Indirect Networks and Multi-Stage Fabrics . . . 112
10.5 Routing Algorithms . . . 113
10.6 Flow Control and Congestion Management . . . 114
10.7 From Store-and-Forward to Virtual Channels . . . 115
10.8 Switch Architecture . . . 116
10.9 What to Carry Forward . . . 117
10.10 Practice Questions . . . 117

A Pipeline Timing-Diagram Interpretation Checklist 119
A.1 How to Use This Checklist . . . 119

B Formula Reference 123

C Glossary of Core Definitions 131

D Advanced Topics 144
D.1 Quantum Computing Architecture . . . 144
D.2 Superconducting Processor Architecture . . . 146
D.3 Final Review Checkpoints . . . 148
D.4 Practice True/False Prompts . . . 149

E Comprehensive Final Review Checklist 150
E.1 Core Topic Checklist . . . 150
E.2 Review Clarifications . . . 152
E.3 Fast Self-Check Prompts . . . 153

F Lecture Sources and Historical Notes 155

Before We Begin

The Thread Through the Book

A fast computer is a machine that spends less time waiting. It keeps useful data nearby, overlaps independent work, and sometimes starts work before it knows whether that work will be needed. Each improvement creates another question: what happens when the guess is wrong, the data is missing, or two processors want to change the same value? Those questions connect the chapters that follow.

This is a course companion, not a replacement for the lectures or the current assignment specification. Read the prose for the mechanism and work through the examples for its consequences. The chapter questions and separate answer sheet test that reasoning; the formula reference, glossary, and final-review checklist help you revisit it.

Stories, Sources, and Simplified Machines

Short lecture examples appear where they explain an architectural choice. An attributed example is tied to the supplied course material in Appendix F; an uncited everyday analogy is an illustration written for this book. Historical machines provide context, while cycle counts and small diagrams describe explicitly simplified models unless identified otherwise. Always carry the stated assumptions into a calculation: forwarding timing, queue width, memory policy, and the meaning of a counter can change the answer.

Choosing a Reading Path

1. *Build the foundation.* Chapters 1 and 2 introduce parallelism and performance. Chapters 3 and 4 show how caches, pipelines, and prediction turn those ideas into a working processor.
2. *Put the pieces together.* Use Chapter 5 alongside Chapters 3, 4, and 6 when interpreting simulator counters or combining cache and pipeline calculations.
3. *Find more overlap.* Chapter 6 lets hardware schedule independent instructions; Chapter 7 gives that job to the compiler. Read them together to compare what each scheduler must know.
4. *Move beyond one core.* Chapters 8 through 10 connect shared data, synchronization, and communication. Coherence answers which copy to read; consistency and synchronization constrain how different accesses fit together.
5. *Review selectively.* Use Appendix E to locate weak concepts, then return to their explanations. Appendix D extends the architectural questions to quantum and superconducting machines.

Chapter 1

Introduction

A processor can have three adders and still finish only one useful addition at a time. If each addition needs the preceding result, the extra hardware must wait. Give those same adders three independent calculations and all three can work together. This small contrast, used in the introductory lecture, is the starting point for the book: performance comes from matching a machine to the work it can actually overlap.[1]

1.1 What Computer Architecture Is

Computer architecture asks how a processor and memory system should be organized to finish useful work efficiently. "Efficiently" might mean a shorter wait for one result, more completed jobs per second, or less energy per calculation. Circuits, fabrication, software, and algorithms set the possibilities; the architect chooses how to arrange the machine within them.

Every architectural decision is a tradeoff among competing goals. A design that minimizes latency for one task may reduce throughput for a server-style workload. A design that aggressively improves performance may consume more energy or require expensive cooling. A design that is elegant on paper may be too costly in area or too difficult to verify at scale. For this reason, architecture is best treated as constrained optimization: choose structures and policies that best match the target application while respecting implementation limits.

1.2 Core Methods: Parallelism, Speculation, and Prediction

Most performance techniques in modern processors can be traced to three closely linked ideas: *parallelism*, *speculation*, and *prediction*. *Parallelism* increases performance by doing multiple useful operations at the same time. *Speculation* increases performance by doing work early, before the machine is fully certain that this work will be needed. *Prediction* estimates uncertain future behavior (for example branch direction or branch target) so speculative work is directed toward the most likely path. Prediction makes speculation more selective: a good estimate directs early work toward a result the program will actually need.

Classifying common optimizations This distinction is useful because many architectural features combine these ideas differently. Packed-word SIMD instructions and multicore processors are primarily forms of parallelism. The course uses *speculation without prediction* broadly for locality-based choices such as retaining cache blocks in case they are needed again. That is a useful teaching category, but a cache hit does not necessarily execute an uncommitted instruction, and ordinary buffering need not involve a guess. Branch prediction is speculation with prediction.

[1] Course handout `n1_intro.pdf`, slide INTRO-6. Lecture-source references use the identifiers collected in Appendix F.

Out-of-order superscalar execution combines both parallelism and speculation because the machine overlaps independent work while also allowing some work to proceed before the final architectural outcome is known.

1.2.1 Speculation Without Prediction

Speculation does not require a learned predictor. A processor can use a fixed optimistic rule rather than a learned estimate. For example, it may execute a load before the addresses of older stores are known, assuming that the load will not read a location one of those stores changes. If the assumption fails, the machine must detect the ordering violation and replay the affected work. No history-based predictor is required, though one could make that decision more selective.

Related idea: a write buffer A write buffer lets the core continue while a store waits to reach the next memory level. The important gain is overlap, and the buffer alone does not prove that the store is speculative. In a machine with precise exceptions, translation and permission faults must be resolved before a store becomes irrevocable. An uncommitted store can wait in a speculative store queue; a committed store can drain from a buffer later without being a guess about whether the instruction should execute.

1.2.2 Speculation With Prediction

When prediction is added, speculation becomes more selective and usually more efficient. The processor estimates likely future behavior, executes along the estimated path, and commits the work only if the estimate is correct.

Scheduling the next crew The first lecture explains the idea through a building project. If the framing should finish today, schedule the wallboard crew for tomorrow instead of waiting for confirmation before making the call.[2] When the estimate is right, the next job starts sooner. If the framing is delayed, the crew can arrive before there is work for it, and the early decision has a cost. The prediction is the estimate about completion; the speculation is acting on it. Merely making the estimate would not shorten the schedule.

Example: branch prediction Branch prediction is the canonical case. If the predictor guesses that a branch is taken, instruction fetch and decode proceed down the predicted target path immediately. A correct guess keeps the front end busy and reduces control-hazard stalls. An incorrect guess forces the machine to squash the speculative instructions and restart on the correct path, which introduces a misprediction penalty. The basic tradeoff is straightforward: high prediction accuracy yields high performance; low accuracy can erase the benefit of speculation.

1.3 Parallelism by Grain Size

Parallelism is best understood by *grain size*, meaning the amount of work grouped into one concurrent unit. At the smallest scale, bit-level parallelism appears in wide datapaths and arithmetic logic that operate on many bit positions at once. A 64-bit adder, for example, processes far more state per cycle than an 8-bit design because the hardware width itself exposes more fine-grained parallel work.

[2] January 13, 2026 lecture, Notability note *1. Syllabus*, *Recording 2*, approximately 14:41–15:25.

At *instruction-level parallelism (ILP)*, the machine overlaps independent instructions within a thread. Superscalar issue, out-of-order scheduling, and pipelining all exploit ILP. Consider two arithmetic instructions: `R1=R2+R3` and `R4=R5+R6`. Because neither depends on the other, a dual-issue machine can dispatch them together. In contrast, `R1=R2+R3` followed by `R4=R1+R6` introduces a data dependence that limits overlap. This illustrates the central ILP limitation: dependencies, control hazards, and hardware resource conflicts bound achievable parallel speedup. Figure 1.1 makes the introductory lecture's comparison explicit.

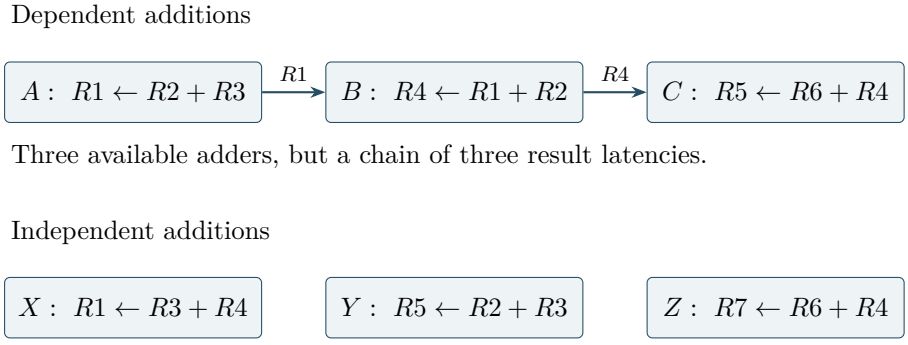


Figure 1.1: Hardware width is useful only when the work is independent. Assume equal add latency, ready inputs, three adders, and enough issue bandwidth; ignore fetch and write-back overhead.

At *thread-level parallelism (TLP)*, multiple threads execute concurrently on multithreaded or multicore hardware. TLP can provide substantial throughput gains, but only if the workload contains enough independent work and synchronization overhead remains controlled. Inter-program parallelism is the coarsest common level, where independent processes run simultaneously under an operating system scheduler. This is the dominant form in desktop multitasking and cloud services, where many programs share the same physical platform.

1.4 A Budget of Transistors

The January 15 lecture turns the power wall into a design question: if the next chip can contain more transistors, but running it faster would exceed a practical cooling budget, what should those transistors do?[3] A larger cache might avoid more memory trips. A wider core might execute more independent instructions. Additional cores might run more threads. These choices spend the same kind of physical resource on different sources of useful overlap.

The industry shift toward energy-efficient multicore designs in the mid-2000s makes that question concrete; Intel's 2006 Core introduction explicitly emphasized performance per watt and multicore operation. It did not mean that instruction-level parallelism or single-core improvement had ended. It meant that more transistors could no longer be treated as a promise of proportionally faster execution for one unchanged program. The available work, the memory system, and the power budget still decide what the extra hardware can accomplish.

[3] January 15, 2026 lecture, Notability note *2. ECE 6100*, *Recording 3*, approximately 06:47–08:05; related slides INTRO-18–19.

1.5 Flynn's Taxonomy

Flynn's taxonomy classifies parallel machines by counting instruction streams and data streams. In *single instruction, single data (SISD)*, one instruction stream acts on one data stream, matching the traditional scalar uniprocessor model. In *single instruction, multiple data (SIMD)*, one instruction stream is applied across many data elements in lockstep, which is highly effective for dense data-parallel computations such as vector math, image processing, and many graphics processing unit (GPU) kernels. The data values may differ across lanes, but the issued operation is the same across the whole SIMD group.

In *multiple instruction, multiple data (MIMD)*, multiple processing elements execute different instruction streams on different data streams. This is the dominant model for modern multicore central processing units (CPUs), distributed servers, and cluster systems because it supports general-purpose, irregular workloads. *multiple instruction, single data (MISD)* exists mostly in specialized contexts, such as certain fault-tolerant designs, and is uncommon in mainstream systems.

1.5.1 SIMD vs. MIMD

SIMD and MIMD solve different problems. SIMD is efficient when the same operation must be applied repeatedly to large collections of data. Its lockstep execution naturally provides synchronization across lanes, which reduces control overhead. If different cores or lanes truly follow different code paths at the same time, the machine is no longer acting like classical SIMD for that region; that behavior belongs to MIMD or to a SIMT-style execution model with divergence handling. MIMD is more flexible because each processor can follow its own control path, but this flexibility introduces synchronization and communication costs that must be managed explicitly by hardware and software.

Modern GPUs often use a hybrid model sometimes called *single instruction, multiple threads (SIMT)*. SIMT provides a SIMD-like execution efficiency for the hardware while allowing a MIMD-like programming model where each thread appears to have its own scalar program counter. The hardware manages the divergence and convergence of these threads, grouping those on the same path into SIMD-style execution units.

Representative workload mappings Simple examples help fix the taxonomy. A small embedded controller such as a thermostat is usually best viewed as SISD. A dense image transform that applies the same arithmetic to many pixels is a classic SIMD workload. A cloud-service cluster is naturally MIMD because many processors or machines execute different control paths for different requests. Some tasks, such as brute-force hash search, can be built in several ways: a lockstep array of identical workers looks SIMD-like, while a farm of largely independent workers looks MIMD. The important step is to identify whether the machine is reusing one instruction stream across many data elements or allowing many instruction streams to proceed independently.

1.5.2 Where Pipelining Fits

Pipelining is orthogonal to Flynn's taxonomy. It is a temporal form of parallelism in which different stages operate on different instructions concurrently. Because pipeline structure is an internal execution organization, pipelining can appear inside SISD cores, SIMD engines, and MIMD processors alike.

1.6 Vectors: Packed-Word SIMD vs. Pipelined Vector Units

Packed-word SIMD and classic vector pipelines both exploit data parallelism, but they do so with different execution structures. Packed-word SIMD divides a fixed-width register into lanes (for example, four 32-bit lanes in 128 bits) and applies one instruction across all lanes in lockstep. An operation such as `ADD_PACKED` performs multiple elementwise additions per instruction, and lane boundaries isolate carries so one lane's overflow does not spill into neighboring lanes. This style underlies CPU SIMD extensions such as streaming SIMD extensions (SSE), advanced vector extensions (AVX), and Arm NEON.

Vector pipelines, in contrast, stream long vectors through deep arithmetic pipelines. A floating-point multiply can be viewed conceptually as operand preparation, multiply, and result normalization/writeback stages. If a pipeline has k stages and processes n elements, a serial implementation needs nk cycles, while a pipelined implementation needs approximately $k + n - 1$ cycles after fill effects are accounted for (Eq. (1.1)):

$$\text{serial cycles} = nk, \qquad \text{pipeline cycles} \approx k + n - 1. \tag{1.1}$$

The result is substantial throughput gain for long vectors even though single-element latency is not eliminated.

Worked vector-throughput example Suppose a vector arithmetic pipeline has 6 stages and the program processes 100 elements. A non-pipelined implementation would spend

$$nk = 100 \times 6 = 600 \text{ cycles} \tag{1.2}$$

to process the vector. The pipelined form instead needs approximately

$$k + n - 1 = 6 + 100 - 1 = 105 \text{ cycles}. \tag{1.3}$$

The difference is the standard vector-pipeline gain: the pipeline still pays 6-stage latency for the first element, but after fill it can accept and complete additional elements at a much higher steady-state rate.

Chaining Vector chaining improves this further. If a second vector operation consumes results from a first operation, the hardware can forward early produced elements directly into the consumer pipeline instead of waiting for the full vector to complete. Chaining reduces producer-consumer delay and improves sustained throughput in vectorized loops.

Why the Cray example belongs here The introductory slides use the Cray-1 to make vector processing concrete.[4] Its vector registers could hold 64 elements, so one vector instruction described a stream of work rather than a single scalar operation. With chaining, an early element could enter the next functional unit while later elements were still passing through the first. The useful comparison is to a line of workstations: the second workstation can start on the first item without waiting for the entire batch. The six-stage calculation above is a teaching model, not a claim about the Cray-1's exact functional-unit latencies.

[4] `n1_intro.pdf`, INTRO-13–14; see also the Cray manual listed in Appendix F.

1.7 Common Chapter 1 Pitfalls

A common mistake is to conflate speculation with prediction. The correct view is that prediction is one way to make speculation more effective, not the definition of speculation itself. Another recurring error is to assume that additional cores guarantee proportional speedup; real speedup is limited by dependencies, communication, and serial regions of code. It is also common to treat SIMD and MIMD as interchangeable, even though they target different workload structures and have different programming implications.

A final source of confusion is the difference between packed-word SIMD and long-vector pipelining. Both process multiple data elements, but packed SIMD exploits lane-level parallel arithmetic within instruction-width registers, while vector pipelines exploit staged streaming over long element sequences. Related to this, pipelining primarily improves throughput rather than the latency of a single operation.

1.8 What to Carry Forward

Look for the work that can overlap, the assumptions that permit it, and the cost of being wrong. Grain size and Flynn's taxonomy describe different aspects of that overlap. Packed SIMD uses lanes; pipelined vectors stream elements through stages. Neither extra lanes nor extra cores can remove a true dependence. Chapter 2 turns these mechanisms into measurable changes in runtime and throughput.

1.9 Practice Questions

The following questions reinforce the main ideas of Chapter 1 and are intended for self-checking after the summary.

1. For each of the following techniques, state whether it primarily exploits *parallelism*, *speculation without prediction*, *speculation with prediction*, or more than one of these ideas: packed-word SIMD, a cache, a branch predictor, and out-of-order execution with branch prediction.
2. Classify each of the following workloads using Flynn's taxonomy, and give a one-sentence justification for each answer: a dense image convolution engine, a cloud-service cluster that handles independent requests, a small thermostat controller, and a brute-force hash search implemented as many independent workers.
3. Explain why pipelining is orthogonal to Flynn's taxonomy. Give one example of a pipelined SISD design and one example of pipelining inside a wider SIMD or MIMD system.
4. A vector arithmetic pipeline has 6 stages and processes a vector of 100 elements. Compute the approximate cycle count for a serial implementation and for a pipelined implementation. Then explain how chaining could reduce producer-consumer delay for two dependent vector operations.
5. Distinguish packed-word SIMD from classic pipelined vector execution. In particular, explain what is fixed by register width in packed SIMD and what is streamed through the pipeline in classic vector execution.

Chapter 2

Performance Evaluation

Imagine choosing between two computers by the number on the clock-speed label. The higher-frequency machine seems the obvious choice until its program executes more instructions or spends more cycles waiting on each one. This chapter gives us a way to follow the work from instructions to elapsed time, so that a convincing sales pitch and a faster computer do not become the same thing.

2.1 Why Performance Evaluation Matters

Architecture decisions are only meaningful if they are evaluated with the right metric. A design that improves one metric can degrade another, and a design that looks fast on one workload can be slow on the workload that actually matters. The goal of this chapter is therefore methodological: define performance correctly, choose representative workloads, and use mathematically correct summaries when comparing alternatives.

2.2 Throughput vs. Runtime

Two primary performance metrics appear repeatedly in computer architecture. *Runtime* (also called response time or turnaround time) measures how long one job takes from start to completion. *Throughput* measures how many jobs complete per unit time. These metrics are related but not interchangeable.

The January 20 lecture makes the distinction with an airline comparison: would you rather take a fast Concorde or operate a slower aircraft that carries many more passengers?[1] The passenger cares about time to the destination. The operator may care about how many passengers the aircraft can move per hour; extra seats can outweigh a longer trip in that comparison. This is a simplified capacity comparison, assuming full flights and ignoring turnaround time, not a complete airline business model.

The same split appears in computing. An interactive user wants one request to finish quickly; a service operator also cares about the rate at which the system completes the whole request stream. Neither metric can stand in for the other.

This distinction also explains why pipelining can be valuable even when individual operation latency does not fall dramatically. A deeper pipeline can increase the number of completed operations per unit time, improving throughput, while leaving single-operation latency unchanged or slightly worse.

[1] January 20, 2026 lecture, Notability note *3. ECE 6100*, *Recording 1*, approximately 02:00–05:15.

2.3 CPU Runtime Equation

The fundamental central processing unit (CPU) time equation used throughout performance analysis is (Eq. (2.1)):

$$\text{CPU time} = \text{Instruction Count (IC)} \times \text{Cycles per Instruction (CPI)} \times \text{Cycle Time (CT)}. \quad (2.1)$$

Using clock frequency $f = 1/\text{CT}$, the same equation is (Eq. (2.2)):

$$\text{CPU time} = \frac{\text{IC} \times \text{CPI}}{f}. \quad (2.2)$$

This decomposition is useful because each factor is controlled by different mechanisms. *Instruction count* is influenced by algorithm choice, compiler quality, and instruction set architecture (ISA) expressiveness. *Cycles per instruction (CPI)* is influenced by microarchitecture, hazard behavior, memory hierarchy behavior, and available parallelism. *Cycle time* is influenced by circuit and pipeline timing constraints.

For instruction mixes, total cycles are computed directly (Eq. (2.3)):

$$\text{Total cycles} = \sum_i (\text{IC}_i \times \text{CPI}_i), \qquad \text{CPI}_{avg} = \frac{\text{Total cycles}}{\text{Total instructions}}. \quad (2.3)$$

Worked example Consider a workload that executes 76 loads at 6 cycles each, 30 stores at 4 cycles each, 300 adds at 1 cycle each, and 58 multiplies at 5 cycles each. Then total cycles are

$$76 \cdot 6 + 30 \cdot 4 + 300 \cdot 1 + 58 \cdot 5 = 1166. \quad (2.4)$$

At 2.2 MHz, runtime is

$$\frac{1166}{2.2 \times 10^6} \approx 5.30 \times 10^{-4}\text{s} = 0.53\text{ms}. \quad (2.5)$$

If multiply latency is reduced from 5 cycles to 3 cycles, new total cycles become

$$76 \cdot 6 + 30 \cdot 4 + 300 \cdot 1 + 58 \cdot 3 = 1050, \quad (2.6)$$

which yields a speedup of

$$\frac{1166}{1050} \approx 1.11. \quad (2.7)$$

The example illustrates a recurring theme: accelerating only one instruction class can help, but total improvement depends on how often that class occurs.

Worked machine-comparison example Now consider three machines that all execute the same source-level program. Machine J runs at 5.1 GHz with $CPI = 3.0$. Machine K runs at 5.5 GHz with $CPI = 3.7$ and the same dynamic instruction count as J. Machine G runs at 6.1 GHz with $CPI = 3.2$ but requires 1.5× as many dynamic instructions. Using Eq. (2.2), the J-versus-K comparison is

$$\frac{T_K}{T_J} = \frac{3.7/5.5}{3.0/5.1} \approx 1.146, \quad (2.8)$$

so K is slower even though its clock is faster. For J versus G, runtimes are proportional to

$$T_J \propto \frac{1.0 \times 3.0}{5.1}, \qquad T_G \propto \frac{1.5 \times 3.2}{6.1}. \quad (2.9)$$

Numerically,

$$T_J \propto 0.588, \qquad T_G \propto 0.787. \tag{2.10}$$

So the faster clock does not rescue Machine G: the larger dynamic instruction count dominates and Machine J is faster overall. This is the central Chapter 2 discipline: compare complete CPU time, not any single factor in isolation.

2.4 RISC vs. CISC Through the Performance Equation

Reduced instruction set computer (RISC) and *complex instruction set computer (CISC)* are best understood as design tradeoffs, not as a simple "better/worse" ranking. Historically, CISC designs emphasized richer, denser instructions to reduce dynamic instruction count and code size. RISC designs emphasized simpler instructions and more regular pipelines, often improving CPI and enabling higher clock rates. In CPU-time terms, this is a direct trade (Eq. (2.11)):

$$\text{IC} \downarrow \text{ may come with} \text{CPI} \uparrow \text{ or} \text{CT} \uparrow, \quad \text{ or vice versa.} \tag{2.11}$$

In modern systems the boundary is less rigid than textbook definitions suggest, and x86 implementations, for example, decode complex instructions into simpler internal micro-operations and then execute them with highly RISC-like back ends. Arm designs, traditionally associated with RISC, also include increasingly rich instruction features to improve efficiency for modern workloads. The practical conclusion is that ISA style should be evaluated by end-to-end performance, power, and software ecosystem impact rather than by labels alone.

2.5 Amdahl's Law

Amdahl's Law quantifies the speedup limit when only part of a workload is accelerated. If fraction f of execution time benefits from a speedup factor s, total speedup is

$$S_{Amdahl} = \frac{1}{(1-f) + \frac{f}{s}}. \tag{2.12}$$

The limit as $s \to \infty$ is

$$S_{\text{upper}} = \frac{1}{1-f}. \tag{2.13}$$

This bound is the key insight: the unaccelerated fraction eventually dominates.

Worked example The performance lecture turns Amdahl's law into a purchasing decision: you simulate jet designs, and a spectacularly large machine promises 100,000 processors.[2] It is tempting to divide the old runtime by 100,000. But suppose only 80% of the original runtime can be parallelized, with ideal acceleration of that part and no extra communication cost. Then

$$S = \frac{1}{0.2 + \frac{0.8}{100000}} \approx 5.0. \tag{2.14}$$

If the old simulation took 100 hours, 20 hours remain serial regardless of the processor count. The other 80 hours shrink to 0.0008 hours, leaving just over 20 hours in total. Even 100 ideal processors would finish in 20.8 hours, already close to that limit. Figure 2.1 shows where the time goes. The purchasing question is now concrete: how much is that last fraction of an hour worth?

[2] `n2_perf.pdf`, PERF-18–20. The numerical baseline here is simplified to 100 hours.

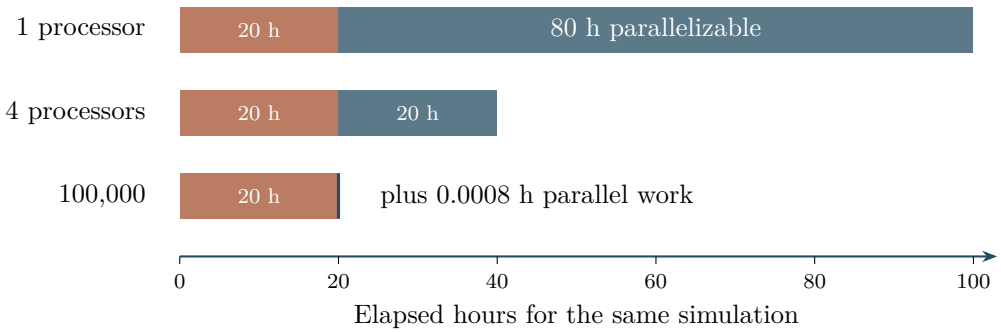


Figure 2.1: Amdahl's fixed-workload limit. The serial 20 hours do not shrink. Assume ideal division of the remaining work and no parallel overhead; the final blue segment is too short to show to scale.

Maximum-speedup interpretation When a problem states that a new mechanism speeds up "25% of a benchmark program," the 25% must be interpreted as a fraction of the *original execution time* of the whole benchmark unless stated otherwise. If that accelerated fraction were made infinitely fast, the maximum possible speedup would be

$$S_{\max} = \frac{1}{1 - 0.25} = \frac{4}{3} \approx 1.33. \tag{2.15}$$

This is the limiting-case form of Amdahl's bound: once the improved portion becomes arbitrarily fast, the remaining 75% of the original runtime dominates completely.

2.6 Gustafson's Law and Why It Does Not Contradict Amdahl

Gustafson's Law addresses a different question: instead of fixing problem size, it assumes the workload scales with available processors. Let α be the sequential fraction measured on an N-processor run. Scaled speedup is

$$S_{Gustafson} = \alpha + (1 - \alpha)N. \tag{2.16}$$

This often grows nearly linearly with N when α is small.

Amdahl and Gustafson do not contradict each other because their assumptions differ. Amdahl is *strong-scaling* analysis (fixed total work). Gustafson is *weak-scaling* analysis (larger total work as resources grow). Both are valid, but they answer different engineering questions.

Example With $\alpha = 0.15$ and $N = 16$,

$$S_{Gustafson} = 0.15 + 0.85 \cdot 16 = 13.75. \tag{2.17}$$

This can exceed Amdahl's fixed-size bound because the scaled workload includes more total useful work.

Return to the jet simulation. Perhaps the real goal is a more detailed airflow model within the same overnight run, rather than finishing the old model as quickly as possible. Additional processors can support that extra parallel work. This is the change of question in the lecture's transition to Gustafson: measure the serial fraction in the scaled parallel run, rather than quietly reusing the old fixed-workload fraction.

2.7 Combining Performance Numbers Correctly

Single-number summaries are useful only when the averaging method matches the quantity being averaged. For times (seconds per task), *arithmetic mean* is appropriate (Eq. (2.18)):

$$\bar{T}_{arith} = \frac{1}{n}\sum_{i=1}^{n} T_i. \tag{2.18}$$

For rates describing *equal amounts of work* in each workload, use the *harmonic mean* (Eq. (2.19)):

$$\bar{R}_{harm} = \frac{n}{\sum_{i=1}^{n} \frac{1}{R_i}}. \tag{2.19}$$

Using arithmetic mean on rates can select the wrong system. Suppose machine A runs two workloads at 4 and 4 transactions/s, while machine B runs them at 2 and 7 transactions/s. Arithmetic means are 4.0 and 4.5, which appears to favor B. But for one transaction of each workload, total throughput is determined by total work divided by total time (Eq. (2.20)):

$$R_A = \frac{2}{\frac{1}{4}+\frac{1}{4}} = 4, \qquad R_B = \frac{2}{\frac{1}{2}+\frac{1}{7}} = \frac{28}{9} \approx 3.11. \tag{2.20}$$

Under equal-work comparison, A is better. This is the lecture's warning about buying the slower machine by averaging the wrong quantity.[3] The result depends on equal work per workload. If instead the machine spends equal *time* at each rate, its total completed work divided by total time is the arithmetic mean of the rates. Start with total work and total time; the appropriate mean follows from what is held equal.

Weighted workload mixes Real benchmark suites are often not uniform. If workload i contributes fraction w_i of the jobs or requests, with $\sum_i w_i = 1$, then the weighted average runtime is

$$\bar{T}_{weighted} = \sum_i w_i T_i. \tag{2.21}$$

For rates under the corresponding fixed-work interpretation, the comparable aggregate rate is

$$R_{eq} = \frac{1}{\sum_i \frac{w_i}{R_i}}. \tag{2.22}$$

These weighted forms are the correct way to combine benchmark results when workloads do not occur equally often.

Worked weighted-runtime example Suppose three workloads occur with weights 0.50, 0.30, and 0.20, and one machine runs them in 4 s, 9 s, and 6 s, respectively. Then the weighted average runtime is

$$\bar{T}_{weighted} = 0.50(4) + 0.30(9) + 0.20(6) = 5.9 \text{ s}. \tag{2.23}$$

The same arithmetic form should *not* be applied blindly to rates. If the reported numbers were throughputs, the fixed-work interpretation would instead require the weighted harmonic form in Eq. (2.22).

[3] `n2_perf.pdf`, PERF-9–11.

2.8 Workload Dependence and Benchmarking

Performance numbers are workload-dependent. Miss ratio, CPI, and even effective clock behavior under thermal constraints depend on the executed program mix. A claim such as "processor X is faster" is incomplete unless it states the benchmark set, input sizes, and aggregation method.

Representative benchmarking therefore requires matching the measured workloads to expected deployment behavior. A web service, a game engine, and a scientific simulation stress very different execution paths and memory behaviors. Chapter 2 methodology demands that architects evaluate on relevant traces and report assumptions explicitly; otherwise conclusions will not transfer to real systems.

2.9 Cost-Aware Performance Comparison

In the January 15 lecture, the question "What is a supercomputer?" leads to a joke about the most expensive machine one could buy in a generation.[4] The joke connects the pursuit of maximum performance to the price of reaching it. An architect still has to ask what useful work that money buys for the intended workload. Raw speedup is therefore only one decision metric. If machine m costs $Cost_m$ and delivers speedup S_m relative to a chosen baseline, two useful value measures are

$$\text{speedup per dollar} = \frac{S_m}{Cost_m}, \qquad \text{cost per unit speedup} = \frac{Cost_m}{S_m}. \tag{2.24}$$

The first form is convenient when maximizing performance for a fixed budget. The second form is convenient when minimizing cost for a required performance gain. Both require the same discipline as every other Chapter 2 metric: use one baseline, one workload definition, and one aggregation method throughout the comparison.

Worked value-comparison example Suppose Server A costs \$4,000 and delivers speedup 1.25, while Server B costs \$5,400 and delivers speedup 1.55. Their speedup per dollar values are

$$\frac{1.25}{4000} = 3.125 \times 10^{-4}, \qquad \frac{1.55}{5400} \approx 2.87 \times 10^{-4}, \tag{2.25}$$

so Server A gives slightly more performance per dollar. Their cost per unit speedup values are

$$\frac{4000}{1.25} = 3200, \qquad \frac{5400}{1.55} \approx 3483.87, \tag{2.26}$$

which leads to the same conclusion under the reciprocal metric. This example is intentionally simple: the point is to show that cost-aware comparisons are still just ratio comparisons once a common baseline is fixed.

2.10 What to Carry Forward

Start with the useful work, then choose the metric: runtime for one job, throughput for a stream of jobs. Track instruction count, CPI, and cycle time together rather than declaring a winner from one factor. Keep the fixed-work and scaled-work questions separate, and choose an averaging rule that matches the workload mix. Those habits matter even more when the next chapter adds a memory hierarchy: a faster core can still spend its time waiting for data.

[4] January 15, 2026 lecture, Notability note *2. ECE 6100, Recording 2*, approximately 06:03–06:32. This retells the teaching point rather than quoting the speaker.

2.11 Practice Questions

The following questions reinforce the main ideas of Chapter 2 and are intended for self-checking after the summary.

1. Machine A executes a workload with $IC = 1.2 \times 10^9$, $CPI = 1.5$, and clock rate 2.4 GHz. Machine B executes the same workload with $IC = 0.9 \times 10^9$, $CPI = 1.9$, and clock rate 3.0 GHz. Compute the runtime of both machines and the speedup of the faster machine over the slower one.
2. Two machines execute the same program. Machine J runs at 5.1 GHz with $CPI = 3.0$. Machine K runs at 5.5 GHz with $CPI = 3.7$. Compute the speedup of K over J. Then consider Machine G, which runs at 6.1 GHz with $CPI = 3.2$ but requires $1.5\times$ as many dynamic instructions as J for the same work. Compute the speedup of G over J.
3. A benchmark mix contains three workloads with weights 0.50, 0.30, and 0.20. On one machine, their runtimes are 4 s, 9 s, and 6 s. Compute the weighted average runtime. Then explain why the same weighted arithmetic mean would be inappropriate if the reported quantities were throughputs instead of runtimes.
4. A proposed optimization accelerates the branch-recovery portion of a program, which currently accounts for 30% of total runtime, by a factor of 4. Use Amdahl's Law to compute the overall speedup. Then compute the maximum possible speedup if that portion were made infinitely fast.
5. A program spends 20% of its time in vectorizable code. Suppose an added accelerator makes only that portion run four times faster. Compute the overall speedup. Then explain why adding more parallel hardware gives diminishing returns when the accelerated fraction is fixed.
6. A parallel machine runs a scaled problem with serial fraction 0.10 on 16 processors. Compute the speedup predicted by Gustafson's Law. Then explain why this result does not contradict Amdahl's Law.
7. For each of the following, state whether arithmetic mean, harmonic mean, or a weighted variant is the correct aggregation method, and explain why: runtimes for an equal-work benchmark suite, throughputs for an equal-work benchmark suite, and runtimes for a production workload mix with known fractions.
8. Two candidate servers cost \$4,000 and \$5,400 and deliver speedups of 1.25 and 1.55 relative to the same baseline. Compute speedup per dollar and cost per unit speedup for both systems. State which system is preferable under each metric.

Chapter 3

Cache Memories

The next value a program needs may already be only a few millimeters from the execution unit, or it may require a much longer trip through the memory system. The program still issues a load in either case. The cache decides how often that simple instruction becomes a long wait. Its success rests on a practical observation: programs tend to return to the same data and to nearby addresses.

3.1 Memory Technology and the Hierarchy

Modern cache design begins with a technology reality: fast memory is expensive and dense memory is slower. *Static random-access memory (SRAM)* cells use more transistors per bit, provide lower access latency, and do not require refresh, so SRAM is the natural choice for on-chip caches. *Dynamic random-access memory (DRAM)* cells are denser and cheaper per bit but slower and refresh-driven, so DRAM is the natural choice for main memory capacity. This is why systems are built as hierarchies rather than as one uniform memory.

A useful mental model is that each level trades speed for size as distance from the core increases. L1 and L2 caches are normally SRAM-based, with L1 optimized most aggressively for low hit latency. Lower levels and main memory prioritize capacity and cost. The architectural problem is to serve enough accesses at the fast levels to justify their cost.

At the circuit level, one practical limiter is wire delay. Longer bit lines have higher effective capacitance and resistance, so sense amplifiers need more time to resolve values. This is one reason very large single-level caches become slow: making arrays taller increases bit-line delay even if logic depth is unchanged.

3.2 Typical Hierarchy Levels

In mainstream systems, the first level is a small L1 cache optimized for minimum hit time. A larger L2 cache is usually next, trading some latency for much lower miss ratio. Many processors add a larger last-level cache, often shared by several cores, before main memory (DRAM). The operating system may also move pages between DRAM and backing storage, such as an SSD. That transfer is managed through virtual memory rather than as another ordinary hardware-cache lookup.

In the simple demand-read model used for the timing calculations, a miss sends the request to the next level. That staged lookup makes average-access-time analysis recursive. Real lower levels also receive write-backs, prefetches, and coherence traffic, so not every transaction there represents a demand miss above.

Think of the few books open on a desk and the many books on a shelf. Keeping a useful book within reach saves repeated trips, but covering the desk with everything makes the working area harder to use. The analogy captures the capacity-versus-access-time tradeoff. The hardware adds a precise rule the desk does not have: each address maps to a particular set of candidate locations.

3.3 Locality: Why Caches Work

Cache behavior is dominated by *locality*. *Temporal locality* means that if data is used now, it is likely to be used again soon. *Spatial locality* means that if one address is accessed, nearby addresses are likely to follow. *Sequential locality* is a common special case of spatial locality in which accesses proceed in order, such as instruction fetch or array traversal.

These properties justify block-based transfer. Instead of moving one byte at a time from DRAM, the system moves an entire cache block, betting that adjacent bytes will be useful soon. The same properties also justify replacement policies that retain recently used data and evict older candidates.

3.4 Cache Geometry and Address Partitioning

To reason cleanly about cache structure, use the triplet (C, B, S) (Eq. (3.1)):

$$\text{cache capacity} = 2^C \text{ bytes}, \qquad \text{block size} = 2^B \text{ bytes}, \qquad \text{associativity} = 2^S \text{ ways}. \tag{3.1}$$

Address fields in (C, B, S) notation In this notation, S is **not** an extra address field beyond tag/index/offset. Instead, S changes the number of sets and therefore the index width.

$$\text{sets} = \frac{2^C}{2^B \cdot 2^S} = 2^{C-B-S}. \tag{3.2}$$

Total line count follows directly from sets and ways (Eq. (3.3)):

$$\text{lines} = \text{sets} \cdot \text{ways} = 2^{C-B-S} \cdot 2^S = 2^{C-B}. \tag{3.3}$$

For direct-mapped caches ($S = 0$), ways $= 1$ so lines $=$ sets. For set-associative caches ($S > 0$), lines $>$ sets because there are multiple ways per set.

$$|\text{Offset}| = B, \qquad |\text{Index}| = \log_2(\text{sets}) = C - B - S. \tag{3.4}$$

Because index plus offset consume $(C - S)$ low-order bits (Eq. (3.5)):

$$|\text{Index}| + |\text{Offset}| = (C - B - S) + B = C - S. \tag{3.5}$$

For an address width of AW bits (Eq. (3.6)):

$$|\text{Tag}| = AW - (|\text{Index}| + |\text{Offset}|) = AW - (C - S). \tag{3.6}$$

So the split remains (Eq. (3.7)):

$$\text{Address} = [\text{Tag} \mid \text{Index} \mid \text{Offset}], \tag{3.7}$$

with

$$|\text{Offset}| = B, \qquad |\text{Index}| = C - B - S, \qquad |\text{Tag}| = AW - (C - S). \tag{3.8}$$

Where the S bits "go" Higher associativity means fewer sets, so index width decreases by S relative to a direct-mapped organization with the same C and B. That is why S appears in index/tag equations but is not carried as an address field.

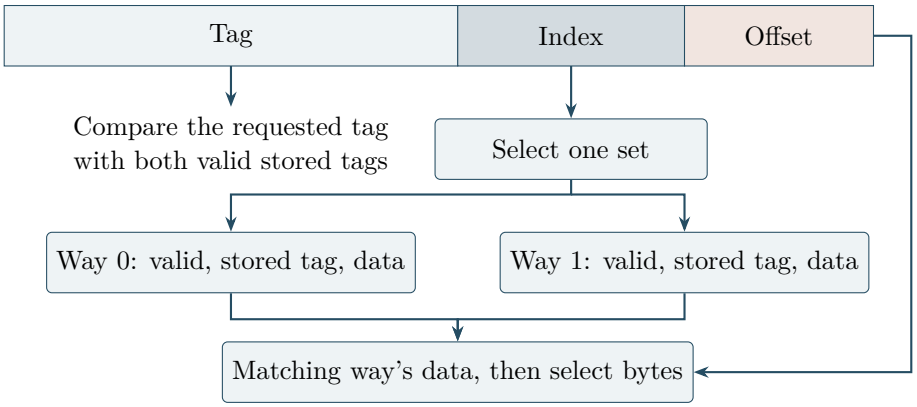


Figure 3.1: A conceptual two-way cache lookup. The index chooses a set, valid tag matches choose the way, and the offset chooses bytes within its block. Field widths are illustrative. If neither valid tag matches, the access misses.

What selects the way? The address tells the cache where to look, but it does not name a way. Figure 3.1 follows one access from the address fields to the matching line. The way is selected by hardware after tag comparison, not by an explicit address field:

1. Index selects one set.
2. Tags for all 2^S ways in that set are compared in parallel.
3. Match logic identifies the hitting way (if any).
4. Internal way-select control drives the data mux (conceptually an S-bit code since $\log_2(2^S) = S$).

Quick example: 32 KiB, 32 B blocks, 4-way With $(C, B, S) = (15, 5, 2)$ (Eq. (3.9)):

$$|\text{Index}| = C - B - S = 15 - 5 - 2 = 8. \tag{3.9}$$

For the same C, B but direct-mapped ($S = 0$) (Eq. (3.10)):

$$|\text{Index}| = 15 - 5 - 0 = 10. \tag{3.10}$$

So moving to 4-way associativity reduces index width by 2 bits (equivalently reduces sets by 4×).

32-bit address specialization If $AW = 32$ (Eq. (3.11)):

$$|\text{Offset}| = B, \qquad |\text{Index}| = C - B - S, \qquad |\text{Tag}| = 32 - (C - S) = 32 - C + S. \tag{3.11}$$

Worked example (metadata sizing) Assume 64-bit cache addresses and $(C, B, S) = (13, 5, 0)$: an 8 KiB direct-mapped cache with 32-byte blocks. The address width here is a stated modeling assumption, not something implied by a processor's 64-bit instruction set. The cache has $2^{13-5} = 256$ blocks and $2^{13-5-0} = 256$ sets. Tag width is $64 - (13 - 0) = 51$ bits. If each block stores one valid bit and one dirty bit, metadata per block is 53 bits. Total tag-store plus metadata is (Eq. (3.12)):

$$256 \times 53 = 13{,}568 \text{ bits}. \tag{3.12}$$

If B is reduced to 3 while C and S are fixed, the number of blocks rises to $2^{10} = 1024$, so tag overhead increases to $1024 \times 53 = 54{,}272$ bits. The data capacity did not change; the extra cost comes from keeping a tag for each of many smaller blocks.

Concrete Tag-Store Accounting The quickest reliable procedure is:

1. determine the number of lines from 2^{C-B},
2. determine tag width from the relevant physical or virtual address width,
3. add only the metadata bits the problem explicitly asks for, and
4. multiply per-line metadata by line count.

For example, suppose a cache is *write-back, write-allocate* and *physically indexed, physically tagged* with $(C, B, S) = (18, 7, 3)$, 64-bit virtual addresses, 4 KiB pages, and 32 GiB of physical memory. Assume physical memory occupies a contiguous byte-addressed range starting at zero with no additional address space to represent. The required physical address width is then 35 bits because $32\,\text{GiB} = 2^{35}$ bytes. Installed RAM capacity alone would not determine the implemented physical address width of a real processor. The cache therefore has $2^{18-7} = 2^{11} = 2048$ lines, and the physical tag width is

$$|\text{Tag}| = 35 - (18 - 3) = 20 \text{ bits}. \tag{3.13}$$

Because the level is write-back, write-allocate, each line also needs one valid bit and one dirty bit, so per-line tag-store plus required metadata is 22 bits. Total tag-store size is therefore

$$2048 \times (20 + 1 + 1) = 45{,}056 \text{ bits}. \tag{3.14}$$

If the problem says to ignore replacement metadata such as LRU state, then it should not be added.

Address partitioning example For a separate $(13, 5, 0)$ example, assume the cache uses 64-bit physical addresses. Take physical address `0000_1234_ABCD_5678`: **Offset (5 bits):** the lowest 5 bits of `78` (`0111_1000`) are `11000`, which is `0x18` (decimal 24). **Index (8 bits):** bits 5 through 12 come from hex digits `5`, `6`, and `7`. In binary these digits are `0101`, `0110`, and `0111`; extracting bits 5–12 yields `1011_0011` (binary), which is `B3`. **Tag (51 bits):** the remaining upper bits are the tag. This direct bit-mapping is the fundamental operation performed by hardware on every memory access. It illustrates how physical cache structure determines exactly which address bits are used for indexing and comparison.

3.5 Direct-Mapped, Set-Associative, and Fully Associative

A *direct-mapped cache* has one way per set, so each block address maps to exactly one location. Lookup is simple and fast, but conflict misses can be severe. A *set-associative cache* allows each index to hold multiple candidate blocks; it reduces conflicts at the cost of parallel tag comparisons and a way-select multiplexer. A *fully associative cache* removes index constraints entirely and minimizes conflict misses, but hardware search complexity and power become substantial for large capacities.

Under (C, B, S) notation, direct-mapped means $S = 0$. Fully associative means one set, so $C - B - S = 0$ and therefore $S = C - B$. This distinction is critical, because confusing these two conditions leads to incorrect address math and wrong simulator behavior.

How each cache is constructed Direct-mapped construction uses one data line and one tag entry per index. The index decoder selects one row, then one tag compare and one word select complete the access. A 2^S-way set-associative cache replicates tag and data entries 2^S times per

index; all tags in the selected set are compared in parallel, and a way-select mux picks the winning line. A fully associative cache has a single set containing all lines, so every access performs an associative search across the entire tag store and relies heavily on replacement metadata.

3.6 Hit, Miss, and Average Access Time

For a cache hit, access latency is the *hit time* HT. For a miss, the processor first spends hit-time logic discovering the miss, then pays *miss penalty* MP to fetch from the next level. The *average access time* is (Eq. (3.15)):

$$AAT = HT + MR \cdot MP, \tag{3.15}$$

where MR is *miss ratio*. Here MP is the *additional* time after miss detection, so hit time is paid once, not twice. This is an average service-time model; an out-of-order processor can overlap some of that time with other work, so AAT is not automatically the stall time per load.

If multiple workloads share one machine and workload i accounts for fraction w_i of the accesses, then weighted average access time is

$$AAT_{weighted} = \sum_i w_i \, AAT_i, \tag{3.16}$$

with $\sum_i w_i = 1$. This weighted form is the correct way to compare cache organizations against a non-uniform workload mix.

Hit time is not a single gate delay. It includes row decoding, word-line drive, bit-line sensing, tag comparison, way selection, and word/byte selection. Cache-geometry choices alter these terms. Taller arrays hurt bit-line delay, wider arrays increase mux and comparator complexity, and higher associativity increases tag-compare and selection cost.

Which change affects which AAT term? When a problem asks which of HT, MR, or MP changes, the most useful first-pass mapping is usually:

1. **data-layout or tiling changes** primarily move *miss ratio* because they change locality and conflict behavior,
2. **VIPT versus PIPT** primarily moves *hit time* because it changes whether TLB lookup can overlap cache indexing, and
3. **write buffers** primarily reduce effective *miss penalty* or visible lower-level service time on writes by decoupling cache progress from memory completion.

Use this classification as a starting point, then check which delays the particular design overlaps or changes.

3.7 The Three C's of Misses

Misses are classed as *compulsory*, *capacity*, or *conflict*. Compulsory misses are first-touch misses and are unavoidable unless data is brought in before first demand access. Capacity misses happen because the working set exceeds what the cache can hold over the reuse interval. Conflict misses happen when mapping restrictions force evictions even though total capacity would have been enough under freer placement. For a cold, demand-only trace, compare with a fully associative cache

of the same data capacity and block size under the stated reference replacement policy, commonly LRU. First references are compulsory; later misses that also occur in that reference cache are capacity misses, while misses avoided by it reveal placement conflicts. State the reference policy, since replacement choices can change the classification.

A full set is not necessarily a full cache The lecture makes the distinction concrete with five blocks competing for one four-way set.[1] Starting cold with LRU replacement, access A, B, C, D, E, A, with all five blocks mapping to that set and no other accesses in between. The first five references are compulsory misses; installing E evicts A, so the final access misses too. It is tempting to call that last miss a capacity problem because the set ran out of room. But if the whole cache can hold at least five blocks, a fully associative cache of the same capacity keeps all five and hits on the final A. The final miss is therefore a conflict miss: the shortage was placement freedom, not total storage.

In practice, this taxonomy provides a diagnosis strategy. If misses are mainly conflict-driven, associativity or victim structures can help. If misses are mainly capacity-driven, larger effective storage or better blocking can help. If misses are mainly compulsory, prefetching and layout changes are primary tools.

Sizing for a repeating working set For a repeating access pattern, the smallest cache that avoids capacity misses must hold every block that remains live across the reuse interval. If N_{live} distinct blocks must stay resident until reuse, then the cache needs at least N_{live} lines. That condition is necessary but not always sufficient in a direct-mapped or low-associativity cache, because several of those live blocks may still collide onto the same set. This is why trace analysis must separate "enough total lines" from "enough placement freedom."

3.8 Replacement Policies and Their Tradeoffs

When a set is full and a miss occurs, one resident block must be evicted. The cache lecture makes the choice concrete with four resident blocks and the access history

$$A,\ B,\ C,\ D,\ D,\ D,\ B,\ C,\ C,\ A,\ A,$$

followed by a request for E.[2] The cache knows exactly what happened, but the best victim depends on what will happen next. *Belady's optimal policy* evicts the block whose next use lies farthest in the future, providing a minimum-miss reference when the future trace is known. Hardware cannot ordinarily know that future. *Least recently used (LRU)* uses the past as a practical guide and evicts D in this example: repeated early uses do not make it recent now. LRU often works well under temporal locality, but it is not guaranteed to match the optimal policy. *First in, first out (FIFO)* ignores reuse and can evict hot lines. If A through D were inserted in that order, FIFO would evict A despite its two recent accesses. *Random* chooses a victim without maintaining a recency order. *Least frequently used (LFU)* tracks frequency but can lag phase changes.

For high associativity, exact LRU state grows costly. Practical designs use approximations, tree-based encodings, or hybrid insertion policies. A notable policy distinction is insertion behavior: a line can be installed as *most recently used (MRU)* (default in many demand-miss paths) or near LRU (useful for speculative prefetch lines to reduce pollution). Related insertion-policy abbreviations are *most recently used insertion policy (MIP)* and *least recently used insertion policy (LIP)*.

[1] January 29, 2026 lecture, Notability note *6. ECE 6100*, *Recording 3*, approximately 03:59–05:58.

[2] `n3_cache_all.pdf`, CACHE-20.

Simulating LRU with timestamps One exact way to simulate LRU is to attach a "last used" timestamp to each line in a set. On every hit, update the touched line's timestamp to the current access time. On a miss, fill an invalid line if one exists; otherwise evict the valid line with the smallest timestamp because that line was touched least recently. This method is conceptually simple and is often the most direct way to hand-simulate LRU on short traces.

Worked cache-trace decode Consider a cache with $(C, B, S) = (6, 4, 1)$. Then the cache has $2^{6-4} = 4$ lines, arranged as $2^{6-4-1} = 2$ sets with 2 ways per set. So the offset is 4 bits, the index is 1 bit, and the remaining upper bits form the tag. Now consider the access sequence `0x24`, `0x34`, `0x2C`, `0x44`. Because the offset is 4 bits, the block numbers are `0x2`, `0x3`, `0x2`, and `0x4`. Using the low-order bit of the block number as the set index, those accesses map to set 0, set 1, set 0, and set 0, with tags 1, 1, 1, and 2, respectively. The first two accesses are compulsory misses, the third is a hit on the already resident block for `0x24`, and the fourth is another miss because it introduces a different tag into set 0. This is the basic hand-simulation loop: decode block, split into set and tag, check the resident ways, then update replacement state.

Worked timestamp-LRU trace Consider a 2-way set with two accesses already resident:

Tag	Timestamp
A	4
B	7

If the next access is a hit on A at time 8, then A's timestamp becomes 8 while B remains at 7. If the following access is a miss at time 9 and no invalid line exists, then B is evicted because it has the smaller timestamp and is therefore the least recently used line. This is the mechanical rule to remember: every hit refreshes exactly one timestamp, and every full-set miss evicts the smallest timestamp.

Exact LRU with a bit matrix Another exact representation for an n-way set is an $n \times n$ recency matrix. Entry $M[i][j] = 1$ means way i is more recent than way j. When way k is accessed, set row k to 1, set column k to 0, and keep the diagonal in its fixed convention. After the update, the least-recently used way is the one whose off-diagonal row bits are all 0, because every other way is marked as newer than it.

Worked bit-matrix update For a 4-way set, suppose the current access is to way 2. Then row 2 is set to 1 against the other ways and column 2 is cleared:

	0	1	2	3
0	–	?	0	?
1	?	–	0	?
2	1	1	–	1
3	?	?	0	–

The question marks keep their previous values because the relative recency order among the untouched ways does not change. After several accesses, the LRU way is the row whose off-diagonal entries are all 0. That row is the one that every other way claims to be newer than.

3.9 Write Policies

Write behavior has two independent decisions. On a write hit, either update the cached copy and defer propagation (*write-back*), or also send the write to the next level (*write-through*). On a write miss, either install the block (*write-allocate*), normally fetching its contents before a partial write, or forward the write without installing it (*write-no-allocate*).

Write-back reduces memory traffic when repeated writes target the same line, but requires a dirty bit and careful eviction handling. Write-through avoids dirty state at that cache level but increases downstream write traffic; a write buffer can absorb some of the immediate delay. Neither policy by itself supplies cache coherence or durable storage. Another core can still hold a stale copy unless the coherence protocol coordinates it, and ordinary DRAM remains volatile even after a write reaches it. Persistence requires an appropriate nonvolatile destination and the system's required ordering and flush operations. The common pairings are write-back plus write-allocate for reuse-heavy data paths, and write-through plus write-no-allocate for streaming paths with little read-after-write reuse.

Reading write traces Write traces are easiest to interpret by separating three questions. First, does the access hit or miss in the current cache state? Second, does the policy allocate the block on a miss? Third, if a valid victim is evicted, is it dirty? In a write-back, write-allocate cache, a write miss usually installs the block and marks it dirty immediately. Each later dirty eviction contributes one writeback event. Dirty lines that remain in the cache at the end of the trace are not counted as writebacks unless the problem explicitly asks for a final flush.

Worked dirty-eviction trace Consider a write-back, write-allocate cache whose relevant set can hold only two lines. Suppose the set is initially empty and the access sequence is

$$W(A),\ W(B),\ W(A),\ R(C),$$

with C mapping to the same full set as A and B. The first write to A is a miss, so A is allocated and marked dirty. The first write to B does the same for B. The second write to A is a hit and leaves A dirty. When C is later accessed, one of the dirty lines A or B must be evicted; that eviction produces a writeback because the victim holds modified data that has not yet reached the next level.

Valid and dirty bits: meaning, need, and size impact Each cache line typically carries at least one *valid bit*. Valid= 1 means the tag/data entry contains a meaningful memory block; valid= 0 means the entry must not hit, even if tag bits happen to match after reset or replacement. A *dirty bit* is needed when that cache level uses *write-back*: dirty= 1 means the line was modified and eviction must preserve that data through a write-back, possibly queued in a buffer; dirty= 0 means eviction can discard the line without write-back traffic. In a pure write-through level, dirty bits are not required because every write is also sent to the next level; any queued write is the buffer's responsibility.

If line count is $N_{lines} = 2^{C-B}$, valid-bit storage is N_{lines} bits. Dirty-bit storage is another N_{lines} bits only when write-back is enabled at that level. Relative to data-array size 2^C bytes, overhead from these two bits alone is (Eq. (3.17)):

$$\text{overhead fraction} = \frac{n_v + n_d}{2^{B+3}}, \tag{3.17}$$

where $n_v = 1$ (valid always) and $n_d \in \{0, 1\}$ (dirty only for write-back). Example: with 32-byte blocks ($B = 5$), valid-only overhead is $1/256 \approx 0.39\%$, and valid+dirty overhead is $2/256 \approx 0.78\%$ (excluding tag, replacement, and coherence bits).

3.10 Reducing Miss Ratio

The first lever is **cache size (C)**. Increasing C can reduce capacity misses because more of the working set fits in the cache. However, larger caches increase hit time due to longer bit lines and decoder delays, and they consume more area and power.

The second lever is **associativity (S)**. Increasing S usually reduces placement conflicts by providing more locations for a given index. The benefit is most significant when moving from direct-mapped to 2-way or 4-way, after which returns diminish. The cost includes increased hit time (due to the way-select multiplexer and parallel tag comparisons) and increased power consumption.

The third lever is **block size (B)**. Increasing B reduces compulsory misses by exploiting spatial locality (fetching more useful data per miss). However, if B becomes too large, fewer lines fit and useful blocks may displace one another more often. More bytes also consume transfer bandwidth and, on a fixed-width serialized path, require more transfer beats. The processor-visible penalty need not increase when critical-word-first returns the requested word before the rest of the block. So the statement "increasing block size cannot improve miss ratio" is false. With a fixed cache capacity, a larger block often improves miss ratio at first by capturing nearby useful data, but the improvement is not monotonic because fewer total lines remain available.

Hardware Prefetching Hardware prefetching tries to move a predictable future access off the demand path. It can reduce demand misses without reducing total memory traffic, because the data still has to be fetched and some predictions are unused. A +1 *prefetcher* fetches block $i + 1$ when block i is accessed. A *stride prefetcher* detects constant deltas (for example $i, i + k, i + 2k$) and fetches the next predicted address. A *Markov prefetcher* uses a correlation table to learn non-linear transitions; if address A is often followed by B then C, the predictor learns and issues those future blocks when A appears. A *hybrid prefetcher* combines a richer predictor such as Markov correlation with a simpler fallback such as next-block prefetching, so the machine can still make a reasonable guess when the correlation table has no useful entry. To prevent *cache pollution* (evicting useful data for speculative data that might not be used), many designs use a *prefetch buffer*. This is a small, separate storage area where prefetched lines sit until they are actually demanded by the CPU. If demanded, they are moved into the main cache.

Prefetch insertion policy matters Prefetched lines are often installed with a low-priority policy such as *least recently used insertion policy (LIP)* rather than as most-recently used lines. The goal is to avoid immediately evicting useful demand data for speculative lines that may never be touched. This helps in set-associative caches, but it does not eliminate the placement limitations of a direct-mapped cache. In a direct-mapped design, a +1 prefetch can still evict useful demand data if the prefetched block maps to the same line.

Trace-driven prefetch statistics Counter names are part of a simulator's contract, not universal definitions. For the model used in Chapter 5, distinguish requests issued, useful prefetched fills, and unused prefetched evictions. *Prefetches issued* count speculative requests launched by the

prefetcher. A *prefetch hit* means a later demand access found a line that was present only because of an earlier prefetch. The model calls an eviction before first demand use a *prefetch miss* or useless prefetch. Other simulators use "prefetch hit" and "prefetch miss" for the prefetch request's own cache lookup instead, so check the specification before comparing reports. These counters matter because a prefetcher can lower demand misses and still be a poor design if it wastes bandwidth or pollutes the cache badly.

3.11 Reducing Miss Penalty

Miss penalty is shaped strongly by transfer mechanics. If a block has 2^B bytes and each bus beat carries up to W bytes, then the number of transfers is $N_{sb} = \lceil 2^B/W \rceil$. In the following examples W divides the block size, giving equally sized sub-blocks. With first-transfer startup latency T_1 and per-following transfer time T_f, block transfer time is (Eq. (3.18)):

$$MP_{full} = T_1 + (N_{sb} - 1)T_f. \tag{3.18}$$

Early restart returns data to the CPU as soon as the requested sub-block arrives rather than after full-block completion. If critical-word position is uniformly distributed and transfer order is fixed from sub-block 0 upward, average CPU-visible penalty becomes (Eq. (3.19)):

$$MP_{ER} \approx T_1 + \frac{N_{sb} - 1}{2} T_f. \tag{3.19}$$

Critical-word-first improves further by requesting the demanded sub-block first, then wrapping to complete the rest in the background. That benefit is only CPU-visible if the machine can resume execution as soon as the critical word arrives. Without early restart or an equivalent partial-consume mechanism, critical-word-first alone does not reduce the stall seen by the processor, because the core would still wait for the full block to complete before continuing.

Non-uniform critical-word distributions The uniform approximation in Eq. (3.19) is only one special case. If sub-block k is demanded with probability p_k and fixed-order transfer would deliver that word after k follow-up transfers, then the early-restart penalty is

$$MP_{ER} = \sum_{k=0}^{N_{sb}-1} p_k \left(T_1 + kT_f\right). \tag{3.20}$$

Under the same simple model, critical-word-first with early restart reduces the CPU-visible penalty to approximately T_1 because the demanded sub-block is requested first and the rest of the block finishes in the background.

Worked miss-penalty comparison Consider a 32-byte block on an 8-byte bus, so the miss repairs 4 words. Suppose each requested word transfer takes 70 ns and transfers are serialized. Baseline repair waits for all 4 words, so the visible miss penalty is

$$MP_{full} = 4 \times 70 = 280 \text{ ns}. \tag{3.21}$$

With early restart under a uniform critical-word position, the demanded word arrives after 1, 2, 3, or 4 transfers with equal probability, so the average visible penalty is

$$MP_{ER} = \frac{70 + 140 + 210 + 280}{4} = 175 \text{ ns}. \tag{3.22}$$

With critical-word-first plus early restart, the demanded word is requested first, so the processor-visible penalty returns to approximately one transfer time, 70 ns, while the rest of the block completes in the background.

Another major lever is adding lower cache levels. If L1 misses probe L2 before DRAM, L1 penalty becomes L2 average access time (Eq. (3.23)):

$$AAT_{L1} = HT_{L1} + MR_{L1} \cdot AAT_{L2}. \tag{3.23}$$

This recursive form extends naturally to deeper hierarchies.

Worked two-level AAT example Suppose $HT_{L1} = 1$ cycle, $MR_{L1} = 0.08$, $HT_{L2} = 8$ cycles, $MR_{L2,read} = 0.25$, and $DRAM_TIME = 60$ cycles. Then

$$AAT_{L2} = 8 + 0.25 \times 60 = 23 \text{ cycles}. \tag{3.24}$$

That lower-level average time becomes the miss penalty seen by L1, so

$$AAT_{L1} = 1 + 0.08 \times 23 = 2.84 \text{ cycles}. \tag{3.25}$$

The main reasoning pattern is recursive rather than additive-by-guess: first solve the lower level, then substitute that average behavior upward.

Inclusive policy: subset, not equality If L2 is *inclusive* of L1, then every line present in L1 must also be present in L2. The reverse is not required: L2 can contain many lines that are not currently in L1. So inclusivity means "L1 is a subset of L2," not "L1 and L2 contain exactly the same data." The subset statement concerns resident block addresses. With write-back private caches, a higher-level copy can be newer than the lower-level data; the protocol must still locate the current value.

Why many L2 ratios are read-only In a hierarchy where L1 uses write-back, write-allocate and L2 uses write-through, write-no-allocate, simulators often report separate L2 read and write traffic. Read hits and read misses form a meaningful reuse ratio because reads probe the resident L2 contents. Writes, by contrast, may simply reflect L1 writebacks or pass-through stores and may not allocate lines on a miss. For that reason, many reports define L2 hit ratio and miss ratio over *reads only* rather than over total L2 traffic.

Victim caches are also miss-penalty tools. A small fully associative buffer captures lines evicted from the primary cache. On a later conflict-driven re-reference, data can be swapped back quickly without going to DRAM. Conceptually, a victim cache provides a small amount of flexible associativity exactly where pressure appears.

Victim cache versus L2 A victim cache and an L2 do not solve exactly the same problem. A victim cache is usually the better value when misses are dominated by a small number of conflict patterns near the L1 boundary, because a few fully associative entries can undo repeated index collisions cheaply. A larger L2 is usually the better value when misses reflect broader capacity pressure or many different miss streams, because it filters a much wider set of misses rather than only recently evicted L1 lines. The right choice depends on whether the workload is mainly suffering from local conflicts or from a deeper shortage of effective capacity.

Write buffers reduce stalls by decoupling cache writes from lower-memory completion. They are especially useful when a following miss would otherwise wait behind pending writes. Correct implementation requires forwarding from the newest matching buffered write and careful ordering semantics.

3.12 Reducing Hit Time

Hit-time optimization frequently conflicts with miss-rate optimization. Smaller arrays and lower associativity usually improve hit latency. However, overly aggressive simplification can raise miss ratio enough to hurt total performance. In general, small caches and direct-mapped organizations are fast on the hit path, but they are often unacceptable unless miss ratio remains controlled by workload behavior or lower-level caches.

Address translation complicates this tradeoff. A naive *physically indexed, physically tagged* path can serialize translation lookaside buffer (TLB) lookup before cache access, increasing L1 hit latency. *Virtually indexed, physically tagged (VIPT)* caches overlap TLB translation with set indexing and preserve physical-tag correctness.

PIPT vs. VIPT timing view With virtual memory, the CPU generates a virtual address split into virtual page number (VPN) and page offset. In a *physically indexed, physically tagged (PIPT)* L1, both index and tag are defined by the physical address. A simple implementation waits for translation before starting the array lookup. Its serialized timing model is

$$HT_{L1,\text{PIPT}} \approx T_{TLB} + T_{L1,\text{index+array+tag}}. \tag{3.26}$$

In a *VIPT* L1, indexing starts immediately using low-order virtual bits while the TLB is translating in parallel. When the physical address arrives, the physical tag compare finalizes hit/miss. A first-order model is (Eq. (3.27)):

$$HT_{L1,\text{VIPT}} \approx \max(T_{TLB},\, T_{L1,\text{index+array}}) + T_{\text{phys_tag_compare}}. \tag{3.27}$$

VIPT can therefore hide much of TLB latency on the hit path. The serialized PIPT expression describes that implementation choice, not an unavoidable lower bound for every physically indexed design. Figure 3.2 shows what the overlap actually removes from the serial path.

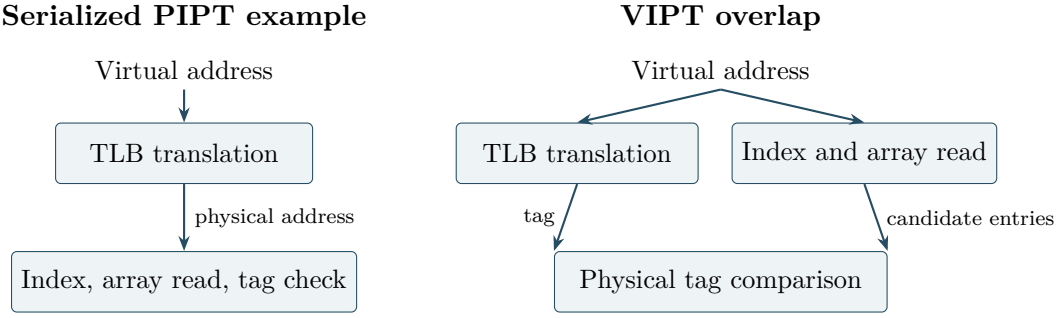


Figure 3.2: Overlapping translation with cache work. VIPT starts the array lookup using untranslated page-offset bits, then checks the physical tag. The simple version requires the entire set index to fit within the page offset.

Why the VIPT constraint exists For VIPT to work cleanly, index-plus-offset bits must fit within page-offset bits (Eq. (3.28)):

$$C - S \leq P, \tag{3.28}$$

where P is page-offset width in bits. If $C - S > P$, then some index bits come from VPN bits. Different virtual addresses that map to the same physical page (*synonyms*) can then index different sets, allowing duplicated physical data in L1. That creates alias-management problems: writes through one synonym may not immediately update the other cached copy, risking stale reads until extra coherence/synchronization machinery repairs it. This constraint often forces minimum associativity for a given L1 size and page size. For example, with $P = 12$ (4 KiB pages) and $C = 15$ (32 KiB L1), one needs $S \geq 3$, i.e., at least 8-way associativity. If $S = 2$ (4-way), then $C - S = 13 > 12$, so VIPT indexing would extend beyond page offset and synonym risk appears.

Concrete VIPT Associativity Check Suppose $C = 18$ and page size is 4 KiB, so $P = 12$. To make the cache VIPT without indexing beyond the page offset, require

$$C - S \leq P \Rightarrow 18 - S \leq 12 \Rightarrow S \geq 6. \tag{3.29}$$

So the minimum acceptable associativity field is $S = 6$, which corresponds to $2^6 = 64$ ways. The arithmetic is simple; the main thing to remember is that the constraint is on *index plus offset*, which in (C, B, S) notation is exactly $C - S$.

Write buffers can also improve effective hit behavior for writes by decoupling write completion from the immediate critical path and by forwarding recent write data when needed. Forwarding can let a load read a preceding buffered store's value without waiting for that store to drain.

3.13 Virtual Memory Technology, Terminology, and Analogy to Caching

Virtual-memory terminology mirrors cache terminology closely. A virtual address is typically split into a *virtual page number (VPN)* and page offset. Translation maps VPN to a *physical frame number (PFN)* through a page table. A *translation lookaside buffer (TLB)* is a cache of recent translations. A TLB miss means the translation is not cached, not that the data page is absent from DRAM. A hardware walker or a software-managed miss handler looks up the page-table entry and checks its permissions. An absent or disallowed mapping can cause a page fault for the operating system to handle. That handling might load a page from storage, allocate a zero-filled page, arrange copy-on-write, or reject an invalid access; it does not always involve disk I/O.

The analogy to caching is direct. VPN behaves like a tag, page offset behaves like block offset, and the TLB behaves like a small associative cache of mapping metadata. Likewise, DRAM behaves as a cache for backing store, while on-chip caches behave as caches for DRAM. This layered view is useful because it applies one conceptual framework to both hardware cache hierarchy and OS-managed virtual memory.

3.14 Compiler and Data-Layout Techniques

Software organization strongly affects cache performance. **Compiler layout of instructions** places frequently executed basic blocks close together in memory to improve I-cache hit rate and reduce

fetch stalls. **Compiler-directed prefetching** issues non-binding prefetch instructions ahead of use, which is effective for predictable array accesses. **Data-layout choices**, including structure-of-arrays (SoA), field reordering, and padding, can reduce cache-line waste and false sharing.

Tiling (blocking) re-organizes loops to process data in smaller cache-resident regions. For a matrix transpose, blocking keeps a small region of the source and destination active together. Assume distinct source and destination arrays. The following pseudocode uses min to handle a final partial tile, so N need not be a multiple of the positive tile size b.

Before tiling

```
for (i=0; i<N; i++) for (j=0; j<N; j++) A[i][j] = B[j][i];
```

After tiling

```
for (ii=0; ii<N; ii+=b) for (jj=0; jj<N; jj+=b)
    for (i=ii; i<min(ii+b,N); i++)
        for (j=jj; j<min(jj+b,N); j++) A[i][j] = B[j][i];
```

Choosing the tile factor is a working-set problem. A footprint calculation gives an upper bound, not a guarantee that the largest fitting tile is fastest. Leave room for all simultaneously active arrays, other live data, and the placement restrictions of the cache. For direct-mapped or low-associativity caches, this means a tile can fail even when the byte count appears to fit, because the active blocks may still alias onto the same few sets. The practical rule is therefore: first check footprint size, then check placement conflicts. This is why tiled matrix multiplication often shows much better cache behavior than a naive matrix multiply, even when both perform the same arithmetic work.

Concrete Tile-Factor Estimate Consider a cache with $(C, B, S) = (10, 6, 2)$, so total capacity is $2^{10} = 1024$ bytes, block size is 64 bytes, and associativity is 4-way. For this simplified footprint calculation, count only TF active rows of one matrix, with 32 eight-byte elements per row. Assume rows are contiguous and block-aligned, and no other data competes for the cache. One row occupies

$$32 \times 8 = 256 \text{ bytes.} \tag{3.30}$$

A footprint estimate for the tile is therefore $256 \times TF$ bytes. The largest tile that still fits in 1024 bytes is

$$256 \times TF \leq 1024 \Rightarrow TF \leq 4. \tag{3.31}$$

There are four sets, and each aligned row occupies four consecutive blocks, one in each set. Four rows therefore occupy four ways in every set, so this particular layout also passes the placement check. A real transpose has both source and destination data active; those additional blocks must be counted before using the same bound.

Review Emphasis: What Tiling Is Really Buying Tiling does not change the arithmetic performed by the program. It changes the order in which the program visits data so that reuse happens while the relevant blocks are still cache-resident. The review-problem reasoning is usually: identify the active data touched by one tile, multiply by bytes per element, require that footprint to fit in the effective cache capacity, and then remember that associativity can still matter. If the

tile fits by byte count but all active blocks map to too few sets, conflict misses can still defeat the transformation.

Loop fusion combines compatible loops that touch related data to improve temporal locality, while preserving their dependences. In the example below, the second computation consumes the value just produced for the same element. Assume the arrays do not overlap. Fusion moves that reuse closer in time.

Separate loops

```
for (i=0; i<N; i++) a[i] = b[i] + 1;
for (i=0; i<N; i++) c[i] = a[i] * 2;
```

Fused loop

```
for (i=0; i<N; i++) { a[i] = b[i] + 1; c[i] = a[i] * 2; }
```

In practice, these compiler and data-layout techniques frequently determine whether hardware cache features help or saturate.

3.15 What to Carry Forward

A cache earns its place by shortening the common access without making the uncommon access too expensive. More capacity, more ways, and larger blocks change different parts of that bargain. Use address geometry to determine where a block can live, a trace to determine which blocks remain there, and AAT to connect the resulting hit and miss behavior to a service-time model.

Software helps by arranging reuse before eviction; prefetching tries to arrange arrival before demand. Neither creates free capacity or bandwidth. When a design improves a counter, follow the consequence through the rest of the hierarchy before declaring it faster. Keeping data nearby addresses one source of waiting; the next chapter asks how to overlap the instruction work around it.

3.16 Practice Questions

The following questions reinforce the main ideas of Chapter 3 and are intended for self-checking after the summary.

1. A physically indexed, physically tagged cache has total capacity 32 KiB, block size 64 B, associativity 4-way, and 48-bit addresses. Compute the number of sets, the number of offset bits, the number of index bits, and the number of tag bits. Then compute the total tag-store size if each line stores a valid bit and a dirty bit in addition to the tag.
2. For each of the following changes, state whether it primarily targets *hit time*, *miss ratio*, or *miss penalty*: adding a victim cache, adding a write buffer, increasing associativity, increasing block size, and using critical-word-first with early restart.
3. Explain the difference between critical-word-first and early restart. Why does critical-word-first by itself not guarantee a reduction in processor-visible miss penalty?

4. A system uses 32-byte blocks, an 8-byte memory bus, and serialized memory transfers. A miss first incurs a 70 ns round trip for the requested word, and each additional word also requires 70 ns. The cache hit time is 10 ns and the miss ratio is 15%. Compute the average access time under three cases: baseline repair with no early use of returned data, early restart with a uniform probability of needing any of the four words in the block, and critical-word-first with early restart.
5. Consider a $(C, B, S) = (6, 4, 2)$ cache with LRU replacement and the access stream `0x589e`, `0xa7a1`, `0x3767`, `0xcce2`, `0xa7aa`, `0xff4d`, `0x376c`, `0x5892`. For each access, determine the set, tag, hit or miss outcome, and final contents of the tag store after the full trace.
6. A 4-way set uses an exact LRU bit-matrix implementation. Starting from a state in which way 0 is least recently used and way 3 is most recently used, update the matrix for the hit sequence $2 \to 0 \to 3 \to 1 \to 2$. Which way is least recently used at the end?
7. A VIPT L1 cache has capacity 64 KiB, block size 64 B, and page size 4 KiB. Compute the minimum associativity required so that the set index fits entirely within the page offset.
8. A miss stream repeatedly alternates between two blocks that map to the same direct-mapped L1 set. Would a small victim cache or a simple +1 prefetcher be the more natural first fix? Explain why.
9. A tiled matrix kernel keeps three $b \times b$ tiles of double-precision data active at once. If the effective L1 capacity available to the tiles is 24 KiB, what is the largest integer tile size b that satisfies the footprint constraint $3b^2 \cdot 8 \leq 24$ KiB?
10. A loop has reuse across a large array, but the full working set is much larger than the L1 cache. Explain why tiling can reduce misses even though the program performs the same loads, stores, and arithmetic overall.
11. A two-level hierarchy has $HT_{L1} = 1$ cycle, $MR_{L1} = 0.08$, $HT_{L2} = 8$ cycles, $MR_{L2,read} = 0.25$, and $DRAM_TIME = 60$ cycles. Compute AAT_{L2} and then AAT_{L1}.
12. A write-back, write-allocate cache receives the sequence: write block A, write block B, write block A again, then access block C, which maps to the same full set as A and B and forces an eviction. Describe when dirty state is created and whether the eviction causes a writeback.

Chapter 4

Pipelining

An instruction passes through several kinds of work: finding the instruction, reading operands, calculating, accessing memory, and recording a result. If each part has its own hardware, the machine need not finish the first instruction before starting the second. Like successive items moving through workstations, instructions can occupy different stages at once. The gain comes from a regular flow of finished work; a dependency or a wrong turn interrupts that flow.

4.1 Pipeline Motivation

Pipelining improves throughput by overlapping instruction execution stages in time. Instead of waiting for one instruction to complete all work before starting the next, the processor places stage registers between major datapath functions and allows multiple instructions to be in flight simultaneously. The key gain is not that one instruction becomes instantly shorter in latency, but that completed instructions per unit time rise after the pipeline fills.

4.2 Five-Stage Example Pipeline

The canonical integer pipeline has five stages (Eq. (4.1)):

$$IF \rightarrow ID \rightarrow EX \rightarrow MEM \rightarrow WB. \tag{4.1}$$

Instruction Fetch (IF) reads the instruction from the instruction memory hierarchy and advances the program counter (PC). Instruction Decode (ID) decodes opcode and operands and reads source registers. Execute (EX) performs ALU work and effective-address calculation. Memory (MEM) performs data-memory access for loads and stores. Write Back (WB) writes results to the architected register file.

Even instructions that do not logically need memory often still occupy the MEM stage in a basic in-order five-stage design so that all instructions remain aligned in a uniform stage progression. This simplifies control and preserves in-order retirement semantics. Figure 4.1 shows the overlap before hazards complicate it.

4.3 Pipeline Speedup Calculation

Suppose an unpipelined design executes one instruction in N stage-equivalent delays, each of delay T_s. Its instruction time is approximately NT_s. A pipelined design with stage registers has clock (Eq. (4.2)):

$$T_{clk,pipe} \approx T_s + T_{reg}, \tag{4.2}$$

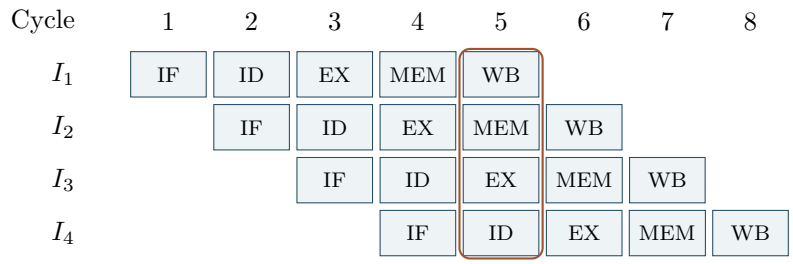


Figure 4.1: Four instructions take eight cycles in an ideal five-stage pipeline. Highlighted cycle 5 contains work from all four instructions. Assume one cycle per stage, no hazards, and separate resources wherever stages overlap.

where T_{reg} is latch/flip-flop overhead. For a long stream, ideal CPI approaches 1, so time per instruction approaches $T_{clk,pipe}$ and ideal speedup becomes (Eq. (4.3)):

$$S_{ideal} \approx \frac{NT_s}{T_s + T_{reg}}. \tag{4.3}$$

If T_{reg} were zero, this approaches N. In practice, register overhead and stalls reduce realized speedup.

Finite-length form For M instructions in an N-stage pipeline (ignoring hazards), total cycles are $N + M - 1$, so (Eq. (4.4)):

$$T_{pipe} \approx (N + M - 1)T_{clk,pipe}, \qquad T_{unpiped} \approx MNT_s. \tag{4.4}$$

This shows fill and drain overhead is more visible for short instruction sequences.

Worked pipeline-speedup calculation Suppose an unpipelined implementation has effective $CPI = 5$, and a 5-stage pipelined version keeps the same clock period. The ideal maximum speedup is therefore 5. If 25% of instructions now experience one extra stall cycle, the pipelined CPI becomes

$$CPI_{pipe} = 1 + 0.25 \times 1 = 1.25. \tag{4.5}$$

The corresponding speedup relative to the original $CPI = 5$ design is then

$$\frac{5}{1.25} = 4, \tag{4.6}$$

which is below the ideal bound because hazards consume part of the pipelined throughput gain.

4.4 CPI Model and Stall Accounting

In steady state without hazards, ideal CPI is 1 (Eq. (4.7)):

$$CPI_{ideal} = 1. \tag{4.7}$$

With hazards, the practical model is (Eq. (4.8)):

$$CPI = 1 + \text{stalls}_{avg}, \tag{4.8}$$

where stalls_{avg} is average stall cycles per instruction. This form is central in Chapter 4: almost every pipeline optimization is either reducing stall frequency, reducing stall duration, or both.

4.5 Program Characteristics Plus Hazards Produce Stalls

Hardware hazards are potential conflicts. A stall occurs only when the running program triggers that conflict. This yields a guiding principle (Eq. (4.9)):

$$\text{Program characteristics} + \text{pipeline hazards} \Rightarrow \text{stalls}. \tag{4.9}$$

For example, a branch hazard matters only when branches occur in the dynamic instruction stream. A structural hazard from a shared memory port matters only when fetch and data access overlap in a conflicting cycle.

4.6 Structural Hazards

Structural hazards arise from insufficient hardware resources. A classic example is a single-ported memory system that must serve both IF and MEM stages. If one instruction needs data memory while another needs instruction fetch in the same cycle, one must wait, causing a stall.

Common avoidance strategies are straightforward architectural provisioning: split L1 into separate instruction and data caches, add ports, duplicate critical units, or schedule around known bottlenecks. In modern designs, many obvious structural hazards are eliminated in hardware because their stall cost is predictable and severe.

CPI impact example If 20% of instructions incur one extra stall cycle due only to a structural hazard, then the average CPI becomes

$$CPI = 1 + 0.2 \times 1 = 1.2. \tag{4.10}$$

This is the same accounting pattern used for other hazards: identify the frequency of the event and multiply by its average stall contribution.

4.7 Data Hazards

Data hazards arise when overlapping instructions could read or overwrite an operand before the program's dependences permit it. The dependence comes from the program; whether it requires a stall depends on where the pipeline produces and consumes the value.

Read After Write (RAW) A *read-after-write (RAW)* hazard occurs when an instruction refers to a result that has not yet been calculated or stored by a preceding instruction. This is the most common hazard in the classic 5-stage pipeline.

```
I1:  ADD R1, R2, R3    I2:  SUB R4, R1, R5
```
(4.11)

In this case, `I2` reads register operands in ID and needs the correct value at its EX input, but `I1` does not write `R1` to the register file until WB. Without intervention, `I2` would read a stale value.

Write After Read and Write After Write *Write-after-read (WAR)* and *write-after-write (WAW)* hazards do not typically occur in the simple 5-stage in-order pipeline because instructions read early (ID) and write late (WB) in a fixed order. However, they become critical in out-of-order execution or pipelines with multiple completion units of varying latencies.

4.7.1 Forwarding (Bypassing)

Forwarding allows a result to be used as an input to a subsequent instruction before it is written to the register file. If the result is already available in a pipeline register (such as EX/MEM or MEM/WB), hardware can forward it directly to the ALU inputs. This eliminates many RAW stalls.

4.7.2 Load-Use Stalls

Forwarding cannot solve every RAW hazard. If a load instruction is followed immediately by an instruction that uses the loaded value, the data is not available until the end of the MEM stage, which is too late for the EX stage of the following instruction.

```
I1:  LW R1, 0(R2)    I2:  ADD R3, R1, R4
```
(4.12)

In this five-stage model, with one-cycle MEM and no earlier load-data path, the pair requires a one-cycle load-use stall even with forwarding. Compilers attempt to schedule independent instructions into these "load-delay" slots to maintain performance.

4.7.3 Same-Cycle Register-File Write Then Read

One subtle pipeline detail is the timing of register-file writes relative to reads in the same cycle. If the register file writes in the early part of the cycle and decode reads in the later part, then an instruction in ID can sometimes see a value written back by an older instruction in WB in that same cycle. If that timing is not supported, the machine needs either an explicit WB-to-ID bypass path or an extra stall. This detail matters when reading timing diagrams because two pipelines with the same five visible stage names can still differ by one cycle on closely spaced dependences.

4.8 Control Hazards

Control hazards occur because the fetch stream does not immediately know the correct next PC for branches and jumps. The branch creates uncertainty in both target and direction (taken/not taken), and the pipeline may fetch wrong-path instructions that must later be squashed.

CPI impact example If branches occur every five instructions on average (branch frequency = 0.2) and each branch introduces one average stall cycle in a simple pipeline, then

$$CPI = 1 + 0.2 \times 1 = 1.2. \tag{4.13}$$

This simple calculation captures why even modest branch penalties have visible throughput cost.

4.8.1 Delay Slots

A *delay slot* ISA semantic says that one or more instructions after a branch execute before control transfer takes effect. The compiler tries to fill these slots with useful independent work. This can reduce explicit branch stalls when suitable instructions exist, but becomes increasingly difficult as pipelines deepen and dependence constraints tighten.

How software fills delay slots Ordinary, non-annulled delay-slot instructions execute whether the branch is taken or not. So a compiler may place an instruction into a delay slot only if doing so is safe on *both* outcomes. The usual search order is:

1. move an independent instruction from before the branch down into the slot,
2. if control-flow structure permits, move an instruction from a successor block only when it is safe for both paths, or
3. leave the slot as a `NOP` if no safe candidate exists.

This is why branch schedules often still leave some slots unfilled: correctness takes priority over slot utilization.

4.8.2 Squash Slots with Likely Bits

Some designs expose *squash-slot* behavior via branch-likely style semantics. The compiler places a candidate instruction near the branch and encodes likely behavior. Hardware keeps or squashes that instruction based on actual branch resolution and likely-bit policy. This improves performance when heuristics are good but complicates ISA semantics and compiler scheduling.

How squash-slot filling differs Squash slots are more permissive than delay slots because wrong-path instructions may be discarded. If the machine assumes a branch is likely taken, the compiler can place instructions from the predicted taken path into those slots. If the prediction is wrong, hardware squashes them. So the key distinction is:

- **delay slot:** instruction must be correct regardless of branch outcome,
- **squash slot:** instruction only needs to be correct on the predicted path because hardware may cancel it.

Cancellation must also prevent wrong-path stores, register updates, and exceptions from becoming architectural effects. The exact condition for annulling a slot belongs to the ISA; it need not match the behavior of another branch-likely instruction.

4.9 Branch Target Buffer (BTB)

The *branch target buffer (BTB)* predicts where a branch goes (target prediction) early enough to guide fetch.

4.9.1 BTB with Tags

A tagged BTB stores branch PC tags and associated targets. On fetch, the BTB is indexed and tag-compared; a valid tag hit yields a predicted target. With enough tag bits to identify the branch

PC, a tag mismatch rejects an entry belonging to a different branch. Partial tags reduce the storage cost but can still produce false matches. Tagging therefore controls destructive *aliasing*, with an area and lookup cost that depends on the implementation.

4.9.2 BTB without Tags

A tagless BTB behaves like a hashed direct lookup table from index to target metadata. It is fast and compact but prone to *aliasing*: unrelated branches may map to the same entry and pollute each other's predictions. This can still be acceptable if the table is sized well and collision rate is low for target workloads, trading accuracy for substantial area savings.

4.9.3 Hardware Return Address Stack

Function returns are indirect branches whose targets are difficult for generic BTB logic. A small hardware *return-address stack (RAS)* records call return PCs and pops on return, producing very high return-target accuracy in common call/return patterns. The stack works because ordinary nested calls return in the reverse order from which they were made. Finite depth, unusual control flow, or a speculative call later squashed can disturb that pattern; speculative RAS updates need recovery just as other prediction state does.

A tiny predictor for deep recursion In the lecture, Conte uses recursive factorial to ask what a small RAS should do when many calls share one return address. Suppose an outside caller saves address C, and four recursive calls at the same instruction each save address R. The real unwind needs R, R, R, R, C, but a three-entry RAS that discards its oldest entry on overflow retains only R, R, R. For this illustration, each pop shifts the remaining entries upward and repeats the old bottom entry at the bottom. It therefore predicts R on all five returns: the recursive returns are right, and the final return to C is wrong. Figure 4.2 separates the lost prediction history from the program's intact call state.

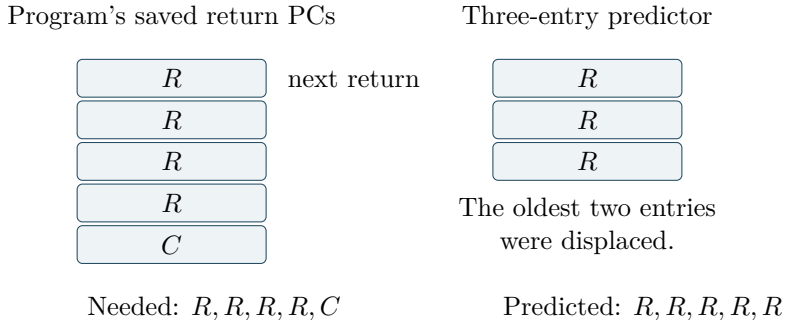


Figure 4.2: Deep recursion need not require a deep predictor to get most returns right. Here overflow drops the oldest entry, and a pop repeats the bottom prediction. Only the final return to the outside caller misses. The program's saved return addresses remain correct.

This is one illustrative refill policy, not a requirement for every RAS. The useful question is not whether a counterexample exists, but whether extra predictor storage avoids enough mispredictions to justify its cost. A RAS predicts where to fetch; it does not replace the program's actual saved return addresses. When a prediction is wrong, resolution of the return redirects execution to the real target.

Worked BTB/RAS trace Assume a direct-mapped BTB indexed by $PC[7:4]$, with full branch-PC checks, initially invalid entries, and an initially empty RAS. Instructions are four bytes long. For this exercise, calls and taken conditional branches allocate or update the BTB at resolution; returns use only the RAS and do not allocate a BTB entry. Now consider the short control-flow sequence

```
0x4000:  CALL 0x4A00
0x4A00:  BREQZ R1,0x4AF0 (taken)
0x4AF8:  RET
```

The call pushes return address `0x4004` onto the RAS and creates or refreshes a BTB entry for the call target `0x4A00`. The conditional branch updates its BTB entry with the taken target `0x4AF0`. Both PCs select index 0, so this update replaces the call's entry rather than storing a second target in that slot. The return then pops the top of the RAS and predicts target `0x4004`. The main bookkeeping distinction is that the BTB remembers branch and jump targets by PC, while the RAS handles the nested call-return discipline explicitly.

BTBs still matter for unconditional control transfers A BTB is not useful only for conditional branches. Even if every control transfer is unconditional, fetch still benefits from knowing the target early. Without a BTB or some equivalent early-target mechanism, the pipeline may need to wait until later decode or execute logic computes the jump target, creating avoidable fetch bubbles.

4.10 Whether-To Branch: Direction Prediction

Target prediction answers where a branch goes. *Direction prediction* answers whether it is taken. A simple static method is software heuristics, especially *backward taken / forward not taken (BTFNT)*: loop back-edges are often taken, many forward conditionals are often not taken. Compilers can also use profile-guided statistics to set likely-direction metadata (for example branch-likely annotations).

Dynamic predictors improve further by learning runtime behavior (for example saturating counters and history-based schemes). Later branch resolution generally increases the cost of a wrong prediction, making direction accuracy more valuable as the recovery distance grows.

A very simple baseline is the *always-taken* predictor. It ignores PC and history entirely and predicts that every conditional branch will be taken. This is crude, but it remains a useful reference point because many loop back-edges are taken most of the time. More sophisticated predictors should therefore be judged not only against one another, but also against simple baselines such as backward taken / forward not taken and always taken.

4.11 Simple Dynamic Direction Predictors

The simplest dynamic predictors store a small amount of state per static branch or per BTB entry.

4.11.1 One-Bit Prediction

A *1-bit predictor* records only the last observed direction for a branch. In a compact BTB-style design, this may appear as a single prediction bit associated with the target entry. If the last outcome was taken, predict taken next time; if the last outcome was not taken, predict not taken. This adapts quickly, but it also flips too easily. For example, a loop-closing branch that is taken

many times and then falls through once on loop exit will usually be mispredicted again on the next loop entry because the last outcome was not taken.

4.11.2 Smith Two-Bit Saturating Counters

A *Smith predictor* replaces the single bit with a small saturating counter, most commonly two bits. The four counter states are commonly interpreted as strongly not taken, weakly not taken, weakly taken, and strongly taken. Prediction comes from the high-order tendency of the counter, while update moves the counter only one step toward the actual outcome. This hysteresis avoids overreacting to one unusual branch result. In the loop example above, a two-bit counter usually mispredicts only on the loop exit rather than on both the exit and the next re-entry.

4.12 Global and Local History Predictors

History-based predictors use recent branch outcomes to select among several learned taken/not-taken tendencies rather than relying on one counter per static branch.

4.12.1 GShare

GShare uses a global history register (GHR) like other global-history predictors, but instead of concatenating PC bits and history bits it combines them with an exclusive-or (XOR). If the predictor uses n index bits, then a common form is

$$index = PC_{low}[n-1:0] \ \oplus \ GHR[n-1:0]. \tag{4.14}$$

Here PC_{low} means the selected PC bits, normally after dropping fixed instruction-alignment bits, not necessarily the literal lowest address bits. The XOR can reduce some destructive aliasing patterns by spreading correlated PCs across the counter table more evenly than a plain concatenation. Conceptually, GShare and GSelect use the same ingredients; they differ mainly in how PC and history are mixed to form the table index.

4.12.2 GSelect

GSelect is a simple global-history predictor. It keeps a *global history register (GHR)* containing the most recent branch outcomes and combines that history with a small hash of the current PC to index a table of saturating counters. GSelect concatenates the p selected PC bits with h history bits, selecting one of 2^{p+h} counters. Prediction comes from the selected counter, usually by treating the upper half of the counter states as taken and the lower half as not taken. After the true outcome is known, the predictor updates that same counter and then shifts the actual outcome into the GHR. The key idea is that one branch can learn differently depending on the recent behavior of other branches.

Worked GSelect trace The following trace makes the lookup order explicit. Assume a 2-bit GHR whose rightmost bit is the newest outcome, a 1-bit PC hash equal to $PC[2]$, and 2-bit Smith counters where `00/01` predict not taken and `10/11` predict taken. Let every counter begin in state `10`, and let the initial GHR be `01`. For the four resolved branches below, prediction always uses the *current* PC hash and the *current* GHR. Only after that prediction is the selected counter updated and the actual outcome shifted into the GHR.

Time	PC	Act.	$PC[2]$	GHR_T	Ctr_T	Pred.	Ctr_{T+1}	GHR_{T+1}
T_0	0x2017	N	1	01	10	T	01	10
T_1	0x2009	T	0	10	10	T	11	01
T_2	0x2017	T	1	01	01	N	10	11
T_3	0x2009	N	0	11	10	T	01	10

After T_3, the final GHR is 10. The visited counter at $(PC[2], GHR) = (0, 10)$ has been trained up to 11, and the counter at $(0, 11)$ has been trained down to 01. This is the mechanical rule to remember: read the counter indexed by the *old* GHR, make the prediction, update that same counter with the actual outcome, then shift the actual outcome into the GHR.

4.12.3 Yeh/Patt

Yeh and Patt's two-level adaptive prediction is a family of organizations using branch history and a table of learned tendencies. The course's *Yeh/Patt* example uses a PC-indexed local-history table and one shared pattern table, a PAg-style organization.[1] Other members of the family use global history or separate pattern tables. A first table, indexed by the branch PC, stores a short per-branch history pattern. That recorded history then indexes a second table of saturating counters. In this organization, GSelect's shared global history is replaced by a local-history lookup before consulting the prediction counters. Two branches that collide in the first table can still interfere with each other's local history. Prediction therefore proceeds in two steps:

1. Use the branch PC to read the branch's recorded history.
2. Use that history to select the saturating counter that predicts taken or not taken.

Update proceeds in the same order in reverse: first train the selected counter toward the actual outcome, then shift the actual outcome into that branch's local history register. This structure is especially useful when different static branches follow different repetitive patterns.

Worked Yeh/Patt trace Now assume a 2-bit history table indexed by $PC[3:2]$ and a shared pattern table of 2-bit Smith counters. Let every history-table entry begin at 00, and let every pattern-table counter begin at 01. The sequence below uses two PCs, 0x3B98 and 0x3B9F, whose hashes are 10 and 11, respectively.

Time	PC	$PC[3:2]$	Hist_T	Ctr_T	Pred.	Act.	Ctr_{T+1}	Hist_{T+1}
T_0	0x3B98	10	00	01	N	N	00	00
T_1	0x3B9F	11	00	00	N	T	01	01
T_2	0x3B98	10	00	01	N	N	00	00
T_3	0x3B9F	11	01	01	N	N	00	10

This trace makes the two-level structure explicit. At T_1, branch 0x3B9F still reads local history 00, so it trains the shared pattern entry for history 00, even though it is a different static branch from 0x3B98. At T_3, however, the same branch now reads local history 01, so it consults and updates a different pattern-table counter. The history table is branch-specific; the pattern table is shared by history value.

[1] See Yeh and Patt's *Alternative Implementations of Two-Level Adaptive Branch Prediction*, listed in Appendix F.

4.13 Perceptron Direction Prediction

Perceptron prediction replaces small counter tables with a simple linear classifier. For one static branch, let the history bits be encoded as

$$x_i = \begin{cases} +1, & \text{if history bit } i \text{ is taken} \\ -1, & \text{if history bit } i \text{ is not taken} \end{cases} \tag{4.15}$$

and let w_0 be a bias weight. The predictor forms

$$y = w_0 + \sum_{i=1}^{n} w_i x_i. \tag{4.16}$$

If $y \geq 0$, it predicts taken; otherwise it predicts not taken. After the real outcome is known, a common training rule updates when the prediction was wrong or when $|y| \leq \theta$. If the target value is $t = +1$ for taken and $t = -1$ for not taken, then the update is

$$w_0 \leftarrow w_0 + t, \qquad w_i \leftarrow w_i + t x_i. \tag{4.17}$$

The exercises use $\theta = \lfloor 1.93n + 14 \rfloor$, where n is history length, following the empirical tuning rule in the original perceptron-prediction study.[2] It is a design choice, not a mathematical requirement. Real weights have finite width; the small examples assume no overflow.

A weight vector grows linearly with history length, while a table containing every distinct history pattern grows exponentially. That makes longer histories attractive, at the cost of computing a weighted sum. A single perceptron still has a limit: one linear decision boundary cannot perfectly represent every pattern, including XOR of two history bits.

Worked perceptron update Suppose the active perceptron has weights $[1, 2, -1, 0, 1]$, where the first value is the bias w_0, and suppose the current 4-bit global history is `1010`. Using taken $= +1$ and not taken $= -1$, that history becomes

$$x = [+1, -1, +1, -1].$$

The perceptron output is therefore

$$y = 1 + 2(+1) + (-1)(-1) + 0(+1) + 1(-1) = 3. \tag{4.18}$$

Because $y \geq 0$, the prediction is taken. If the true outcome is not taken, then the prediction is wrong and training must occur with target $t = -1$. The updated weights become

$$w_0 \leftarrow 1 - 1 = 0, \qquad [w_1, w_2, w_3, w_4] \leftarrow [2, -1, 0, 1] + [-1, +1, -1, +1] = [1, 0, -1, 2].$$

After the same outcome is shifted into the history register, the new GHR becomes `0100` if the newest bit is placed on the right.

[2] Daniel A. Jiménez and Calvin Lin, *Dynamic Branch Prediction with Perceptrons*; see Appendix F.

4.14 Hybrid and Tournament Predictors

A *hybrid predictor* combines two or more component predictors and uses a chooser to decide which component to trust for a branch context. A common *tournament predictor* stores a small table of saturating counters indexed by branch PC or by a short hash of PC and history. When the chooser leans one way, the machine uses one predictor family; when it leans the other way, it uses the competing family.

The usual training rule is simple. If the component predictors disagree and one is correct while the other is wrong, the chooser is updated toward the correct component. If both are correct or both are wrong, the chooser often remains unchanged because that outcome did not distinguish which component was better for that context. This organization is useful when one predictor handles short repetitive patterns well while another captures longer correlations better. For example, a tournament design may arbitrate between a Yeh/Patt predictor and a perceptron predictor.

Predictor-quality metrics Two branch-prediction ratios appear repeatedly in simulator output. For one measurement window, let N_{inst} count retired instructions and N_{branch} count the conditional branches receiving a direction prediction. Let $N_{correct}$ count their correct direction predictions. Then

$$f_{branch} = \frac{N_{branch}}{N_{inst}} \tag{4.19}$$

is the dynamic branch fraction and

$$Acc_{branch} = \frac{N_{correct}}{N_{branch}} = 1 - \frac{N_{mispred}}{N_{branch}} \tag{4.20}$$

is branch-prediction accuracy. These metrics answer different questions: branch fraction tells how often control-flow prediction matters, while prediction accuracy tells how often the predictor was correct when it mattered.

Prediction and update timing Prediction happens near fetch, but the true branch outcome becomes known only later when the branch resolves or retires. A pipelined predictor therefore cannot always train itself using only its *current* tables, because the history pattern, perceptron sum, or chooser state seen at update time may differ from the state that existed when the prediction was originally made. Save the prediction-time index, history, and other needed metadata until update. Train the selected counter's current value rather than blindly restoring its old value and losing intervening training. The serial worked traces predict one branch, resolve it, and update before the next prediction. Machines with several unresolved branches may instead shift predicted outcomes into history speculatively and restore or repair that history after a misprediction. Those timing rules must be stated before using a trace to model the hardware.

4.15 Timing-Diagram Interpretation

Timing diagrams are not just bookkeeping artifacts. They are often enough to infer what forwarding paths, register-file timing rules, and structural resources a pipeline must have. A reliable reading order is:

1. Identify true, anti-, and output dependences from the program text before looking at the diagram.

2. Mark repeated stage occupancy, distinguishing a stalled instruction from a normally multi-cycle operation.
3. For each stall candidate, decide whether the cause is data, structural, or control.
4. Infer which hardware feature would be required to remove that stall.

Common structural cues If one instruction is fetching while another is accessing data memory in the same cycle without conflict, then the design must have either separate instruction and data paths or enough memory-porting to support both operations. If fetch pauses exactly when a load or store reaches MEM, a shared memory path is the likely cause.

Common forwarding cues If a dependent ALU instruction reaches EX before the producer reaches WB, some bypass path must exist. If a load-use pair still suffers one bubble but longer ALU-to-ALU chains do not, the machine likely has EX/MEM-style forwarding but still cannot make load data available early enough for the immediately following EX stage. If a consumer in ID proceeds only when the producer is in WB, the design may be relying on WB-to-ID forwarding or on same-cycle register-file write-then-read timing rather than on an earlier bypass path.

Register-file timing cue Suppose a producer writes back in cycle k and a dependent consumer reads registers in ID in that same cycle k. If the consumer does not stall further, then the machine must support either same-cycle write-before-read behavior or an equivalent WB-stage bypass into decode. If the consumer waits one more cycle, absence of those mechanisms is one possible explanation; rule out other hazards before inferring the hardware.

Control-hazard cue Wrong-path instructions that appear briefly and are then crossed out or squashed indicate speculative fetch. A gap after a branch with no wrong-path work suggests conservative branch resolution rather than aggressive prediction. Likely-bit or squash-slot behavior is visible when an instruction after the branch is fetched speculatively but only conditionally allowed to complete.

A fuller checklist remains in Appendix A, but these rules are the minimum set needed to infer pipeline structure from the kinds of timing traces used throughout this text.

4.16 What to Carry Forward

A pipeline overlaps work without removing the program's dependences. Follow a value from its producing stage to its consuming stage before deciding whether forwarding is enough or a bubble is necessary. Follow a branch from prediction to resolution before counting the cost of a wrong path. The stage names alone do not specify either timing rule.

Prediction keeps the front end supplied with plausible work; saved context lets the predictor learn from that decision later. The next scheduling question is what to do when one instruction waits but another is ready. After the integrated examples, Chapter 6 develops machinery that can choose that ready work without losing precise program behavior.

4.17 Practice Questions

The following questions reinforce the main ideas of Chapter 4 and are intended for self-checking after the summary.

1. An unpipelined design has $CPI = 5$. A new implementation uses five pipeline stages. What is the maximum possible speedup if the pipelined clock period is unchanged? Then compute the new CPI if 25% of instructions suffer a single-cycle stall.
2. A scalar pipeline has base $CPI = 1$. Structural hazards add an average of 0.05×1 cycles per instruction, load-use hazards add 0.12×1, and branch mispredictions add 0.15×2. These are additional, non-overlapping contributions per retired instruction. Compute the final CPI.
3. For the sequence below, list the RAW, WAR, and WAW dependences. Then identify which forwarding path is most important for reducing stalls.

```
LW R1, 0(R2)
ADD R3, R1, R4
SUB R5, R3, R6
SW R5, 0(R7)
```

4. What hardware is implied when (a) instruction fetch overlaps a data-memory access, (b) a dependent ALU consumer reaches EX before its producer reaches WB, and (c) decode reads a value in the cycle an older instruction writes it back?
5. Explain the distinct roles of the branch target buffer (BTB), return-address stack (RAS), and direction predictor. Why does a BTB remain useful even when the branch direction is trivial?
6. Use the worked trace's direct-mapped, fully tagged BTB indexed by `PC[7:4]`: four-byte instructions, initially invalid entries, allocation for calls and taken conditional branches, RAS-only returns, and an initially empty RAS. Simulate `0x4000: CALL 0x4A00`, `0x4A00: BREQZ R1,0x4AF0` (taken), `0x4AF8: RET`, `0x4004: ADD`. State each control instruction's BTB effect and the final RAS top.
7. A GSelect predictor uses a 2-bit global history register and indexes the predictor table with `PC[2]` and the current GHR. The row for `PC[2]=0` is `[01,10,11,00]` for GHR `00,01,10,11`, the row for `PC[2]=1` is `[10,01,00,11]`, and the initial GHR is `01`. Simulate the following branch sequence in order: `(0x2004,T)`, `(0x2008,N)`, `(0x2004,T)`, `(0x200C,T)`. For each access, give the selected table entry before update, the prediction, the updated counter, and the new GHR.
8. A Yeh/Patt predictor uses `PC[3:2]` to select a 2-bit local-history register. Explain how its history-table and pattern-table lookups differ from GSelect. Give one case where it distinguishes branches that simple global-history indexing aliases together.
9. A student claims that if every Yeh/Patt local-history register is initialized to `00`, then the first prediction for every branch must be not taken. Is that claim correct? Explain which structure determines the actual first prediction and why the history table alone is not enough.
10. A perceptron predictor uses $G = 4$ bits of global history. The active perceptron has weights $[1, 2, -1, 0, 1]$, the current GHR is `1010`, and the next actual branch outcome is not taken. Compute the output y, state the prediction, determine whether training occurs, and give the new GHR after update. You may assume the threshold rule $\theta = \lfloor 1.93G + 14 \rfloor$ used in this text.

Chapter 5

Integrated Synthesis and Cache Simulation Concepts

A simulator reports fewer cache misses after a change, yet the modeled program runs more slowly. That result is possible: the larger cache may take longer to access, or the new policy may generate expensive traffic elsewhere. To explain it, we need to follow an event through the whole machine instead of treating each counter as a separate verdict.

5.1 From an Event to an Explanation

Begin with one demand access. Ask which level receives it, whether a miss causes a fill or write-back, and which counters record each event. Then ask whether the resulting delay actually stalls the processor or overlaps other work. This sequence connects the cache model to runtime and prevents a common mistake: interpreting more activity as more useful work.

5.2 Integrated Themes from Chapters 1 and 2

The first two chapters establish the measurement framework used throughout the rest of the book. An architectural change should first be classified by what it is trying to improve: parallelism, speculation, or some combination of the two. Its performance impact must then be evaluated with the correct metric, usually runtime for one job or throughput for a stream of jobs. The central CPU-time equation,

$$\text{CPU time} = \text{IC} \times \text{CPI} \times \text{CT},$$

turns that reasoning into a concrete tradeoff among instruction count, pipeline efficiency, and clock period. Within that framework, Amdahl's law bounds fixed-size speedup, Gustafson's law explains scaled-size behavior, and the choice between arithmetic and harmonic means determines whether a summary is mathematically meaningful.

Worked check Consider an optimization that accelerates 30% of runtime by 2×. The total speedup is

$$S = \frac{1}{(1 - 0.3) + \frac{0.3}{2}} = \frac{1}{0.85} \approx 1.176. \tag{5.1}$$

The unaccelerated fraction remains in the denominator and bounds overall speedup.

Worked check: maximum possible speedup Consider a mechanism that accelerates 25% of the original benchmark runtime and makes that portion arbitrarily fast. The maximum total speedup is

$$S_{\max} = \frac{1}{1 - 0.25} = \frac{4}{3} \approx 1.33. \tag{5.2}$$

This is the standard way to interpret the *maximum* speedup of a partial optimization.

5.3 Integrated Themes from Chapter 3

Chapter 3 adds the memory-hierarchy side of the performance model. That means translating a cache specification such as (C, B, S) into sets, ways, and tag/index/offset fields; distinguishing compulsory, conflict, and capacity misses; and understanding how replacement and write policies alter observed traffic. It also means using

$$AAT = HT + MR \cdot MP$$

as a compact summary of hierarchy behavior while remembering that cache size, block size, and associativity can influence all three terms in different ways.

Worked check Consider a cache with $HT = 1.8$ ns, $MR = 0.06$, and $MP = 28$ ns (Eq. (5.3)):

$$AAT = 1.8 + 0.06 \cdot 28 = 3.48 \text{ ns}. \tag{5.3}$$

Worked check: repeated trace reasoning Consider a direct-mapped L1 with $(C, B, S) = (8, 4, 0)$, write-back/write-allocate, and the repeated access sequence

$$R\ 0x41F1,\ R\ 0xB54A,\ W\ 0x4204,\ W\ 0xCA3B,\ R\ 0x420E,\ W\ 0xB537. \tag{5.4}$$

With $B = 4$, the six accesses touch the five blocks $\{0x41F, 0xB54, 0x420, 0xCA3, 0xB53\}$ because $0x4204$ and $0x420E$ lie in the same block. The first iteration therefore has 5 compulsory misses. The key structural fact is that blocks $0xCA3$ and $0xB53$ map to the same direct-mapped set, so after warm-up the sequence continues to suffer two conflict misses per iteration. Over iterations 2–10, that produces 18 additional misses. Those 18 misses are the non-compulsory misses for the full run, and because the conflicting writes repeatedly evict dirty lines, the full sequence produces 19 writebacks if dirty lines are *not* flushed after the final access.

Victim cache and +1 prefetch contrast That same trace separates two common remedies cleanly. A small victim cache can eliminate the recurring two-block ping-pong at one set; after warm-up, the combined L1+victim-cache miss count can drop to zero because each conflicting block is recovered from the victim structure instead of DRAM. By contrast, a +1 prefetcher is not automatically a cure. In this trace it can prefetch block $0x420$ early enough to remove one first-iteration demand miss, but it does not remove the recurring conflict between $0xCA3$ and $0xB53$. This is the key design lesson of the trace: next-block prefetching and victim caching address different miss mechanisms.

5.4 Integrated Themes from Chapter 4

Chapter 4 connects pipeline structure to observed performance. Reading a timing diagram is therefore an exercise in inference: the reader is reconstructing hazards, bypass paths, memory behavior, and branch-handling policy from visible stage occupancy. Static direction baselines include always taken and backward taken / forward not taken. One-bit and Smith-counter schemes, GShare, GSelect, local-history predictors, perceptrons, and tournament predictors add runtime learning.

Pipeline Speedup

Assume a 5-stage design where each stage has 250 ps logic delay and pipeline-register overhead is 30 ps. Unpipelined instruction time is $5 \times 250 = 1250$ ps. Pipelined clock is $250 + 30 = 280$ ps. Ideal long-run speedup is (Eq. (5.5)):

$$S \approx \frac{1250}{280} \approx 4.46. \tag{5.5}$$

This is below 5 because register overhead is non-zero.

CPI with Branch Stalls

If branch frequency is 18% and each branch contributes an average of 1.6 stall cycles (Eq. (5.6)):

$$CPI = 1 + 0.18 \times 1.6 = 1.288. \tag{5.6}$$

This illustrates how even moderate branch penalties raise CPI above the ideal value of 1.

Integrated CPI Accounting

Suppose a scalar core has base $CPI = 1$. Loads are 30% of all instructions, the L1 data-cache miss ratio on loads is 5%, and each such miss adds 20 cycles. Branches are 18% of all instructions, the misprediction rate is 10%, and each misprediction adds 2 cycles. Assume the stated penalties are additional to base execution and do not overlap one another. Then the added CPI from load misses is

$$0.30 \times 0.05 \times 20 = 0.30, \tag{5.7}$$

while the added CPI from branch mispredictions is

$$0.18 \times 0.10 \times 2 = 0.036. \tag{5.8}$$

So the total CPI estimate is

$$CPI \approx 1 + 0.30 + 0.036 = 1.336. \tag{5.9}$$

This kind of combined accounting is the most useful way to compare bottlenecks: all stall mechanisms are translated into the common unit of cycles per retired instruction.

Software Branch Handling

When a branch has delay slots, every instruction moved into those slots must be safe regardless of whether the branch is taken. When a branch instead has squash slots under an “assume taken” policy, the compiler may place instructions from the predicted taken path there, because wrong-path entries will be canceled. The resulting placement rule is:

1. fill delay slots only with path-independent instructions,
2. fill squash slots from the predicted path when profitable, and
3. leave `NOP`s when correctness would otherwise be uncertain.

5.5 Cache Simulation Conceptual Model

The cache simulation framework models hierarchy behavior at the level of addresses, tags, and policy events rather than payload data values. A common parameterization is (Eq. (5.10)):

$$(C_1, B, S_1, C_2, S_2, \{\text{MIP,LIP}\}, \{\text{None,+1,Markov,Hybrid}\}, r). \tag{5.10}$$

In this notation, C controls capacity (2^C bytes), B controls block size (2^B bytes), and S controls associativity (2^S ways). Here *most recently used insertion policy (MIP)* and *least recently used insertion policy (LIP)* are common insertion-policy options.

5.6 Hierarchy Semantics in the Model

The hierarchy model treats L1 accesses separately from L2 internals, while L2 requests are viewed as arriving from an upper-level requester. This separation makes it easier to reason about statistics at each level without conflating local policy with downstream timing.

Write strategy also changes the meaning of those statistics. *Write-back, write-allocate (WBWA)* behavior in L1 emphasizes dirty-state tracking and deferred propagation, whereas *write-through, write-no-allocate (WTWNA)* behavior in L2 forwards every write to the next level and does not allocate on write misses. Forwarding a write does not imply that another processor can already observe it or that it has reached persistent storage.

The ordering of read-miss repair and writeback events can also matter. Different servicing orders change replacement-state evolution and therefore alter downstream hit, miss, and writeback counts. A disabled cache level requires an explicit modeling rule. If requests bypass it entirely, its lookup contributes no hit time; substitute the next active level's service time directly. If the specification instead models an always-miss lookup stage, retain only the latency that specification assigns to it.

5.7 Prefetching Concepts

+1 *prefetching* predicts the next sequential block ($X + 1$) after a demand miss on block X. *Markov prefetching* instead models miss-stream transitions as a finite-state table, associates each source miss block with a bounded set of likely successors and their frequencies, and selects a prediction from the highest-frequency successor under a deterministic tie-break rule. *Hybrid prefetching* combines correlation-based prediction with sequential fallback when correlation information is absent.

To interpret prefetch behavior correctly, the simulator must also distinguish demand-filled lines from prefetched lines. A prefetch metadata bit provides that distinction. In this model, the first demand use of a prefetched line counts as a prefetch hit, while eviction before first demand use counts as a prefetch miss. These are usefulness counters; another simulator may use the same names for lookup outcomes instead.

Worked prefetch-counter interpretation Suppose a simulator reports 120 prefetched lines, 70 prefetch hits, and 25 prefetched lines evicted before first use. The 70 prefetch hits show that many speculative fills were in fact useful: later demand accesses avoided ordinary misses because of earlier prefetching. The 25 unused evictions show that some prefetched lines consumed bandwidth and space without helping demand accesses. The remaining prefetched lines are ambiguous without more context, because some may still be resident and not yet demanded. This is why prefetch usefulness is not judged by one counter alone; both benefit and waste must be read together.

5.8 Statistics and AAT Interpretation

Typical hierarchy statistics include access counts, hit/miss counts, hit/miss ratios, writeback counts, and prefetch effectiveness counters. In two-level analysis (Eq. (5.11)):

$$AAT_{L1} = HT_{L1} + MR_{L1} \cdot MP_{L1}, \qquad MP_{L1} = AAT_{L2}. \tag{5.11}$$

$$AAT_{L2} = HT_{L2} + MR_{L2,read} \cdot DRAM_TIME. \tag{5.12}$$

$$DRAM_TIME = DRAM_AT + DRAM_AT_PER_WORD \cdot WORDS_PER_BLOCK. \tag{5.13}$$

Here $DRAM_AT$ is startup time before the per-word transfers. It differs from T_1 in Chapter 3's transfer model, which already includes delivery of the first sub-block. That definition explains why one formula multiplies by the full word count and the other uses $N_{sb} - 1$ follow-up transfers. This form emphasizes that higher-level penalty is the lower-level average access behavior, not a fixed constant.

5.9 Conceptual Tradeoffs Across Parameters

Increasing C generally reduces capacity misses but tends to raise hit time. Increasing S reduces conflict misses but increases lookup complexity. Increasing B can exploit spatial locality but may also increase pollution and miss penalty. More aggressive prefetching can reduce compulsory misses yet increase bandwidth pressure and useless fills.

5.10 What to Carry Forward

A useful simulator report tells a causal story: a demand access misses, a fill or write-back changes the hierarchy, and some part of the resulting delay becomes lost processor progress. Check the denominator, the event definition, and the timing assumptions before comparing any two counters. A lower miss ratio is evidence about one part of that story, not a complete performance verdict.

This method extends to the queues and reorder buffers in the next chapter. The names of the structures change, but the question remains where useful work waits and what prevents it from moving. Appendix B collects the formulas once those modeling choices are clear.

5.11 Practice Questions

The following questions reinforce the integrated ideas of Chapter 5 and are intended for self-checking after the summary.

1. A hierarchy reports the following parameters:

   $$HT_{L1} = 1, \quad MR_{L1} = 0.10, \quad HT_{L2} = 7, \quad MR_{L2,read} = 0.20, \quad \text{DRAM_TIME} = 80.$$

   Compute AAT_{L2} and AAT_{L1}. Then explain why the L1 miss penalty is modeled as the lower-level *average* access time rather than as a fixed constant.

2. A simulator reports 120 prefetched lines, 70 prefetch hits, and 25 prefetched lines evicted before first use. What do these three counters say about prefetch usefulness and prefetch waste? What additional information would you want before concluding that the prefetcher is a net win?
3. In a hierarchy that uses write-through, write-no-allocate behavior at the lower level, why is it often more meaningful to report the lower-level miss ratio over *reads only* rather than over all accesses?
4. Two cache configurations have the same capacity and associativity. Configuration A has lower hit time but higher miss ratio. Configuration B has higher hit time but lower miss ratio. Explain why the better design cannot be chosen correctly without computing or estimating average access time.
5. Explain why the servicing order of read-miss repair and writeback traffic can change later hierarchy statistics even when the high-level policy choices are unchanged.
6. A scalar core has base $CPI = 1$. Loads are 30% of all instructions, the L1 data-cache miss ratio on loads is 5%, and each such miss adds 20 stall cycles. Branches are 18% of all instructions, the branch misprediction rate is 10%, and each misprediction adds 2 cycles. Estimate the total CPI and identify which source of stalls contributes more.

Chapter 6

Dynamic ILP: Superscalar Scheduling and Out-of-Order Execution

A load misses in the cache. The next instruction needs its result, but a later instruction could add two values that are already available. Must the whole machine wait? Out-of-order execution begins with that question and earns its complexity by answering no, while still making the final result look as though the program ran in order.

6.1 Why Cycles per Instruction Can Go Below One

In basic single-issue pipelines, designers often treat cycles per instruction (CPI) near 1 as a practical lower bound. Superscalar execution allows more than one instruction to start execution in a cycle, rather than merely overlapping different stages of successive instructions. This broader goal is usually described as instruction-level parallelism (ILP). In this context, instructions per cycle (IPC) is the natural companion metric, as shown in Eq. (6.1):

$$IPC = \frac{1}{CPI}. \tag{6.1}$$

So achieving $CPI < 1$ is exactly equivalent to achieving $IPC > 1$. Over a sustained run, useful retirement throughput cannot exceed the supply and service capacities of the preceding stages:

$$IPC_{retire} \leq \min(IPC_{fetch},\ IPC_{dispatch},\ IPC_{execute},\ IPC_{state\ update})\,. \tag{6.2}$$

If any major stage can only handle one instruction per cycle on average, then the whole machine cannot retire two instructions per cycle on average, even if another stage is wide enough to do so. In other words, the minimum term in Eq. (6.2) is the throughput bottleneck. This is a sustained-rate bound, not a rule for every short measurement window: retirement can briefly drain already completed work faster than the front end supplies new instructions.

Each term corresponds to a different possible limit. IPC_{fetch} asks how many useful instructions the front end can supply per cycle. $IPC_{dispatch}$ asks how many decoded instructions can be prepared and sent onward. $IPC_{execute}$ asks how many instructions the function units (FUs) and memory system can actually complete. $IPC_{state\ update}$ asks how quickly the machine can accept completed work into precise architectural state. The retire rate cannot exceed the weakest of these rates, because retirement is the final visible outcome of all the earlier pipeline activity.

6.2 Historical Motivation

The lecture introduces Tomasulo's algorithm through a constraint of the IBM System/360 Model 91: long floating-point operations had to share an architecture with only four floating-point registers.[1]

[1] `n5_ilp_part1.pdf`, ILP-14; R. M. Tomasulo's original paper is listed in Appendix F.

More arithmetic hardware alone would leave two problems unsolved. A calculation could wait on a genuinely unfinished input, or an independent calculation could be held back simply because the program reused a register name. Reservation stations let ready work wait near the execution units; tags distinguish which unfinished calculation will produce each value. That historical pressure makes the modern distinction memorable: a dependency on a value must survive, while a collision between names can be removed.

6.3 Superscalar Pipeline View

A practical superscalar model can be described with five main logical units plus a final retirement or commit action. Figure 6.1 provides a compact visual summary of how these units interact in the overall datapath. In that figure, the boxed labels denote the main structures or stages, while the smaller labels on the right identify the dominant action on the adjacent arrow bundle: fetch, dispatch, fire or issue, complete, and retire or commit. The front end is drawn as separate Branch Handling and Instruction Fetch boxes, but together they still represent the single logical instruction fetch (IF) unit described below.

1. Branch Handling and Instruction Fetch (IF) This unit performs instruction fetch (IF) while consulting branch-handling structures such as direction predictors, branch-target mechanisms, and return-address prediction. Its goal is to produce multiple candidate instructions per cycle.

2. Dispatch and Decode (DISP) This unit decodes incoming instructions in program order and dispatches them toward the scheduling structures. Dispatch is where program order is still explicit, which matters for precise recovery.

3. Scheduling Scheduling assigns ready instructions to available FUs in time. This is where execution can diverge from program order to exploit parallelism.

4. Execute (EX) FUs perform arithmetic, memory, and control operations with operation-specific latencies. Multiple FUs allow parallel progress.

5. State Update (SU) State update gathers completed results, handles squash or recovery, and prepares instructions for precise architectural completion. The final outgoing flow from this unit is the retirement or commit action, where results become architecturally visible in program order. In Fig. 6.1, retirement is represented by the final outgoing arrows below State Update rather than by a separate boxed stage.

6.4 Dependency Types and Why They Matter

Superscalar scheduling must respect three dependency classes.

Read-after-write (RAW) is a true flow dependency: a later instruction needs a value produced by an earlier one, so executing them in the wrong order would use stale data and produce the wrong result.

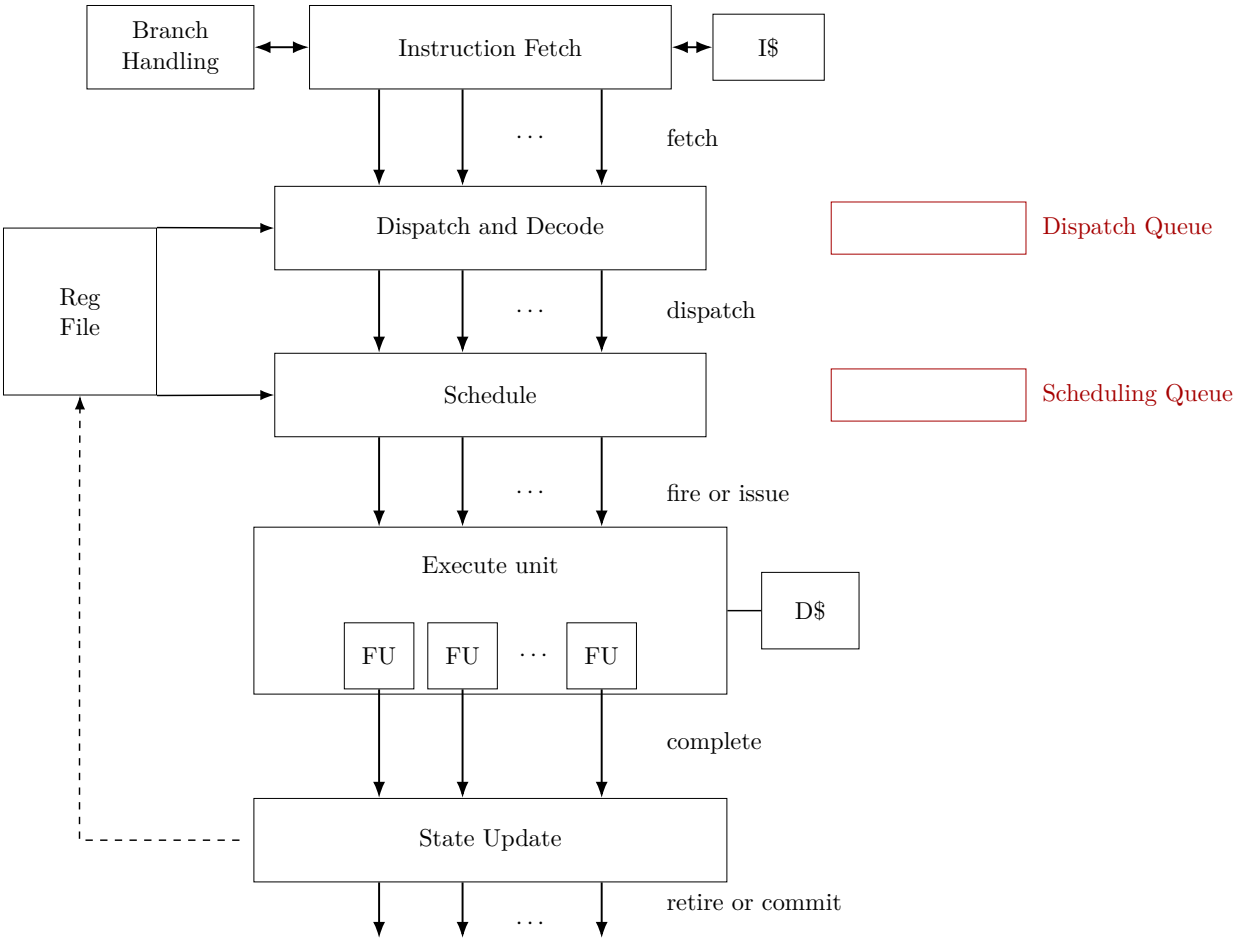


Figure 6.1: Logical superscalar pipeline view illustrating one representative organization. The figure is not specific to FICI, FICO, or FOCO; those classifications depend on the firing and completion policy. Boxed elements denote major structures or stages, while the right-side annotations denote the adjacent flow actions.

Write-after-read (WAR) is an anti-dependency: a later instruction must not overwrite a register before an earlier instruction has finished reading it, even though the two instructions do not share a true data value.

Write-after-write (WAW) is an output dependency: two instructions write the same destination, so their writes must appear in the correct order or the final architectural value will be wrong.

For arbitrary input values, a true RAW dependency must be respected even when register names change. WAR and WAW are name dependencies that arise from register reuse and can be reduced or removed by renaming the register.

6.5 Fire/Complete Taxonomy

A compact way to classify designs is by fire order and complete order.

Fire in order, complete in order (FICI) is the classic simple-pipeline behavior. Instructions begin execution in program order and also finish in program order, which simplifies control but limits how much parallelism the machine can exploit.

Fire in order, complete out of order (FICO) is a simpler superscalar step. Instructions still begin

execution in order, but different latencies allow later instructions to finish first, so some parallel speedup is possible without fully out-of-order scheduling.

Fire out of order, complete out of order (FOCO) is more aggressive and offers higher potential ILP. The scheduler may start and finish instructions whenever operands and hardware are ready, which exposes more ILP but requires more machinery to preserve correct architectural behavior.

6.6 Scoreboarding Intuition (FICO Style)

A scoreboard tracks whether source operands, destinations, and required function units are currently available. A simplified FICO-style fire condition is shown in Eq. (6.3).

$$\text{Fire if } \neg busy(src_1) \ \wedge \ \neg busy(src_2) \ \wedge \ \neg busy(dest) \ \wedge \ \neg busy(FU_{needed}). \tag{6.3}$$

When an instruction fires, destination and function-unit busy state are set. When the instruction completes, those busy states are cleared according to the implementation timing rules. Here $busy(FU)$ means that the unit cannot accept another operation, not simply that some older operation remains inside its pipeline. This simplified in-order-fire design reads source values before younger writers can pass them; a general out-of-order scoreboard needs additional WAR protection. The result is correct execution with moderate hardware complexity, but some parallel opportunities remain unused compared with more aggressive out-of-order firing.

6.7 Function Units and Latency

Superscalar benefit depends on the latency mix and on the number of available units. In many designs, arithmetic logic unit (ALU) operations have short latency, multiply operations take more cycles, and memory accesses vary widely depending on cache behavior. The key point is that latency and throughput are related but not identical. An operation may have a long latency and still support good throughput if the corresponding FU is pipelined and can accept a new instruction every cycle. By contrast, a long-latency operation on a scarce or non-pipelined FU can create backpressure quickly, because younger ready instructions may have no place to execute. Memory operations are especially important here. A cache hit may behave like a modest fixed-latency operation, while a cache miss can hold resources for many extra cycles and expose whether the machine can continue doing useful independent work during the delay.

Unit count matters for the same reason. If the machine has only one ALU, one multiplier, and one load/store path, the scheduler has few choices. Adding more execution resources, including additional FUs such as ALUs, multipliers, or more load/store capacity, gives the scheduler more flexibility to overlap different instruction classes and reduce structural conflicts. Designers therefore optimize both unit availability and scheduling policy, not just raw clock frequency. The practical goal is not merely to make individual operations fast, but to ensure that the mix of latencies, FU counts, and pipeline structure supports high average IPC on realistic workloads. Table 6.1 gives the course's illustrative cycle counts, not universal timings for those instruction classes or fixed hardware stage boundaries. Whether divide is rare depends on the workload. What matters for scheduling is its stated latency and how often its unit can accept a new operation.

Type	Model latency	Interpretation
Floating-point (FP) multiply	3	Illustrative pipelined multiply, including exponent and significand handling.
FP add	2	Align significands, add, then normalize and round.
Load	2	An assumed cache-hit latency; misses take longer.
Store	1	Admission to a buffer, not completion at every observer.
Divide	Variable	Iterative or pipelined implementations have different latency and initiation intervals.
ALU	1	Most common operation, so short latency is especially valuable.

Table 6.1: Illustrative function-unit latencies. Actual designs and workloads differ; the one-cycle store entry denotes buffer admission rather than global visibility.

6.8 Worked FICO Timing Example

The following seven-instruction sequence illustrates the scoreboard idea in a concrete schedule:

$$
\begin{aligned}
A &: \texttt{ADD}\ R1, R2, R3 \\
B &: \texttt{MUL}\ R5, R1, R4 \\
C &: \texttt{ADD}\ R3, R1, R2 \\
D &: \texttt{ADD}\ R1, R5, R4 \\
E &: \texttt{MUL}\ R6, R7, R8 \\
F &: \texttt{ADD}\ R1, R3, R3 \\
G &: \texttt{ADD}\ R4, R3, R5
\end{aligned}
$$

Assume a fetch rate of two instructions per cycle, one ALU with latency one cycle, one multiplier with latency three cycles, and a simplified *state update (SU)* stage used to make completion visible. Both units accept one new operation per cycle. DISP scans at most two instructions in program order, stopping at the first blocked instruction. During cycle k, it may see busy bits cleared by EX in that cycle and select operations to start EX in cycle $k+1$. Sources are captured at that selection point, and SU has enough bandwidth for two results per cycle. Table 6.2 shows one compact timing schedule for this example. The stage labels are *instruction fetch (IF)*, *dispatch (DISP)*, *execute (EX)*, and *state update (SU)*. Repeated DISP entries indicate that the instruction is blocked in the in-order fire logic, not that it is being redispatched from scratch.

Walkthrough Cycles 0–3 establish the basic pattern. A and B are fetched together, but B cannot leave DISP immediately because it needs the value of $R1$ from A. Once A executes, B can move forward, and C can also reach EX shortly afterward because it depends on the same produced value. This is why B spends an extra cycle in DISP in Table 6.2: the machine is still firing instructions in program order, so the scoreboard must wait until A has released the needed state.

Instr.	Operation	0	1	2	3	4	5	6	7	8	9
A	ADD $R1, R2, R3$	IF	DISP	EX	SU						
B	MUL $R5, R1, R4$	IF	DISP	DISP	EX	EX	EX	SU			
C	ADD $R3, R1, R2$		IF	DISP	EX	SU					
D	ADD $R1, R5, R4$		IF	DISP	DISP	DISP	DISP	EX	SU		
E	MUL $R6, R7, R8$			IF	DISP	DISP	DISP	EX	EX	EX	SU
F	ADD $R1, R3, R3$			IF	DISP	DISP	DISP	DISP	EX	SU	
G	ADD $R4, R3, R5$				IF	DISP	DISP	DISP	DISP	EX	SU

Table 6.2: Timing table for a simplified FICO schedule. Repeated DISP entries indicate that an instruction remains blocked in the in-order dispatch-and-fire path while waiting for data or hardware to become available.

Cycles 4–6 expose the main weakness of FICO. D is genuinely blocked because it needs $R5$ from B, so its repeated DISP entries reflect a real RAW dependence. E, however, is independent of D and could use the multiplier, yet it still cannot fire early because the machine will not let younger instructions pass an older blocked instruction in the dispatch path. That is the key limitation of fire-in-order scheduling: some stalls come from true dependencies, but others come from the policy itself.

Cycles 7–9 show both overlap and continued restrictions. E eventually occupies the multiplier while F and G reach the ALU later, so the machine does achieve some parallel execution and some out-of-order completion. At the same time, the single ALU and single multiplier keep the scheduler's choices narrow, and the in-order fire rule prevents the machine from exploiting all available independence. The example therefore illustrates why FICO is attractive as a first superscalar step—it is much simpler than full FOCO—but also why more aggressive out-of-order firing and renaming are needed to capture more ILP.

6.9 Register Renaming as the Next Step

Register renaming is the introduction of additional physical register identities so later writes do not falsely block earlier reads or writes. Conceptually, renaming transforms many WAR/WAW hazards into non-hazards while preserving program meaning. The dependency pressure that remains is primarily RAW, which reflects true data flow. In a typical implementation, the registers named by the instruction set architecture (ISA) are the *architected registers*, while the larger hidden pool used internally by the machine is the *physical register file*. The architected registers are the names the programmer sees in assembly language and that the ISA defines as part of the visible machine state. The physical register file contains additional hidden storage used internally by the processor to hold renamed versions of those values.

This distinction matters because several in-flight instructions may all refer to the same architected destination at different moments in time. Rather than repeatedly reusing one architectural name and creating false WAR/WAW conflicts, the machine can assign a different physical register to each new write while still preserving the illusion of a small fixed architected register set. Older instructions can continue reading the older physical version, while younger instructions are redirected to the newer one. A *register alias table (RAT)* maps each architected register name to the physical register

that currently holds its most recent value. More precisely, the speculative RAT names the latest *renamed producer*, whose value may still be unfinished. When dispatch renames a destination, it allocates a fresh physical register and updates the RAT so younger instructions will refer to the new location rather than the older one. Ready bits in the physical register file then indicate whether each renamed value has actually been produced yet. Some architectures describe the same underlying status with *busy bits* instead. The terminology varies, but the idea is the same: one convention marks a value as ready to use, while the other marks it as still busy or unavailable. Figure 6.2 follows two versions of one name.

Program order	Architectural names	One renamed form
A	`MUL R1,R2,R3`	`MUL P10,P2,P3`
B	`ADD R4,R1,R5`	`ADD P11,P10,P5`
C	`ADD R1,R6,R7`	`ADD P12,P6,P7`

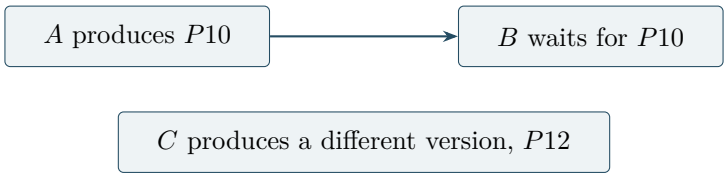


Figure 6.2: A register name is not a value's identity. Assume the incoming sources map to the shown physical registers and new destinations receive $P10$, $P11$, and $P12$. C may finish before A, but B still consumes A's $P10$, not C's $P12$.

6.10 Tomasulo's Algorithm

Tomasulo's algorithm is a classic out-of-order execution method that implements *FOCO* by allowing instructions to fire when their operands are ready rather than strictly in program order. Its central idea is to combine renaming with *reservation stations*, which are scheduling entries that hold an instruction until the required operands and execution resources are available. Historically, this approach is associated with Bob Tomasulo's work on the IBM System/360 Model 91, and it remains the conceptual foundation behind many modern out-of-order schedulers.

Main structures A simplified Tomasulo-style machine contains a dispatch unit, a set of reservation stations, one or more FUs, a result-broadcast network, and a renamed register file. Dispatch still runs in program order, but once instructions occupy reservation stations they may wake up and fire out of order. Each reservation-station entry records the required FU type, the destination tag or physical register, and for each source either a ready value or a tag identifying which older instruction will eventually produce that value.

Dispatch, fire, and complete At dispatch time, each source operand is examined. If the needed value is already ready, the value is copied into the reservation station. If it is not ready, the reservation station records the producing tag instead. The destination is renamed so younger instructions refer to the new producer rather than to the older architectural name. Later, when both sources are ready and a matching FU is available, the instruction can fire even if an older unrelated instruction is still waiting somewhere else in the machine. Producing a result and publishing it are

separate events: an instruction may finish computation but wait for the common data bus (CDB) before broadcasting its tag and value. The broadcast wakes dependents and updates the appropriate value storage; in this example, *complete* means that publication has occurred.

Illustrative four-instruction trace The following short trace captures the key behavior. Instruction A is a load that misses in the cache. Instruction B is an add that depends on A. Instruction C is an independent load that hits in the cache. Instruction D is an add that depends on C. Because the machine is out of order and the cache is non-blocking, C and then D can make progress while A is still waiting on the miss. Use one pipelined load/store unit that accepts one request per cycle, one ALU, and one CDB. Take the load-response times as given: C's result is available for broadcast in cycle 4, and A's in cycle 6. An add that fires in cycle k is eligible to broadcast in cycle $k+1$. Dependents may fire in the cycle after their source broadcasts, and the oldest eligible instruction wins a CDB conflict. Figure 6.3 follows these rules, matching the sequence in the lecture handout.[2]

Instruction	Dispatch	Fire	Result eligible	CDB broadcast
A: load miss	0	1	6	6
B: add using A	0	7	8	8
C: load hit	0	2	4	4
D: add using C	0	5	6	7

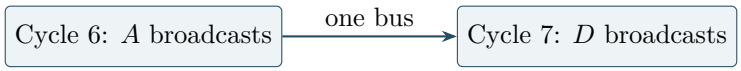


Figure 6.3: Computation can finish before publication. A and D are both eligible in cycle 6, but the single CDB serves only A; D waits until cycle 7. No cycle broadcasts two results.

Cycle 0. A, B, C, and D dispatch into reservation stations. Sources that are already available are copied in as values; missing sources are recorded as tags. Destinations are renamed, and the corresponding destination entries are marked not ready.

Cycle 1. A fires and encounters its cache miss.

Cycle 2. Independent load C fires while B remains blocked on A. This is the gain from out-of-order scheduling and hit-under-miss support: the older miss does not freeze independent work.

Cycle 3. Nothing completes; B still waits for A, and D for C.

Cycle 4. C completes first and broadcasts its tag and value on the common data bus (CDB). Reservation stations compare that tag against all waiting operands, so D wakes up and becomes ready to fire.

Cycle 5. D fires and begins its add using the value from C.

Cycle 6. A finally completes its miss and D also wants to complete. With only one CDB, both cannot broadcast at once, so the usual rule is to let the older instruction win. A wins this cycle, and B becomes ready for the next cycle. D retains its result while waiting for the bus.

Cycle 7. B fires, and D broadcasts its delayed result.

Cycle 8. B broadcasts its result.

[2] Conte, `n5_ilp_part1.pdf`, slides ILP-23–24.

This trace shows why it is important to distinguish *fire* from *complete*. Firing sends operands into an FU and begins execution. Completing publishes the result so the rest of the machine can observe it.

Non-blocking caches This Tomasulo-style example also motivates a *non-blocking cache*. Such a cache is often described as supporting *hit under miss*. If A misses in the cache, the machine does not necessarily want every younger memory access to stop. Instead, independent cache accesses may continue to probe the cache pipeline. That allows C to hit and complete while A is still waiting on a lower-level response. That improves throughput, but it also makes control more complex because the cache and scheduler must now track multiple outstanding requests correctly.

6.11 Result Buses and Physical Register Files

Basic Tomasulo has an obvious bottleneck: one common data bus can publish only one result per cycle. For instructions that each require one such broadcast, the sustained completion rate is bounded by Eq. (6.4).

$$IPC_{complete} \leq N_{result\ buses}. \tag{6.4}$$

Adding more result buses raises the completion ceiling, but it also increases wakeup complexity because each reservation station must compare against more simultaneous broadcasts. This is a bound on that class of result producers, not necessarily on every instruction: stores and branches may use different completion paths.

An explicit physical register file (PRF) stores values at locations selected by physical register number. The scheduler need not hold a full copy of every operand value, but it still has to learn when each source becomes ready. Conventional designs may retain associative wakeup comparisons on physical-register tags. The course also considers directly polling a ready-bit table; that is a scheduling choice, not an automatic consequence of using a PRF.

Separate the bandwidth for values from the bandwidth for readiness. If at most W_{issue} two-source operations read their operands each cycle, a simple unbanked PRF provisioned for full bandwidth needs approximately

$$N_{data\ read\ ports} \approx 2W_{issue}, \qquad N_{write\ ports} \approx N_{result\ buses}. \tag{6.5}$$

Polling both source-ready bits for every one of N_{sched} waiting instructions instead requires up to $2N_{sched}$ one-bit status checks per cycle, not that many full-width data reads. Banking, clustering, bypass paths, and operand caching change the port requirements. Neither the storage organization nor its power cost can be judged from the presence of physical registers alone.

6.12 RATs, Ready Bits, and Dispatch in Renamed Machines

Once the machine uses an explicit physical register file, dispatch must translate instruction set architecture (ISA) register names into physical register numbers. That basic renaming step is summarized by Eq. (6.6).

$$P_{src1} = RAT[A_{src1}], \qquad P_{src2} = RAT[A_{src2}], \qquad RAT[A_{dest}] \leftarrow P_{free}. \tag{6.6}$$

The first two mappings identify the current physical locations of the source values. The third mapping allocates a fresh physical destination so younger instructions depend on the new producer rather than on an older architectural name.

In this style of design, the fire rule is usually expressed with *ready bits* on physical registers rather than with *busy bits* on architected registers. The same logical condition is shown in Eq. (6.7).

$$fire_i \Leftarrow ready(P_{src1}) \wedge ready(P_{src2}) \wedge free(FU_i). \tag{6.7}$$

Each scheduling-queue entry is a reservation station that records the FU type, the destination physical register or tag, and either ready source values or the physical-register identities that will eventually supply them. When a producing instruction completes, the result bus updates the physical register file and wakes any waiting entries that need that value.

Do not equate a PRF with a *future file*. A future-file organization keeps a latest speculative value per architectural name and a separate committed data file; a PRF keeps multiple physical versions, with maps selecting speculative and committed views. Both separate speculative progress from precise state, but they store and recover that state differently. Register renaming also does not settle whether a load and store refer to the same memory address; memory-dependence tracking remains a separate job.

6.13 Precise State and Recovery

Aggressive out-of-order execution creates a second problem beyond scheduling: the machine may complete instructions in an order that does not match the sequential state the programmer expects. If an interrupt, page fault, or other exception occurs at the wrong moment, software still needs a coherent restart point. That requirement is usually stated as the *precise exception* property in Eq. (6.8).

$$\begin{aligned} \text{restartable fault at } i \Rightarrow\ & \text{all older instructions are committed} \\ & \wedge\ i \text{ and all younger instructions have no committed effects} \end{aligned} \tag{6.8}$$

This requirement was not always imposed in the same way. In Tomasulo's original floating-point setting on the IBM System/360 Model 91, the out-of-order machinery covered a narrower part of the processor, and the machine could tolerate *imprecise exceptions* in situations that modern general-purpose cores cannot. General-purpose ISAs and operating systems normally require precise restart for faults such as a page-faulting load. The faulting instruction itself must not commit its result; trap-like events can define a different architectural boundary.

6.14 Reorder Buffers and Retirement

The standard answer to precise-state recovery is the *reorder buffer (ROB)*. Dispatch allocates ROB entries in program order. Execution may finish out of order, but retirement occurs in order from the ROB head, which creates a clean boundary between speculative work and architecturally committed state.

A ROB entry tracks destination information, completion status, exception state, and the information needed to recover the correct program counter (PC). In a data-in-ROB organization it

also holds a result value; in an explicit-PRF organization the value lives in the PRF, so the ROB can hold register identities and recovery metadata instead. In simple form, retirement follows Eq. (6.9).

$$retire \Leftarrow ready(ROB_{head}) \wedge \neg exception(ROB_{head}). \tag{6.9}$$

If the machine can retire at most W_{retire} instructions per cycle, then one cycle may commit up to the oldest W_{retire} consecutive ready non-excepting entries from the ROB head, but it may not skip over an older unready or excepting entry. Retire width therefore limits architecturally visible IPC even when execution hardware has already completed more work internally. If the instruction at the ROB head is marked as an exception, the machine instead flushes younger speculative work and transfers control to the retry or handler PC, as shown in Eq. (6.10).

$$\begin{aligned} exception(ROB_{head}) \Rightarrow \text{flush younger instructions} \\ \wedge \text{ transfer control to the retry or handler PC} \end{aligned} \tag{6.10}$$

For example, a page-faulting load is usually retried after the operating system resolves the fault, whereas a trap-like event may resume at the following instruction.

ROB with bypass or forwarding Some designs let dispatch search the ROB from youngest to oldest for the most recent in-flight producer of a source register. If the matching entry is ready, dispatch can copy the value directly. If it exists but is not ready, dispatch copies the tag or ROB identity and waits. If no matching entry exists, the value must already have retired, so dispatch reads it from the architected register file. This is often described as a ROB with bypass or forwarding. It is effective, but the search itself is associative and can become part of the critical timing path.

Conte connects this search to the forwarding paths in the five-stage pipeline: both find the newest older producer instead of blindly reading an outdated architectural register. The difference is what can be forwarded. A ready producer supplies a value; an unfinished producer supplies an identity to wait for. The ROB search generalizes the earlier pipeline's bypass question from a few known stages to a window of instructions that may finish in a different order. It must still select the youngest matching producer *older than the consumer*, not a younger write to the same register.

Worked tag-based ROB-bypass trace The following tag-based example makes the lookup rule explicit. Assume the architected register file initially holds committed ready values for $R1$, $R2$, and $R3$, and that dispatch allocates ROB tags in program order. Consider:

A: LW $R1, 0(R3)$
B: ADD $R3, R2, R1$
C: ADD $R1, R3, R2$

At dispatch, the source resolution is:

Instr.	Dest reg/tag	Src #1 at dispatch	Src #2 at dispatch
A	$R1$/A	$R3$ value from architected RF	n/a
B	$R3$/B	$R2$ value from architected RF	waits on tag A from ROB entry A
C	$R1$/C	waits on tag B from ROB entry B	$R2$ value from architected RF

The key rule is that dispatch always takes the newest in-flight producer. So B does not read the old committed $R1$ value; it waits on tag A. Likewise, C does not read the old committed $R3$

Instr.	Dest reg/tag	Src #1	Src #2	Queue status
A	$R1$/A	$R3$ value ready	n/a	ready to fire
B	$R3$/B	$R2$ value ready	tag A not ready	waiting on A
C	$R1$/C	tag B not ready	$R2$ value ready	waiting on B

ROB entry	Dest reg	Ready	Value source
A	$R1$	0	pending load result
B	$R3$	0	pending add result
C	$R1$	0	pending add result

Table 6.3: Tag-based Tomasulo plus ROB bypass after dispatch of A, B, and C.

value; it waits on tag B. After all three instructions have dispatched, the in-flight structures look like Table 6.3.

From this point onward, execution follows the dependence chain directly. A completes first and marks ROB entry A ready, which wakes B. B then completes and marks ROB entry B ready, which wakes C. C completes last. Retirement still occurs strictly in order A, then B, then C, so after the last retirement both the scheduling queue and the ROB are empty. After retirement, the architected register file holds the retired results, with $R3$ containing the value produced by B and $R1$ containing the younger value produced by C. For any such trace, once the total elapsed cycles are known, fragment IPC is retired instructions divided by elapsed cycles over the chosen measurement interval.

Freeing physical registers Renamed machines also need a rule for reclaiming old physical registers. The exact policy differs across implementations, but an old physical register cannot return to the free list while any surviving reader or recovery path still needs that version. Common implementations describe this with use counters, pending-free bits, or retirement-based reclamation. For example, if an architected register was formerly mapped to physical register $P23$ and rename later redirects that architected name to $P27$, the old $P23$ mapping cannot be recycled immediately. It becomes reclaimable only after no older instruction can still read it and the newer mapping has safely become the committed architectural version. The details vary, but all correct schemes delay reuse until the older mapping is no longer architecturally or speculatively required.

When the architectural view changes Completion and retirement are not the same event. In an explicit-PRF design, completion writes the destination physical register and marks the matching ROB entry ready. Retirement updates the committed mapping and permits reclamation of the previous physical version; it need not copy the value into a second data file. In a data-in-ROB design, retirement instead copies the result into a separate architected register file. This separation is what allows the machine to execute aggressively while still presenting precise sequential state to software. A retiring store may remain in a committed store buffer until it can drain under the memory system's ordering rules; retirement is not immediate global visibility.

Worked PRegs/RAT/ROB end-state trace The following short trace makes the bookkeeping easier to follow. Assume the initial mappings are $R0 \to P0$, $R1 \to P1$, $R2 \to P2$, and $R3 \to P3$, with $P4$ through $P7$ free. Assume two-wide fetch and dispatch, two-wide retirement, two 1-cycle ALUs, and one 1-cycle LSU. Evaluate each cycle in the order Retire, Complete, Schedule, Dispatch, Fetch. An instruction firing in cycle k completes in cycle $k + 1$; Schedule can use a result published

earlier in that same cycle, but Retire has already run and must wait until the next cycle. These are this example's conventions; the preceding CDB example uses next-cycle wakeup instead. Consider the following sequence:

$$\begin{aligned} A&: \text{ LW } R3, 0(R1) \\ B&: \text{ ADD } R1, R3, R2 \\ C&: \text{ ADD } R2, R2, R1 \end{aligned}$$

One legal cycle-by-cycle interpretation is:

Cycle 1. Fetch A and B.

Cycle 2. Dispatch A and B. A allocates $P4$ as the new destination for $R3$ and records old mapping $P3$ in its ROB entry. B therefore reads $P4$ as its first source, allocates $P5$ as the new destination for $R1$, and records old mapping $P1$ in its ROB entry. Fetch C.

Cycle 3. A fires on the LSU. Dispatch C, which now reads $P2$ and $P5$, allocates $P6$ as the new destination for $R2$, and records old mapping $P2$ in its ROB entry.

Cycle 4. A completes, marking $P4$ ready and its ROB entry ready. Because that completion is visible before scheduling in the same cycle, B can now fire.

Cycle 5. A retires, so the committed mapping of $R3$ becomes $P4$ and old mapping $P3$ can be freed. B completes, marking $P5$ ready, and C can then fire.

Cycle 6. B retires, so the committed mapping of $R1$ becomes $P5$ and old mapping $P1$ can be freed. C completes, marking $P6$ ready.

Cycle 7. C retires, so the committed mapping of $R2$ becomes $P6$ and old mapping $P2$ can be freed.

Different simulator orderings can change the cycle numbers, but not the required data dependences or final committed mapping for this allocation sequence.

Instr.	FU	Dest physical reg.	Src #1	Src #2	Final status
A	LSU	$P4$	$P1$	n/a	completed and retired
B	ALU	$P5$	$P4$	$P2$	completed and retired
C	ALU	$P6$	$P2$	$P5$	completed and retired

ROB entry	Dest architected reg.	Prev physical reg.	Final status
A	$R3$	$P3$	retired
B	$R1$	$P1$	retired
C	$R2$	$P2$	retired

Architected reg.	Final physical reg.
$R0$	$P0$
$R1$	$P5$
$R2$	$P6$
$R3$	$P4$

Physical reg.	Ready	Free
$P0$	1	0
$P1$	1	1
$P2$	1	1
$P3$	1	1
$P4$	1	0
$P5$	1	0
$P6$	1	0
$P7$	0	1

Three rules determine the outcome. Dispatch allocates the next free physical destination and records the previous architectural mapping in the ROB. Completion makes the speculative destination ready for younger consumers. Retirement is the event that both updates the architected view and frees the previous physical mapping. The final mapping table is the committed view, not evidence of a separate copy of every value. Ready bits on free registers are irrelevant stale state and must be initialized appropriately when those registers are allocated again.

6.15 Future Files and Checkpoint/Repair

The course's *future-file* organization uses two data views: a latest speculative value per architectural name and a separate file of committed values. Producer tags ensure that an older completion does not overwrite a newer version of the same name. Retirement copies the correct in-order result into the committed data file; on recovery the speculative view can be rebuilt from committed state. An explicit-PRF design solves the same problem with multiple physical versions and recoverable maps instead of requiring that second value copy. Keeping these organizations distinct makes the recovery mechanism easier to follow.

Comparison with tag-only Tomasulo A physical-register plus RAT design does not automatically increase the amount of ILP that the scheduler can discover. If a tag-based Tomasulo machine and a physical-register machine have the same function units, the same wakeup/select policy, the same retire width, and enough result buses that completion bandwidth is not the bottleneck, then they can expose the same execution schedule and the same IPC on a given trace. The main differences are in bookkeeping cost, lookup style, recovery organization, and physical-register reuse rather than in the dependence constraints themselves. Performance diverges only when one design changes a practical limit such as result-bus count, port bandwidth, wakeup timing, or retirement bandwidth.

The ROB is not the only recovery method. *Checkpoint/repair* schemes periodically capture a coherent snapshot of register state and use that snapshot as the recovery point. One common description uses three conceptual views of the state: a messy current copy, a working copy being built toward coherence, and a golden copy representing the last known coherent sequential state.

The key control concept is the *instruction boundary*. Dispatch chooses a dynamic boundary, often denoted by markers such as IB1 and IB2. Instructions older than the current boundary in *program order* contribute to both the messy state and the working copy. Younger instructions contribute only to the messy state, even if they finish earlier in wall-clock time. Version tracking must preserve the youngest writer within each view when multiple instructions write the same architectural name. Once every instruction older than the boundary has completed without an unhandled fault, and each register reflects its youngest writer in that prefix, the working copy may be promoted to the new golden copy. The machine then chooses a new boundary and repeats the process.

If an exception occurs, the machine restores the golden copy and continues from that last coherent point. It may then re-execute younger instructions sequentially or under tighter control until it reaches the faulting instruction and handles the event precisely. This can reduce some of the timing pressure associated with aggressive ROB-bypass logic, though it introduces its own bookkeeping and copying costs. Restoring registers alone cannot undo a younger store that has escaped to memory or a device. Stores and other externally visible effects must therefore be deferred, buffered, or otherwise recoverable until the chosen recovery boundary is safe.

6.16 Speculation Without and With Prediction

Superscalar pipelines use speculation to keep FUs busy. Some designs can speculate without high-quality prediction (for example, eager execution ideas), but the useful work fraction is then limited. High-accuracy branch handling usually gives better effective IPC because fewer speculative instructions are squashed.

6.17 Security Implications of Speculative Execution

Precise exceptions solve the architectural recovery problem, but they do not automatically solve the microarchitectural side-effect problem. An instruction that is later squashed may still have changed caches, predictors, or other timing-visible structures before the machine discovered that the instruction should not retire. That is the core issue behind speculative side channels.

The best-known examples are Spectre- and Meltdown-style attacks. At a high level, the attacker causes the processor to execute work that will later be squashed or faulted. Even though the architectural state rolls back correctly, the speculative work may still have fetched a cache line or trained a predictor. The attacker then measures timing differences to infer information that should never have become visible through the ISA.

Shared-core execution can make this worse. With *hyperthreading*, or more generally simultaneous multithreading, two threads share parts of one physical core. That increases the opportunity for one thread to observe side effects caused by another thread's speculative activity. The security lesson is broader than any single named vulnerability: architectural rollback is not a complete isolation mechanism. Mitigations may need to prevent unsafe speculative accesses or isolate their observable effects, not merely try to clean up afterward.

6.18 Implementation Pressure

Once out-of-order execution moved from isolated examples into full commercial processors, these mechanisms had to satisfy real timing, area, power, and verification constraints. That practical pressure is one reason the field explored multiple recovery styles rather than converging immediately on a single universal design.

The architectural lesson is that elegant scheduling ideas are not sufficient by themselves. ROB bypass logic, associative searches, speculative-state tracking, and recovery control all stress timing closure and verification. As out-of-order windows grow wider and deeper, state management and recovery logic become first-order design problems rather than secondary implementation details.

6.19 Interpreting Out-of-Order Processor Statistics

Trace-driven superscalar simulators often report more than cycles and IPC. Those counters are useful only when their meanings and denominators are clear.

Fetched versus retired instructions The architecturally meaningful IPC uses retired instructions, not fetched instructions. *Trace instructions* count the dynamic instructions in the workload itself. In an execution-driven model, *fetched instructions* can include wrong-path work fetched before a branch misprediction was repaired. *Retired instructions* count only the instructions that reached

precise architectural state. Some trace-driven models never fetch actual wrong-path instructions and instead impose a modeled penalty, so their counters need not behave this way. For a fully drained run over a committed-path trace, retired instructions should match the trace count; warm-up boundaries and partially drained windows require separate accounting.

Miss and misprediction counters Branch-misprediction counts measure control-flow repair events. Instruction-cache misses reduce front-end supply, while data-cache misses usually delay load or store completion and can starve dependent instructions. These counters therefore explain why IPC may fall even when the machine nominally has ample fetch width and many function units.

Queue-pressure counters Maximum and average dispatch-queue, scheduling-queue, and ROB occupancy show where pressure accumulates. If ROB usage remains near its limit and dispatch is frequently blocked because the ROB is full, then commit bandwidth, long-latency instructions, or recovery delays are throttling forward progress. Lightly used scheduling queues with many *no-fire cycles* do not by themselves identify the cause. Check whether the queue is empty because fetch is starved, occupied by instructions awaiting operands, or blocked by unavailable execution units.

Isolating control and memory effects Many simulators also offer a no-miss or no-misprediction mode. Such a configuration does not model real execution, but it is analytically useful because it isolates pure width, scheduling, and dependence limits from penalties caused by cache misses and poor control-flow prediction.

6.20 What to Carry Forward

A younger instruction can make useful progress while an older one waits, provided the machine keeps the right value versions and respects true dependences. Renaming removes conflicts caused by reused names; it does not manufacture a missing operand or settle a memory alias. Result bandwidth determines when completed computations can publish their values, and in-order retirement determines when their effects become committed state.

Read a timing table with those separate events in mind: dispatch, fire, result production, publication, and retirement. State the within-cycle convention before counting cycles, and distinguish value-storage ports from scheduler readiness checks. The next chapter moves scheduling work toward the compiler and asks how much of this overlap can be arranged before the program runs.

6.21 Practice Questions

The following questions reinforce the main ideas of Chapter 6 and are intended for self-checking after the summary.

1. A FOCO processor fetches at most 6 instructions per cycle, dispatches at most 4 per cycle, has three 1-cycle function units that can each accept one instruction per cycle, and retires at most 5 per cycle. Assuming the window always contains enough independent ready work of the right types, what is the maximum IPC? Which resource creates the bound?
2. A dependence graph has nodes A through G and edges $A \to C$, $B \to C$, $B \to D$, $C \to E$, $D \to F$, $E \to G$, and $F \to G$. If the dependence height of a node is the length of the longest path from that node to an exit, compute the height of every node.

3. In a FICO-style machine with one LSU, one ADD unit, two-cycle loads, and one-cycle adds, trace the sequence `A: LW R1,0(R2)`, `B: ADD R3,R1,R4`, `C: ADD R5,R6,R7`, `D: ADD R8,R3,R9`. Assume dispatch is two-wide in program order starting in cycle 1; a dispatched instruction can fire no earlier than the next cycle. Fire is at most two-wide in program order, the ADD unit accepts one instruction per cycle, and results become usable in the cycle after completion. A two-cycle load firing in cycle 2 completes at the end of cycle 3. Ignore fetch, queue-capacity limits, and completion-bandwidth limits. Fill in a cycle table for dispatch, fire, and complete. Then state whether adding a second ADD unit removes all stalls, and explain which stalls remain due to the in-order fire rule.
4. Consider Tomasulo with tags and a ROB with bypass for the sequence `LW R1,0(R3)`; `ADD R3,R2,R1`; `ADD R1,R3,R2`. When the second and third instructions dispatch, which of their source operands come from the register file and which come from the youngest older ROB producer?
5. Using the same three-instruction sequence, write the contents of the scheduling queue or reservation stations immediately after dispatch of all three instructions. For each entry, identify the destination tag or physical register, which source operands are already ready, and which operands are waiting on older tags.
6. Suppose the machine from the previous question has two ADD units, one LSU, and enough result buses that completion bandwidth is not the bottleneck. What is the earliest completion order of the three instructions, and in what order do their results become architecturally visible?
7. In a renamed machine, the current mapping of architected register `R1` points to physical register `P10`. A new instruction `ADD R1,R2,R3` dispatches and receives destination physical register `P24`. What mapping change occurs immediately at dispatch, and why can `P10` not be freed at that same moment?
8. Distinguish the event of broadcasting a result on the common data bus from the event of updating precise architected state at retirement. Why is this distinction essential once execution and completion occur out of order?
9. For each statement below, answer true or false and justify briefly: (a) with enough physical registers, WAR and WAW dependences can be removed, but RAW dependences remain; (b) if a value is present in the future file or physical register file, then precise architected state has already been updated; (c) using fetched instructions instead of retired instructions can overstate IPC on a run with many squashes.
10. A simulator reports 1,600 fetched instructions, 1,200 retired instructions, 1,000 cycles, 220 no-fire cycles, and 90 ROB-blocked dispatch cycles. Compute IPC using the architecturally meaningful instruction count. Then explain what the gap between fetched and retired instructions suggests and what the ROB-blocked count suggests.

Chapter 7

Compiler-Scheduled ILP: VLIW and Software Pipelining

The previous chapter asked the processor to find ready work while the program ran. For a regular loop, much of that work can be planned earlier. The compiler already knows which operations use each result and which functional units the target machine provides. It can arrange a schedule in advance, including an overlap in which one iteration loads data while another computes and a third stores its result. This chapter follows that alternative, along with the uncertainty that a static plan cannot remove.

7.1 VLIW as a Compiler-Scheduled Alternative

One response to dynamic-scheduling complexity is to shift more of the scheduling work into the compiler. As superscalar windows grow, the hardware would like a larger scheduling queue so it can search farther ahead for independent work, but larger queues also make wakeup, select, and operand-readiness checks harder to close at a short cycle time. This tradeoff motivates the *very long instruction word (VLIW)* approach, historically associated with Josh Fisher. In a VLIW machine, one instruction word encodes several operations intended to execute in parallel, so the compiler determines the schedule at compile time rather than leaving that work to large runtime scheduling hardware.

In this context, an individual encoded action is an *op*. A set of ops intended to execute in the same cycle is a *multi-op*. VLIW reduces runtime hardware complexity because the machine no longer needs a large associative scheduling queue, aggressive wakeup logic, or much of the dynamic issue machinery used in FOCO-style processors. The tradeoff is that the compiler must do much more work up front, and the resulting code is less adaptable to runtime variation such as cache misses or changing branch behavior. This tradeoff is attractive in domains such as *digital signal processors (DSPs)*, where workloads are regular and low-power execution is important.

Compiler design must balance three competing goals: code quality, compilation time, and maintainability. The acceptable search cost depends on block size, optimization level, and whether compilation happens ahead of time or during execution. Optimal scheduling with general dependence and resource constraints is computationally difficult; exhaustive search grows rapidly with block size. Practical VLIW compilers therefore rely on heuristics rather than exact search for every block. List scheduling is one of the standard heuristics for this job. The broader performance point is that widening fetch and scheduling machinery yields diminishing returns: useful IPC usually grows sublinearly with front-end width, while the cost of dynamic scheduling hardware grows quickly.

Instr.	Operation	True dependences	Latency
A	LW $R2, X$	None	2
B	LW $R3, Y$	None	2
C	LW $R4, Z$	None	2
D	ADD $R5, R1, R2$	A	1
E	MUL $R6, R2, R3$	A, B	3
F	MUL $R7, R1, R3$	B	3
G	ADD $R8, R7, \#1$	F	1
H	SW $R8, X$	G	1
I	ADD $R9, R2, R4$	A, C	1

Table 7.1: Illustrative basic block for VLIW list scheduling.

7.2 Dependence Graphs and List Scheduling

The compiler begins with a *data dependence graph*. True flow edges capture read-after-write dependencies, while anti-dependence and output-dependence edges capture false constraints caused by register reuse. True dependences cannot be removed merely by renaming; false dependences must also be respected unless a transformation actually removes them. A basic block's program-order dependence graph is acyclic. A compact loop graph can contain cycles because its edges also represent dependences across iterations. Constructing the graph by hand is mechanical. Draw one node per op, then add all true flow edges from each definition to the later uses that require that value. Next add anti-dependence edges from a read to a later write of the same architectural name, and output-dependence edges between two writes of the same name. An ordering edge may be omitted only if the remaining paths enforce all of its timing requirements.

Table 7.1 gives a compact block that will be used for the scheduling example. Assume one pipelined load/store unit, one pipelined arithmetic logic unit (ALU), and one pipelined multiplier. Each unit can accept a new op every cycle; the listed latencies describe when results become available, not how long the unit remains occupied. Also assume the compiler has already assigned registers so that the remaining constraints shown in the table are dominated by true dependences rather than by avoidable name conflicts. Treat X, Y, and Z as distinct, nonvolatile memory locations with nonfaulting accesses in this simplified model. The load from X must still precede the store to X; that memory anti-dependence is separate from the register edges. Figure 7.1 shows the same block in a more visual form: the original operations, the resulting data dependence graph, and one priority list ordered by dependence height.

One common priority heuristic is *dependence height*, which estimates how much latency remains below a node on the longest dependent path. For an acyclic graph, a heuristic that includes the current instruction's latency is Eq. (7.1).

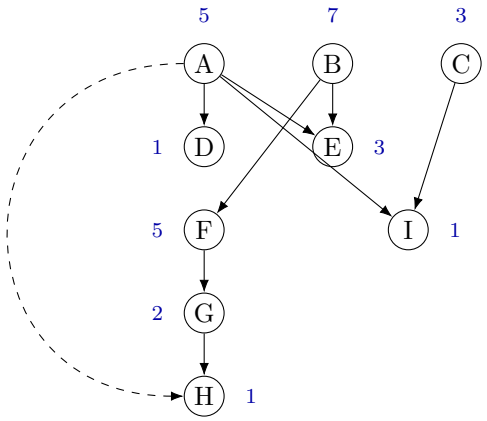
$$height(i) = \begin{cases} lat(i), & succ(i) = \emptyset \\ lat(i) + \max\limits_{j \in succ(i)} height(j), & \text{otherwise} \end{cases} \tag{7.1}$$

Higher-height instructions receive higher priority because delaying them is more likely to lengthen the critical path. To annotate a graph by hand, start from the sinks and assign each sink its own

Original operations

A: LW $R2, X$
B: LW $R3, Y$
C: LW $R4, Z$
D: ADD $R5, R1, R2$
E: MUL $R6, R2, R3$
F: MUL $R7, R1, R3$
G: ADD $R8, R7, \#1$
H: SW $R8, X$
I: ADD $R9, R2, R4$

Data Dependence Graph



List ordered by dependence height

B (7)
A (5)
F (5)
C (3)
E (3)
G (2)
D (1)
I (1)
H (1)

Figure 7.1: Operations, dependence graph, and a height-based priority list. The dashed $A \to H$ edge preserves the load-before-store order at X. Numbers indicate dependence height; ties may be broken arbitrarily.

latency. Then sweep backward, adding the current node latency to the maximum height among its successors. The priority list sorts nodes from highest height to lowest, with an explicit tie rule such as program order. At each scheduling step, only nodes whose predecessors have been placed and whose timing constraints permit issue are ready candidates. For the block in Table 7.1, the longest true-dependence chain is $B \to F \to G \to H$, so the heuristic tends to prioritize F ahead of E once both become candidates for the single multiplier.

List scheduling then places the highest-priority ready instructions into each cycle, subject to both dependence timing and resource availability. A simple legality condition is shown in Eq. (7.2).

$$cycle(i) = \min\{c \mid \forall j \in pred(i),\ c \geq cycle(j) + \delta_{ji},\ \text{resources are legal in } c\}\,. \tag{7.2}$$

Here δ_{ji} is the required issue separation on edge $j \to i$. For a RAW edge in this fixed-latency model, it is $lat(j)$; WAR, WAW, and memory edges depend on when operands are read and effects occur. The resource test asks whether a unit can *accept* an operation, not whether its pipeline is entirely empty. This is not an optimal algorithm, but it is practical and captures the central VLIW idea: schedule as much parallel work as possible before runtime while still respecting dependencies and resource limits.

Table 7.2 shows one possible compiler-produced schedule for the block running on a machine with one arithmetic logic unit (ALU), one multiplier, and one load/store unit. The point is not that this schedule is unique, but that the compiler can choose the cycle packing in advance and encode it directly. For illustration, this schedule keeps the three initial loads in source order even though a greedier dependence-height policy could start B before A and shorten the block by one cycle. That choice leaves one explicit empty cycle later, which makes the pause-field encoding example concrete. The schedule highlights two points. First, the compiler only packs operations together when both the dependence constraints and the functional-unit constraints allow it. Second, the absence of an operation in a slot does not require the hardware to infer what to do dynamically; the compiler has already encoded the intended schedule. In particular, cycle 6 is intentionally empty because G cannot start until the three-cycle multiply F has finished producing $R7$. Table 7.3 shows the same schedule in a field-level packing format closer to an actual multi-op layout. This table defines which operands occupy each unit's fields in each cycle; it does not, by itself, define the serialized op order

in memory.

7.3 Multi-Op Encoding, Tail Bits, and Pause Fields

VLIW raises an encoding question: how should a machine represent the boundaries of a multi-op when not every cycle uses every functional unit? One simple approach is to fill every unused slot with an explicit no-op, but that wastes code space. An alternative is to encode only real ops and add a small amount of grouping information. Each encoded op still carries its ordinary fields such as opcode, destination register, and source registers; the additional VLIW-specific fields only describe grouping and delay.

One useful scheme adds a *tail bit* to each encoded op. If the tail bit is 0, the current op is not the end of the current multi-op. If the tail bit is 1, the current op ends the current multi-op. The following op then begins a new multi-op. This lets the instruction stream represent a variable number of ops per cycle without requiring every cycle to contain the same number of explicit no-op fillers.

A related field is a *pause field*, which inserts a specified number of empty issue cycles before the next multi-op. Thus a multi-op issued in cycle k with pause p is followed by the next one in cycle $k + p + 1$. In the simple scheme used here, the pause value is interpreted only on the op whose tail bit ends the current multi-op; pause values on earlier ops in the same multi-op are ignored. This is a compact way to represent silence between multi-ops while keeping each op's encoding regular. Table 7.4 illustrates the idea using the schedule from Table 7.2. It shows one legal serialized op order for the same multi-ops that appear in Table 7.3. Some implementations instead use simple interlocking at the multi-op level, which can remove the need for an explicit pause field. That saves encoding space, but it also gives up some of the compiler's direct control over exactly when the following multi-op may begin.

7.4 Software Pipelining and Cyclic Scheduling

Loops create another opportunity for compiler-scheduled parallelism. Instead of scheduling only one copy of a loop body, the compiler can overlap operations from different iterations so that a later iteration begins before an earlier one has fully finished. The older name for this idea is *polycyclic scheduling*; the more common modern name is *software pipelining*.

Cycle	Load/Store Unit	ALU	Multiplier
1	A		
2	B		
3	C	D	
4			F
5		I	E
6			
7		G	
8	H		

Table 7.2: One legal VLIW-style schedule for the block under fixed functional-unit limits.

Cycle	IALU (1 cycle)			MUL (3 cycles)			Load (2 cycles)		Store (1 cycle)	
	d	s_1	s_2	d	s_1	s_2	d	addr	addr	s
1							$R2$	X		
2							$R3$	Y		
3	$R5$	$R1$	$R2$				$R4$	Z		
4				$R7$	$R1$	$R3$				
5	$R9$	$R2$	$R4$	$R6$	$R2$	$R3$				
6										
7	$R8$	$R7$	#1							
8									X	$R8$

Table 7.3: Field-level packing of the schedule from Table 7.2. Rows 3 and 5 are multi-ops because they contain more than one op in the same cycle.

Encoded op	Tail	Pause	Meaning
A	1	0	Single-op multi-op for cycle 1.
B	1	0	Single-op multi-op for cycle 2.
C	0	0	First op of the cycle-3 multi-op.
D	1	0	Ends the cycle-3 multi-op.
F	1	0	Single-op multi-op for cycle 4.
I	0	0	First op of the cycle-5 multi-op.
E	1	1	Ends the cycle-5 multi-op and inserts one pause cycle before the next multi-op.
G	1	0	Single-op multi-op for cycle 7.
H	1	0	Single-op multi-op for cycle 8.

Table 7.4: Illustrative VLIW encoding using tail bits and a pause field rather than explicit no-op padding.

Conceptually, the compiler behaves as if it had unrolled the loop many times, compacted the overlapping schedule, and then retained only the repeating pattern. That repeating pattern is the *kernel*. The code that fills the pipeline before the kernel reaches steady state is the *prologue*. The code that drains the remaining work after loop exit is the *epilogue*. The distance in cycles between the start of successive iterations is the *initiation interval (II)*. Once the schedule reaches steady state, the loop begins a new iteration every II cycles, so the throughput is $1/II$ iterations per cycle.

Unrolling view before compaction One useful way to understand software pipelining is to look first at a small explicit unrolling. Suppose a loop body consists of five ops A, B, C, D, E, where E is the loop-closing branch. A two-times unrolled form can be sketched conceptually as

$$A,\ B,\ C,\ D,\ E^*,\ A',\ B',\ C',\ D',\ E',$$

where E^* is the inverted early-exit branch that leaves the loop if the second copy should not begin. This makes the scheduling opportunity visible: once the loop is unrolled, the compiler can interleave A' with still-pending work from the first copy instead of waiting for the whole first iteration to drain first. Software pipelining then takes the next step and *compacts* that overlapping pattern into a

finite prologue, a repeating kernel, and a finite epilogue rather than materializing a large explicitly unrolled body.

Which dependences matter most in the first sketch In early cyclic-scheduling sketches, it is common to draw only true dependences and the loop-carried recurrences that constrain them. Anti-dependence and output-dependence edges are still real constraints in the full compiler problem, but they are often omitted from the first visualization because software renaming can frequently remove them. The scheduling bottlenecks that determine $RecII$ come from the true cyclic constraints, so those are the ones worth emphasizing first.

7.5 Iterative Modulo Scheduling

A practical way to construct these schedules is *Iterative Modulo Scheduling (IMS)*. The compiler first estimates a lower bound on II, then attempts to build a modulo schedule at that value. If the attempt fails, it increments II and tries again.

The first bound comes from resource pressure. For disjoint resource classes whose units each accept one operation per cycle, a loop with N_t operations of type t and U_t compatible units has the bound

$$ResII = \max_t \left\lceil \frac{N_t}{U_t} \right\rceil \tag{7.3}$$

This is the resource-constrained initiation interval. For a nonpipelined unit, shared port, or multi-cycle resource reservation, count occupied resource slots instead of merely counting operations.

The second bound comes from loop-carried recurrences. If a dependence cycle c has total required edge latency $Lat(c)$ and total iteration distance $Dist(c) > 0$, then

$$RecII = \max_c \left\lceil \frac{Lat(c)}{Dist(c)} \right\rceil \tag{7.4}$$

This is the recurrence-constrained initiation interval. If the dependence graph has no loop-carried cycles, then the practical lower bound is $RecII = 1$.

The schedule must satisfy both limits, so the compiler begins from

$$MII = \max(ResII, RecII), \qquad II \geq MII. \tag{7.5}$$

Many texts call this lower bound the *minimum initiation interval (MII)*. IMS follows a simple pattern:

1. Start with $II = MII$.
2. Attempt a legal modulo schedule at that value.
3. If the attempt fails, increment II and retry.

A modulo reservation table for a chosen II has exactly II rows. An operation that would occupy cycle x in a longer unrolled schedule is placed into row $x \bmod II$, provided the dependence and resource constraints remain legal.

Instr.	Operation	Key dependences	Latency
A	ADD $R2, R2, \#4$	loop-carried on $R2$	1
B	LW $R3, 0(R2)$	A	2
C	SW $R0, 0(R3)$	B	1
D	ADD $R10, R10, \#-1$	loop-carried on $R10$	1
E	BRGTZ $R10$, LOOP	D; branch must close the loop body	1

Table 7.5: Loop used to illustrate software pipelining and modulo scheduling.

Review Emphasis: Why IMS Starts at a Lower Bound The naive version of "try every initiation interval" can become too expensive because each scheduling attempt is itself a scheduling problem. List scheduling is often treated as roughly $O(n^2)$ in the number of operations, so repeatedly trying many impossible values of II can push the compile-time cost much higher. The point of $ResII$ and $RecII$ is to skip initiation intervals that cannot possibly work. A failed heuristic attempt at MII does not prove that MII is impossible, and the bound itself need not be attainable. Practical implementations limit search effort and may abandon pipelining when register pressure, code growth, or other legality constraints make it unattractive.

In most textbook treatments, the underlying machine is modeled with a point-latency abstraction: an op occupies one issue slot, and its result becomes ready after a fixed latency. This keeps the scheduling problem manageable. More detailed models that explicitly schedule register-file ports, result buses, and intermediate pipeline stages are possible, but they are much more expensive to optimize and are usually avoided in introductory treatments.

7.6 Worked Software-Pipelining Example

The following loop illustrates the idea. Assume two arithmetic logic units, two memory units, one-cycle add/branch latency, and two-cycle load latency. Every unit accepts one operation per cycle, and a result issued in cycle k is usable in cycle $k + latency$. Let $R10$ start at a positive trip count N; the loop decrements it and continues while it remains positive. Use a four-byte address step for aligned 32-bit words, and assume that the pointer table read by B is disjoint from every location written by C. Accesses are nonvolatile and nonfaulting in this scheduling model; real code motion must preserve exception and memory semantics too. These address and counter conventions adapt the handout's schematic loop into a finite example.[1]

The true data-dependence view contains $A_i \to B_i$, $B_i \to C_i$, and $D_i \to E_i$, plus the recurrences $A_i \to A_{i+1}$ and $D_i \to D_{i+1}$. The branch must also resolve in time to decide whether to start iteration $i + 1$. Operand capture and register allocation must preserve each B_i result until C_i consumes it; distinct iteration versions or rotating registers can make that bookkeeping explicit.

For this loop, the resource bound is

$$ResII = \max\left(\left\lceil \frac{3 \text{ ALU/branch ops}}{2 \text{ ALUs}} \right\rceil, \left\lceil \frac{2 \text{ memory ops}}{2 \text{ memory units}} \right\rceil\right) = \max(2, 1) = 2.$$

The data recurrences on $R2$ and $R10$ each have latency 1 and distance 1, giving a data-only $RecII = 1$. If the graph also includes the non-speculative control constraint $D_i \to E_i \to D_{i+1}$,

[1] `n5_vliw_spipe_handout_s26-1.pdf`, ILP-86–87.

Modulo slot	ALU 0	ALU 1	MEM 0	MEM 1
0	A_i	D_i		
1	E_i		B_i	C_{i-1}

Table 7.6: One modulo reservation table for the loop in Table 7.5 with $II = 2$.

Phase	Operations
Prologue	A_1, D_1 then B_1, E_1^*
Kernel	repeat every $II = 2$ cycles: A_i, D_i then B_i, C_{i-1}, E_i
Epilogue	one empty issue cycle, then C_{last}

Table 7.7: Prologue, kernel, and epilogue for the software-pipelined loop.

that cycle gives $RecII = 2$. Either way, the combined lower bound is $MII = 2$, and the following schedule resolves each branch before starting the next iteration.

One legal modulo reservation table appears in Table 7.6. The first row begins iteration i. The second row finishes the current iteration's control work while also performing the store from iteration $i - 1$. That overlap is the essence of software pipelining.

The emitted code must still handle loop entry and loop exit correctly. Table 7.7 separates the schedule into prologue, kernel, and epilogue form. The early-exit branch E^* is the inverted form of the loop branch; it lets the machine leave the prologue cleanly if the loop should terminate before the steady-state kernel is entered. Both exits go through the epilogue so the last required store is not lost. If zero iterations are possible, an entry test must bypass the entire transformed loop. The empty epilogue cycle matters: the final load issues with the exit branch and its result is not usable until two cycles later. Figure 7.2 shows the complete three-iteration timing, including that drain interval.

Cycle	ALU/branch issue	Memory issue	Phase
0	$A_1,\ D_1$		Prologue
1	E_1^*	B_1	Prologue
2	$A_2,\ D_2$		Kernel
3	E_2	$B_2,\ C_1$	Kernel
4	$A_3,\ D_3$		Kernel
5	E_3 (exit)	$B_3,\ C_2$	Kernel
6			Drain: B_3 not ready
7		C_3	Epilogue

Figure 7.2: Three iterations start two cycles apart, but the final store still has to wait for its own load. B_1, B_2, and B_3 become usable in cycles 3, 5, and 7. The branch decides when to stop starting iterations; it does not erase unfinished stores.

This example also shows why software pipelining is attractive for VLIW machines. The compiler exposes the repeating parallel structure explicitly, so the hardware can execute the steady-state kernel with very little dynamic scheduling logic. At the same time, general-purpose superscalar machines still benefit from dynamic scheduling because real loads do not always return in a fixed

number of cycles and because binaries may need to run across several microarchitectural variants.

Review Emphasis: Prologue, Kernel, and Epilogue The prologue is the startup code needed before the repeating pattern is fully populated. The kernel is the compact repeated pattern that behaves like many overlapped loop iterations. The epilogue drains work that was started by the final kernel iterations but has not yet completed. If nothing wraps across the compact kernel boundary, the prologue or epilogue may become trivial, but the conceptual split is still useful because it explains how a rerolled schedule preserves the behavior of the unrolled loop.

7.7 Memory Disambiguation

Loads and stores create a special scheduling problem. If a compiler moves a load ahead of an older store to the same address, the load may read the wrong value. The compiler therefore needs *memory disambiguation*: it must prove that the references do not overlap, preserve the required order, or use a supported runtime alias check with a correct fallback or recovery path. Adding a fence does not make a reordered aliasing load read the value from a store that has not executed yet. Potential faults, volatile accesses, device effects, and interthread semantics impose additional constraints even when addresses differ.

This is one of the limits of VLIW compared with dynamic scheduling hardware. An out-of-order processor may speculate and recover if it guesses wrong. A VLIW compiler must usually be more conservative unless it has strong alias information. That is one reason VLIW works best when memory behavior is regular enough for compile-time reasoning to be reliable.

7.8 What to Carry Forward

Static and dynamic scheduling must obey the same true dependences; they differ in when they discover and arrange the overlap. A compiler can make a regular loop's parallel structure explicit, but a legal steady-state kernel is only part of the result. Startup, final draining, register lifetimes, memory aliases, and exceptional behavior must still preserve the original program.

Use MII as a lower bound, not a promise of an achievable schedule. Then check every recurring resource slot and every dependence, including the work that crosses iteration boundaries. The next chapter widens the view again: instead of finding parallel instructions within one core, it coordinates processors that share data.

7.9 Practice Questions

The following questions reinforce the main ideas of Chapter 7 and are intended for self-checking after the summary.

1. For the sequence `LW R1,X`; `ADD R2,R1,R3`; `ADD R1,R4,R5`; `MUL R6,R2,R1`, draw the data dependence graph and label the pure, anti-, and output dependences.
2. Using load latency 2, add latency 1, and multiply latency 3, compute the dependence height of each instruction in the previous question. Keep the false dependences and use a conservative predecessor-latency separation for every edge in this exercise. Then list the instructions in descending order of dependence height.

3. Assume a VLIW machine with at most one ALU op, one MUL op, and one memory op per multi-op. Use the timing conventions of Question 2, with each unit able to accept one operation per cycle. Produce one legal list schedule for the instruction sequence in Question 1.
4. A VLIW schedule contains the following multi-ops: row 1 has a single load; row 2 has an add and a multiply; row 3 is empty because of a dependence delay; row 4 has a store. Explain how a tail-bit plus pause-field encoding can represent this schedule without inserting explicit no-op instructions into every unused slot.
5. A loop contains 3 ALU/branch operations, 1 multiply, and 3 memory operations per iteration. The machine has 1 ALU, 1 MUL, and 1 memory unit. Each unit accepts one operation per cycle and uses a separate resource class. Compute $ResII$. Then explain how a loop-carried recurrence could still force MII above that resource bound.
6. For the loop in Question 5, suppose the strongest loop-carried recurrence has total latency 4 and distance 1. Compute $RecII$ and MII.
7. Explain why a modulo reservation table for initiation interval II has exactly II rows, and explain the role of the mapping from cycle x in the unrolled schedule to row $x \bmod II$.
8. Why does IMS begin at MII instead of always starting at $II = 1$? Explain the compile-time cost argument and the correctness argument.
9. In a software-pipelined loop, what work belongs in the prologue, what work belongs in the kernel, and what work belongs in the epilogue?
10. Under what conditions may a compiler legally move a load ahead of an older store? Explain why this question is fundamentally a memory-disambiguation problem rather than only a functional-unit scheduling problem.

Chapter 8

Multiprocessors and Cache Coherence

Private caches solved a single core's distance problem by keeping useful data nearby. Put two such cores beside one shared memory and the solution creates a new problem: both cores can keep a copy, but one may change it. Which copy should the next reader trust? Coherence is the machinery that makes those private copies behave like one shared location.

8.1 From Instruction-Level to Thread-Level Parallelism

Instruction-level and thread-level parallelism are complementary: a machine can run several threads while each core overlaps instructions within its own stream. This moves the discussion from single-core scheduling into *multiprocessor* design. At this point the most visible form of parallelism is *thread-level parallelism (TLP)*: the programmer or runtime system must decide how to divide work across multiple processing elements. Unlike instruction-level parallelism, thread-level parallelism is visible to software and therefore directly affects correctness, synchronization, and programmability.

Historically, architecture texts often used the term *processing element (PE)* for one independently executing processor in a larger machine. Modern commercial systems more often use *core*, but the older PE terminology remains useful because it applies equally well to CPUs, accelerator lanes, and other specialized execution agents. Table 8.1 places multiprocessors in the broader grain-size view of parallelism.

At the architecture-classification level, two familiar Flynn terms continue to matter: *Single instruction, multiple data (SIMD)* machines apply the same operation to multiple data elements at once. *Multiple instruction, multiple data (MIMD)* machines allow different PEs to execute different instructions on different data at the same time. Modern shared-memory multiprocessors are usually discussed as MIMD machines. That does not exclude SIMD units inside each core; it only means that the overall machine permits several independently executing instruction streams. Do not confuse SIMD with *single program, multiple data (SPMD)*. SPMD workers run the same program on different data, but they can take different branches and advance independently, so an SPMD application can run on MIMD hardware.

Parallelism level	Typical mechanism	Programmer visibility
Instruction-level	superscalar issue, out-of-order execution, VLIW	usually hidden
Data-level	SIMD vectors, packed operations, vector pipelines	partly visible through data layout or intrinsics
Thread-level	multiple threads or PEs executing concurrently	directly visible

Table 8.1: Common grain sizes of parallelism. Multiprocessor programming is primarily concerned with the thread level.

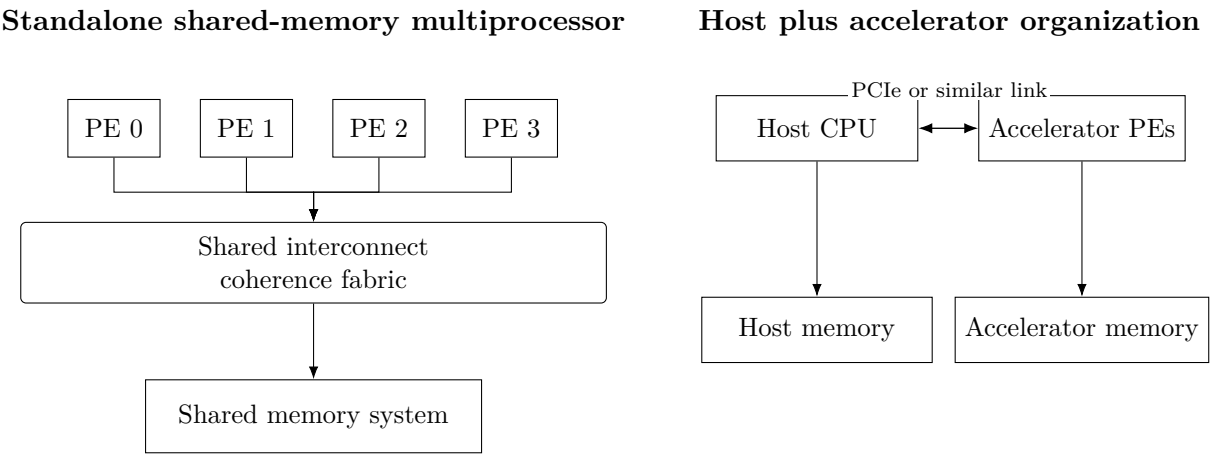


Figure 8.1: Two common multiprocessor organizations. Left: one shared-memory system. Right: a host plus accelerator with distinct memory, so software must move data explicitly.

8.2 Memory Organization and System Structure

A multiprocessor can be classified along another axis besides SIMD versus MIMD: how the memory system is organized. Two useful communication models are *shared memory* and *message passing*; neither alone specifies the physical placement of memory.

In a *shared-memory* machine, participating PEs can access common memory locations. Threads in one process normally share a virtual address space, but separate processes can map the same physical storage at different virtual addresses. Communication occurs when one PE writes data that another PE later reads. This model is attractive because it is easy to think about, but it immediately raises the cache-coherence problem once private caches are introduced.

In a *message-passing* program, participants exchange data through explicit operations such as `send()` and `receive()`. They may run on separate-memory nodes or on one shared-memory machine; the latter can still generate cache-coherence traffic underneath the messaging library. The *partitioned global address space (PGAS)* model presents globally addressable data with an explicit affinity to particular participants, making locality part of the programming model.

Figure 8.1 contrasts two common organizations. A standalone shared-memory multiprocessor integrates several PEs around a common memory system and runs the operating system and application code directly. An accelerator organization instead separates a host processor from a specialized parallel engine; the host launches work on the accelerator and explicitly copies data to and from accelerator memory.

Accelerators include graphics processors, machine-learning accelerators, and many other specialized engines. Their memories are often kept separate for protection, compatibility, and implementation reasons. The cost of that separation is explicit data motion: the host must copy inputs to the accelerator, launch the parallel work, and copy results back. This is the separate-memory organization drawn in the figure, not a requirement for every accelerator. Other designs expose shared or managed memory; a common address does not by itself guarantee equal access cost or remove synchronization requirements.

One Address Space, Unequal Distances

Uniform memory access (UMA) means that the underlying main-memory access cost is approximately uniform across processors and locations, apart from effects such as caches and queueing. With *non-uniform memory access (NUMA)*, placement matters: a core can reach nearby memory through its local controller, while remote memory requires additional interconnect travel. Both can implement shared memory. In *cache-coherent NUMA (ccNUMA)*, coherence preserves the shared-location abstraction even though physical access costs differ.[1] Figure 8.2 separates those two ideas.

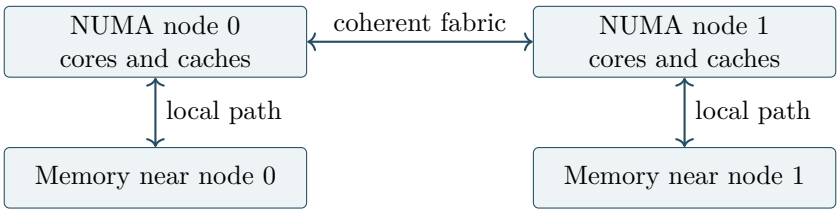


Figure 8.2: Shared does not mean equally near. A core in node 0 can access memory near node 1, but that remote path also crosses the inter-node fabric. Placement affects cost without changing which physical locations are shared.

Placing threads near the pages they use can reduce remote traffic; moving a thread without moving its data can do the opposite. NUMA is therefore a reason to care about data placement, not a synonym for message passing or incoherence.

8.3 Interconnects: Shared Buses and Point-to-Point Links

The interconnect between PEs matters just as much as the PE count. In this text, a *bus* means a *shared observable broadcast medium*: all agents attached to the bus can observe the transactions sent across it. That observability is the key property used later by snooping coherence.

A *point-to-point interconnect* behaves differently. A packet is sent from one endpoint to another over a dedicated link, and unrelated agents do not automatically see that traffic. Modern peripheral links such as *PCI Express (PCIe)* fit this model even though informal conversation sometimes calls them buses. Separate links can carry independent traffic instead of contending for one shared wire bundle, but a packet on one link is not automatically visible everywhere. Broadcast snooping can still be built over point-to-point links by explicitly distributing requests; it is a fabric service, not an inherent property of the physical link.

This distinction matters because some coherence mechanisms require every cache to observe every relevant request. Snooping works naturally on a shared observable medium. Directory-based coherence does not require broadcast visibility because a separate directory structure explicitly tracks who has each block.

8.4 The Cache-Coherence Problem

Shared memory becomes difficult as soon as each PE receives a private cache. Without additional coordination, different caches can hold different copies of the same memory location and update them independently. That breaks the programmer's illusion of one shared value.

[1] `n6_multiproc v1.1.pdf`, MP-10; see also the Linux NUMA documentation in Appendix F.

Step	Event	Locally visible value of x
0	Initial memory value	memory holds 1
1	P_0, P_1, and P_2 all cache x	each cache now holds 1
2	P_0 writes $x \leftarrow 22$ in its cache	P_0 sees 22
3	P_1 writes $x \leftarrow 44$ in its cache	P_1 sees 44
4	P_2 executes `x++` using its stale cache copy	P_2 sees 2

Table 8.2: An incoherent-cache scenario. Each PE can observe a different value of the same logical variable.

Consider a memory location x whose initial value is 1. Assume three PEs each read x and therefore bring a copy into their private caches. Now suppose synchronization orders these operations across the three PEs:

1. P_0 writes $x \leftarrow 22$.
2. P_1 writes $x \leftarrow 44$.
3. P_2 executes `x++`.

If the caches do not coordinate, then P_2 may still read its stale local copy of 1, increment it, and write back 2. With that ordering, the increment should read 44 and produce 45, but an incoherent cache system can instead use the stale value and produce 2. Program order alone orders instructions within one PE; it does not establish this sequence across different PEs. Table 8.2 summarizes the failure.

Simply using write-through caches does not solve the problem. Write-through propagates a write to the next level, possibly through a buffer; it neither guarantees an immediate DRAM update nor tells peer caches that their copies are stale. The real issue is not whether memory eventually receives the write; it is whether all cached copies remain mutually consistent.

A coherent cache system is normally expected to satisfy three properties:

1. **Read-after-write semantics for one processor.** If a PE writes a value and no one else writes that location in the meantime, a later read by the same PE should return the value it wrote.
2. **Eventual visibility to other processors.** If one PE writes a location and another PE later reads it, then after sufficient time has passed and assuming no intervening writes, the reader must eventually observe the value that was written.
3. **Consistent ordering of writes.** All PEs must agree on one order of writes to the same location.

The second property is intentionally phrased in terms of *sufficient time* rather than instant propagation. Physical systems cannot propagate signals across large chips or boards instantaneously. Coherence therefore promises eventual agreement and a consistent write story, not instantaneous zero-time updates.

8.5 Coherence Solution Families

Architecturally, there are three standard ways to solve the coherence problem.

State	My copy	Other caches	Memory copy
M	dirty and writable	no one else has a valid copy	stale
S	clean	others may also hold clean copies	correct
I	not valid in this cache	unspecified	unspecified

Table 8.3: Meaning of the three MSI states for one cache block.

The first is a *shared-cache* design in which all PEs communicate through one common cache structure. If it is the only cached copy accessible to the PEs, this avoids inconsistency among private copies. Sharing only a last-level cache does not remove the need to coordinate any private caches above it. The weakness is scalability: once the shared cache becomes the bottleneck, adding more PEs does not produce linear speedup even for embarrassingly parallel workloads.

The second is *directory-based coherence*. A directory tracks which caches currently hold each block and coordinates invalidations, forwarded data, or ownership transfers. This can scale better than a pure broadcast bus because the system does not require every cache to observe every request.

The third is *snooping coherence*. Here each cache has a *snooper*: logic that knows both what is currently in that cache and what transactions are appearing on the shared interconnect. If the interconnect is a shared observable medium, the snooper can react immediately when another PE requests a block that this cache currently owns or shares. The remainder of this chapter develops the classic snooping approach because it exposes the central state-machine ideas clearly.

8.6 Snooping Coherence and the MSI Protocol

A classic starting point for snooping coherence is the *MSI* protocol. Each cached block is marked as *modified (M)*, *shared (S)*, or *invalid (I)*. The meaning of those states is summarized in Table 8.3.

Two kinds of bus requests are especially important:

- `GetS`: request shared read permission for a block.
- `GetM`: request exclusive access with intent to write.

A dirty eviction produces a `WriteBack` transaction. Snoopers may also perform two high-level actions:

- *abort*: tell the requester to wait or retry because another cache still owns the necessary data or permission.
- *intervene*: supply the data directly rather than letting memory respond.

Table 8.4 summarizes the basic bus-level vocabulary.

From the requesting PE's perspective, the stable-state behavior is simple. If a block is invalid and the PE reads it, the cache issues `GetS` and moves to `S` when the data arrives. If a block is invalid and the PE writes it, the cache issues `GetM` and moves to `M` when ownership arrives. If a block is already shared and the PE wants to write, the cache must first upgrade through `GetM` so that all other copies are invalidated. Table 8.5 gives one compact summary.

From the snooper's perspective, the same bus requests mean something different. If a cache in state `S` sees another PE request `GetM`, it must invalidate its own copy and move to `I`. If a cache in state `M` sees `GetS`, it must ensure that the most recent data becomes visible and then downgrade to

Transaction or action	Meaning
`GetS`	Read miss requesting a clean shared copy.
`GetM`	Read-for-ownership requesting exclusive writable access.
`WriteBack`	Dirty data is returned to the backing store, for example on eviction or an MSI shared downgrade.
abort	Requester must wait or retry because another cache is still handling the block.
intervene	A snooping cache supplies the current data value directly.

Table 8.4: Common bus transactions and snooper responses in a snooping MSI system.

Current state	Local event	Bus action	Next state
I	read miss	`GetS`	S
I	write miss	`GetM`	M
S	read hit	none	S
S	write hit	`GetM`	M
M	read hit	none	M
M	write hit	none	M
S	replacement	none	I
M	replacement	`WriteBack`	I

Table 8.5: Stable-state MSI behavior from the requesting cache's perspective.

`S`. If it sees `GetM` while in `M`, it must give up ownership entirely and move to `I`. Table 8.6 summarizes the stable-state snooping behavior.

8.7 Transient States and a Worked Two-Cache Trace

Real buses and coherence fabrics are not instantaneous, so caches often need *transient states* while waiting for a request to complete. A common notation concatenates the path of the transition. For example, `IM` means that the cache started in `I`, issued a request whose destination is `M`, and is still waiting for that request to complete. Likewise, `SM` means a shared block has requested an upgrade to modified state but has not yet received final ownership.

Table 8.7 walks through a short two-cache scenario that mirrors the classic classroom introduction to MSI. It uses an abort/writeback/retry implementation; direct data handoff is a different valid implementation and need not insert the retry:

1. P_1 has a read miss and obtains a shared copy.
2. P_2 then has a read miss and also obtains a shared copy.
3. P_1 writes the block and upgrades from `S` to `M`.
4. P_2 then tries to write the same block, initially receives an abort, waits in `IM`, and retries after P_1's writeback has made the current data available.

The important insight is that the coherence protocol is maintaining one simple invariant: a

Current state	Snooped event	Action	Next state
I	sees `GetS` or `GetM`	none	I
S	sees `GetS`	none	S
S	sees `GetM`	invalidate local copy	I
M	sees `GetS`	make backing copy current, downgrade	S
M	sees `GetM`	provide or write back data, invalidate	I

Table 8.6: Stable-state MSI behavior from the snooper's perspective.

Step	Event	Bus msg.	P_1 state or action	P_2 state or action
1	P_1 read miss	`GetS`	$I \to S$	$I \to I$
2	P_2 read miss	`GetS`	$S \to S$	$I \to S$
3	P_1 write hit on shared block	`GetM`	$S \to M$	snoops `GetM`: $S \to I$
4	P_2 write miss while P_1 owns block	`GetM`, abort	write back before relinquishing ownership; then $M \to I$	$I \to IM$
5	P_2 retries after ownership transfer	retry `GetM`	$I \to I$	$IM \to M$

Table 8.7: One worked MSI trace showing stable states, a transient `IM` state, and a write-ownership transfer between two caches.

writable dirty copy must be unique. Before one cache acquires writable `M` permission, all competing readable copies must be invalidated. Allowing another cache to remain in readable `S` would violate the write-invalidate protocol, even if its current bits happened to match.

8.8 MESI, MOESI, and MOESIF Refinements

MSI captures the essential coherence idea, but practical protocols often add extra stable states that reduce unnecessary traffic. The most common additions are *exclusive (E)*, *owned (O)*, and *forward (F)*. These states do not change the basic correctness goal. They refine who may write, whether memory already has the current value, and which cache should respond to later requests.

Exclusive State and the Hit Line

The `E` state means that one cache has the block, the block is clean, memory is correct, and no other cache currently holds a copy. Its main advantage is that a later local write can move silently from `E` to `M` without first broadcasting `GetM`. That saves an upgrade transaction when a PE reads a block and then soon writes it.

The system needs some way to determine whether the requester is truly the only cache holding the block. In a snooping bus, a common mechanism is a *hit line*: when a cache issues `GetS`, any other cache that already has the block asserts one shared indication line. If no cache asserts the line, the requester may enter `E`; otherwise it must enter `S`. Without that shared-presence indication, the protocol cannot safely distinguish "only clean copy" from "one of several clean copies."

State	Local access	May share?	Backing-store and response responsibility
M	read/write	no	Backing store may be stale; this cache owns the current value.
O	read	yes	Backing store may be stale; the owner must preserve and supply the current value.
E	read/write	no	Backing store is current; a local write changes E to M.
F	read	yes	Backing store is current; this is the designated clean responder.
S	read	yes	Current in MSI/MESI; may be stale when an O owner holds responsibility for dirty data.
I	none	–	No usable local copy.

Table 8.8: Stable-state permissions and responsibilities. "May share" permits other copies; it does not prove they currently exist. In particular, an S copy does not guarantee current backing memory in a protocol with O.

Owned and Forward States

The `O` and `F` states refine what happens after a cache-to-cache transfer. Suppose a cache holds a dirty block in `M` and another PE requests `GetS`. In MSI, the owner typically writes the block back or otherwise makes memory current before both copies become shared. The `O` state avoids that immediate writeback. Instead, the original owner keeps a dirty shared copy, memory remains stale, and that owner becomes the designated responder for later requests. The cost is that `O` does not permit silent local writes: other sharers may exist, so the cache must obtain exclusive permission first.

The `F` state plays an analogous role for clean data. Suppose a cache has a block in `E` and another PE issues `GetS`. In the illustrative policy used here, the original cache supplies the data and becomes `F`, while the requester enters `S`. Other protocols can choose the new requester as forwarder instead; the state name alone does not determine that choice. On a later shared read miss, the `F` holder can respond instead of memory, avoiding ambiguity among multiple clean sharers.

Table 8.8 summarizes the permissions and responsibilities in this model.

The subtle case is an S copy beside an O owner. Both caches hold the current data, but only the owner owes the eventual dirty writeback; the S holder can discard its copy without writing it back. Figure 8.3 makes that distinction visible.

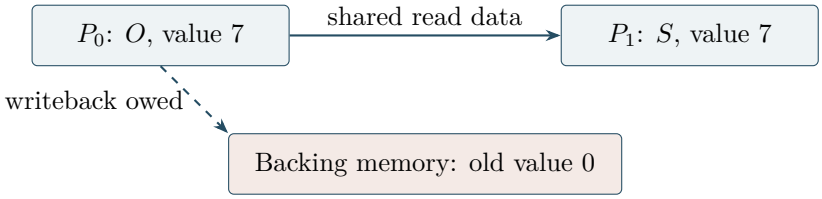


Figure 8.3: After P_0 writes 7 and supplies a shared read, MOESI can leave the data in O/S while memory still holds 0. The S copy is readable and current; it does not carry the owner's writeback responsibility. The dashed arrow is an obligation, not a transfer occurring now.

When solving coherence traces, it is useful to track three kinds of information separately after each access:

1. the stable or transient state of the block in each relevant cache,
2. whether main memory was read or written,
3. and whether the data arrived from memory or from another cache through intervention.

That bookkeeping separates a *state-machine* question from a *traffic* question. Two protocols can end in similar visible states yet generate different memory actions and cache-to-cache transfers along the way.

One useful way to think about these states is as a permission hierarchy. `M` grants read and write permission and carries responsibility for the current dirty value. `E` grants read and write permission but for a clean exclusive copy. `O`, `F`, and `S` are read-only states from the local PE's perspective; they differ mainly in whether that cache is responsible for serving future requests and whether memory is already correct. `I` grants no rights at all.

Protocol Families

Different stable-state sets lead to familiar protocol families:

- *MOSI*: `M`, `O`, `S`, `I`. This family can capture dirty sharing without requiring an exclusive-state detector.
- *MESI*: `M`, `E`, `S`, `I`. This family adds silent clean-exclusive upgrades.
- *MOESI*: `M`, `O`, `E`, `S`, `I`. This combines dirty-sharing optimization with silent clean-exclusive upgrades.
- *MESIF*: `M`, `E`, `S`, `I`, `F`. This adds a designated clean responder without an owned dirty-sharing state.
- *MOESIF*: `M`, `O`, `E`, `S`, `I`, `F`. This further designates one clean shared responder.

The presence of these states does not fully determine a protocol. The precise transition rules are an implementation choice. The important architectural question is what optimizations the state set makes possible.

Worked State-Refinement Examples

Three short examples capture the intent of the extra states:

1. **MSI versus MESI on a private read followed by a write.** A read miss issues `GetS`. If no other cache indicates possession, MESI places the block in `E`. A later local write becomes `E`→`M` with no bus upgrade. MSI would have placed the block in `S`, so the write would require `GetM`.
2. **MOESI optimization for a dirty owner receiving `GetS`.** Suppose P_1 holds a block in `M`, and P_2 issues `GetS`. Instead of writing the block back immediately and making both copies clean, P_1 may supply the data directly and become `O` while P_2 becomes `S`. Memory is still stale, so P_1 remains responsible for that value until eviction or another ownership change.
3. **MOESIF optimization for a clean exclusive owner receiving `GetS`.** Suppose P_1 holds a block in `E`, and P_2 issues `GetS`. P_1 can supply the clean value directly, become `F`, and let P_2 enter `S`. Memory remains correct, but P_1 is now the designated clean responder for later shared requests.

Field	Purpose
Presence bits	One bit per cache or PE indicating which agents currently hold the block.
Dirty bit	Indicates whether one cache currently holds the dirty up-to-date value.
Optional owner/forward ID	Identifies the cache responsible for responding to requests in protocols with `O` or `F`.

Table 8.9: Conceptual fields in a directory entry for one memory block. Real systems may compress or distribute this information, but these are the core ideas.

8.9 Directory-Based Coherence

Snooping assumes that every interested cache can observe every relevant transaction. That assumption is natural on a shared broadcast bus but becomes difficult to maintain as systems grow larger or move to point-to-point interconnects. Directory-based coherence removes the need for one globally visible snooping medium. Instead, a separate directory structure records who has each block and coordinates the messages required to preserve coherence.

Why a Directory Exists

Two jobs must still be done even without a snooping bus:

1. The system must know which caches currently hold a copy of the block.
2. The system must serialize ownership-changing writes so that all PEs observe one consistent write order.

In a directory protocol, the directory performs both roles. It tracks sharers and acts as the ordering point for conflicting requests to that block. Two processors may request writable ownership close together in time, but the controller cannot grant overlapping write permissions. Different blocks can have different home controllers and independent transaction orders; this is not one total order across all memory addresses.

Directory Entry Structure

At a high level, directory metadata is associated with memory blocks. It must identify or conservatively cover every possible holder so no needed invalidation is missed; false-positive sharers cost traffic but need not break correctness. For an MSI-style protocol, a useful minimal conceptual format is:

- a *presence bit vector* with one bit per cache or PE,
- a *dirty bit* indicating whether some cache has the only current dirty copy.

The presence vector distinguishes one sharer from many sharers. The dirty bit distinguishes a clean shared situation from a case in which one cache has the current modified value. Protocols with `O` or `F` often also record which cache is the current designated responder.

Table 8.9 shows a conceptual directory entry layout.

Directory Messages

The vocabulary shifts slightly in a directory protocol because requests no longer appear as one shared visible bus transaction. Instead, caches and the directory exchange explicit messages. Typical messages include:

- `GetS`: request a shared readable copy.
- `GetM`: request exclusive writable ownership.
- `Ack`: acknowledgment that a requested action completed.
- `Nack`: negative acknowledgment, typically meaning the requester must retry later.
- `Invalidate`: command to discard a clean shared copy.
- `Recall`: command to return a dirty value and surrender ownership.

The distinction between `Invalidate` and `Recall` is important. An invalidated clean sharer can simply discard the block and acknowledge. A recalled dirty owner must return data because memory does not yet have the current value.

Agent View versus Directory-Controller View

In a controller-oriented implementation, each cache contains an *agent* that handles processor-facing requests and local state transitions, while a home *directory controller* serializes permission changes for its blocks. The controller may be associated with a memory node or a cache slice; it need not reside in DRAM itself. The agent is responsible for actions such as servicing a local `LOAD` hit, turning a `STORE` miss into `GetM`, and holding transient states while replies are still in flight. The directory controller is responsible for sharer tracking, issuing invalidations or recalls, deciding when memory must be accessed, and determining when a requester may finally receive data or permission.

This viewpoint often expands the message vocabulary beyond the abstract `GetS`/`GetM` pair alone. It is common to distinguish:

- processor-side requests such as `LOAD` and `STORE`,
- network-visible permission requests such as `GETS`, `GETM`, and sometimes `GETX`,
- and controller responses such as `DATA`, `DATA E`, `REQ INVALID`, `INVACK`, `RECALL GOTO I`, and `RECALL GOTO S`.

`GETX` is especially useful in simulator-oriented MESI-like protocols. Architecturally, a clean exclusive copy could often upgrade silently from `E` to `M`. In the teaching model described here, the controller records the clean-to-dirty change explicitly. The agent then issues `GETX` and may wait in a transient such as `EM`, meaning "I believe I am the exclusive clean holder, and I am waiting for the directory to confirm my writable upgrade." This is not the only implementation choice, but it is a practical one because it keeps the directory's view synchronized with the cache's permissions. Other directory protocols deliberately record an exclusive holder as *possibly dirty*, allowing the $E \to M$ write to remain truly silent.

Simulation-Oriented Coherence Accounting

Many coherence simulators do not model the payload data values themselves. They model only state, message flow, and where the up-to-date value logically comes from. That can answer control

and traffic questions: was memory read, was dirty data written back, and did another cache supply the block? It does not alone prove that every load receives the correct payload value; data-value or invariant checking is needed for that stronger claim.

For the same reason, some pedagogical directory simulators route a cache-to-cache transfer through the directory controller even though the value logically originated in another cache. Architecturally, that still counts as a cache-to-cache transfer because memory was not the source of truth for the requester. The most useful counters in that style of analysis are therefore memory reads, memory writes, cache-to-cache transfers, and silent-upgrade-style events. Some teaching simulators count an $E \to M$ event in a "silent-upgrade-style" category even when they model an explicit `GETX/ACK` exchange. That counter means no peer invalidation or data fetch was needed; it does not mean zero network messages.

Worked MSI-Style Directory Transactions

Consider a block whose directory entry currently shows either a clean shared state or one dirty owner.

Shared read request (`GetS`). Suppose P_2 issues `GetS`. If the directory sees no dirty owner, then memory already has the correct value. The directory sets the presence bit for P_2 and returns the block so that P_2 enters `S`. If it sees a dirty owner, this MSI example recalls the value, makes the backing store current, downgrades the old owner to S, and records both readable holders before completing the transaction. The result is that the requester receives a correct shared copy and the directory's state matches reality.

Exclusive write request (`GetM`). Now suppose P_2 instead issues `GetM`. The directory must ensure that all competing copies disappear before granting writable ownership. If another cache has the block dirty, the directory sends `Recall` to that owner and collects the returned data. If other caches merely hold clean shared copies, the directory sends `Invalidate` to each one and waits for all acknowledgments. The requester can use writable ownership only after every required invalidation is acknowledged and the current data is available. Data and acknowledgments can arrive in either order in a more detailed implementation. If P_2 already has a current shared copy, it needs permission but not necessarily another data transfer. The final entry identifies P_2 as the sole writable owner, enforcing the single-writer invariant and a consistent order for this block.

Why Directories Scale Better

Directories remove the requirement that every coherence event be globally broadcast. Only the requester, the directory, and the currently relevant sharers need to participate in a transaction. That scales more naturally to larger systems and to topologies in which point-to-point hops replace one visible shared bus. The cost is storage and bookkeeping overhead. Directory state can be large, which is why real designs use compressed, sparse, hierarchical, or distributed variants. The conceptual model, however, remains the same: a directory tracks who has the block and coordinates legal ownership changes.

8.10 Directory Scaling, Limited Directories, and Distributed Organizations

The conceptual directory in the previous section is easy to explain because it behaves like one central hardware record of all sharers. The immediate drawback is size. For a dense directory covering every block of 2^m bytes of memory with 2^v-byte blocks, the number of entries is

$$\frac{2^m}{2^v} = 2^{m-v}.$$

Each entry then needs enough information to identify the sharers and encode control state. For a full bit-vector directory with N PEs, that means roughly N sharer bits plus a small number of control bits. As N grows, that metadata becomes large; a centralized controller can also become a throughput bottleneck. One useful way to reason about review-style metadata questions is therefore:

1. compute the number of directory entries from memory size and block size,
2. estimate the number of bits per entry from the sharer encoding plus control bits,
3. multiply the two.

For a full MSI-style bit-vector directory, the per-entry cost is roughly "one sharer bit per PE plus a few control bits." For a bounded-pointer directory, the sharer field becomes "K IDs of width $\lceil \log_2 N \rceil$ plus an overflow bit," which reduces metadata only as long as K stays small enough.

Limited Directory Schemes

One way to reduce directory size is to track only a bounded number of sharers explicitly. Instead of keeping one presence bit for every PE, the directory may keep K *presence IDs*, each wide enough to name one PE. Each ID therefore needs about $\lceil \log_2 N \rceil$ bits, so the sharing record becomes proportional to $K\lceil \log_2 N \rceil$ rather than N. This saves space when $K \ll N$, but it introduces an overflow problem: what happens when more than K sharers want the block at once?

Two broad responses are common:

1. **Broadcast on overflow.** The directory sets an overflow bit and stops tracking every sharer exactly. A later invalidation must then be sent to every PE, or at least to every PE in the relevant coherence domain, and the requester must wait for all necessary acknowledgments. This preserves correctness but increases latency and traffic.
2. **Limit sharing aggressively.** The directory evicts or invalidates one existing sharer when the pointer list is full. This preserves a bounded metadata size but can be harmful for read-mostly or read-only data, where many readers would ideally coexist.

The basic tradeoff is unavoidable: either metadata grows with the possible sharer count, or the protocol pays additional traffic or invalidation cost once that count is exceeded.

Hierarchical and Distributed Directories

The next step is to remove the single centralized directory bottleneck. One option is a hierarchy of coherence domains. For example, a machine may group several L1 caches under a private or local lower-level cache and maintain one local directory for that group. A higher-level directory then tracks which lower-level domains hold the block. An *inclusive* lower-level cache simplifies this

Organization	Main advantage	Main drawback	Typical coherence implication
One shared lower-level cache	Simple to explain and easy to share data	Becomes a bandwidth bottleneck quickly	L1 coherence still required; shared cache alone does not remove the need for directory or snooping logic
Private lower-level caches with higher directory	Scales better and supports hierarchical domains	Duplicates some lower-level data across domains	Higher directory tracks which domains, not just which individual L1s, hold a block
Sliced distributed lower-level cache and sliced directory	Balances traffic and avoids one lower-level bottleneck	Requires an address-to-slice hash and a more complex on-chip interconnect	Different addresses can be processed by different directory slices in parallel

Table 8.10: Common lower-level cache and directory organizations for scalable shared-memory multiprocessors.

tracking: every block held above is represented below. The lower-level bytes need not be current when an upper cache owns a modified copy, so inclusion does not remove the need to track dirty responsibility. Then the higher directory can reason in terms of the lower-level caches rather than every individual L1.

This style is sometimes described as a *last-level private* organization, because each cluster keeps its own lower-level cache and directory knowledge. The cost is duplication: if several domains need the same block, copies may appear in several lower-level caches. That is a reasonable exchange when it simplifies lookup and scales more cleanly than one fully shared cache.

Shared Last-Level Caches and Sliced Organizations

A single shared L2 or last-level cache is not a complete solution by itself. If the L1 caches are still private, coherence information is still needed for those L1s. Worse, one shared lower-level cache quickly becomes a bandwidth bottleneck. Even if two PEs want unrelated blocks, they still contend for the same shared cache ports and internal bandwidth.

Modern designs therefore tend to partition the last-level cache into *slices*. A hash of the physical address determines which slice owns the block and which directory slice manages coherence for that address. This distributes both data and directory traffic across the on-chip interconnect. Two unrelated blocks can then proceed through different slices in parallel. Table 8.10 summarizes the major tradeoffs.

The inclusive / exclusive / non-exclusive distinction also matters here. An *inclusive* hierarchy guarantees presence of upper-level blocks below, not necessarily identical up-to-date bytes while an upper level holds dirty ownership. An *exclusive* hierarchy forbids duplication between adjacent levels and maximizes effective capacity. A *non-exclusive* hierarchy allows some overlap without requiring it in all cases. Different choices change how much coherence metadata must be stored at each level and how much traffic is required when ownership changes.

Sharing pattern	Typical behavior	Coherence consequence
Private	One PE accesses the data	No true sharing traffic is required
Read-only	Many PEs read, but no PE writes	Sharing is cheap once the copies are installed
Producer-consumer	One PE writes a value that another later reads	Ownership moves in an ordered handoff pattern
Migratory	A writable variable moves from one PE to another over time, often under lock protection	The writable copy repeatedly changes owners
Read-write shared	Multiple PEs repeatedly read and write the same location	Coherence traffic can become intense, especially for synchronization variables

Table 8.11: Common sharing patterns and their coherence implications.

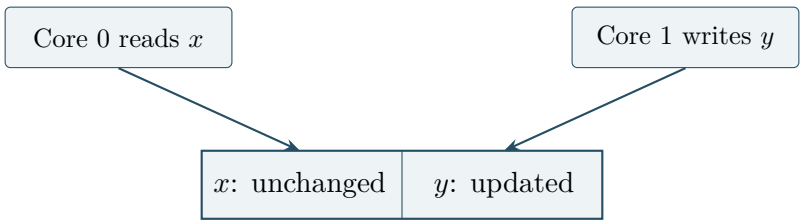
8.11 Sharing Patterns and False Sharing

Not all shared data behaves the same way. Architecturally, it is useful to classify a block by its *sharing pattern* because that pattern strongly influences coherence traffic. Table 8.11 summarizes several common cases.

The cache operates on blocks, not on individual source-language variables. That fact creates the classic *false-sharing* problem. Suppose one block holds two variables:

- x, which is effectively read-only, and
- y, which is migratory because different PEs write it under lock protection.

Each time y changes owners, the coherence protocol invalidates or transfers the whole block. That also invalidates x, even though no PE wrote x at all. The miss on the next access to x is therefore caused by block placement, not by true semantic sharing of x itself. Figure 8.4 isolates the placement problem.



One cache block: coherence permissions cover both variables.

Writing y invalidates Core 0's copy of the whole block.
Its next read of x can miss even though x did not change.

Figure 8.4: False sharing comes from the unit of coherence. Placing x and y in different blocks can remove this interference, at the cost of extra space and possibly a larger working set.

False sharing can cause *coherence misses*, often grouped informally as a "fourth C" beyond compulsory, capacity, and conflict misses. True sharing can cause coherence misses too; the

distinguishing feature of false sharing is that different variables unnecessarily share one coherence block. The cure is usually structural rather than algorithmic: avoid placing variables with very different sharing patterns into the same cache block. Read-only data, migratory lock-protected data, and heavily read-write shared synchronization variables are best separated when layout control is available. In trace-based reasoning, the symptom is usually easy to spot: when two variables share one block, writes to one variable increase invalidations, transfers, and misses for the other even though the source program did not truly share both variables semantically.

8.12 Programming Models: Shared Memory and Message Passing

The architecture choices above are reflected directly in parallel programming models. A shared-memory program typically divides the data among threads, lets each thread compute a local partial result, and then uses explicit synchronization to merge those results safely. The following example uses a lock to protect a critical section that updates a global sum:

```
float X[64];
float globalSum = 0;

Thread 0: sum X[0:15] locally, then lock(L); globalSum += local0; unlock(L);
Thread 1: sum X[16:31] locally, then lock(L); globalSum += local1; unlock(L);
Thread 2: sum X[32:47] locally, then lock(L); globalSum += local2; unlock(L);
Thread 3: sum X[48:63] locally, then lock(L); globalSum += local3; unlock(L);
```

The lock prevents a lost update in the critical section. Without it, two threads might read the same old value of `globalSum`, add their local sums independently, and then overwrite one another. That kind of latent synchronization bug is one reason shared memory is powerful but dangerous. The lock fixes lost updates, not floating-point reproducibility: a different order of partial sums can still change rounding in the final low bits.

A message-passing version of the same idea avoids shared writable state:

```
Thread 0: compute local0; send(local0, 4);
Thread 1: compute local1; send(local1, 4);
Thread 2: compute local2; send(local2, 4);
Thread 3: compute local3; send(local3, 4);

Thread 4:
  globalSum = 0;
  for (i = 0; i < 4; i++)
    globalSum += receive(i);
```

In this formulation, synchronization is implicit in the blocking `receive()` operation. The collector thread waits until the required messages arrive. The advantage is portability: this model works across independent machines with separate memories. The cost is explicit communication overhead and a more demanding programming model.

8.13 Synchronization Primitives and Coherence Traffic

Coherence preserves the illusion of one shared memory, but it does not remove the need for synchronization. If several threads can update the same variable concurrently, the program still needs explicit coordination such as locks or barriers. A *mutex* or lock grants one thread exclusive access to a critical section. A *barrier* forces a set of threads to wait until the last participant arrives.

At the architectural level, the key question is how the hardware supports these operations efficiently. A lock can be represented by a single shared variable, for example:

- `lock = 0`: free
- `lock = 1`: held

The challenge is acquiring that lock atomically so that two PEs cannot both believe they succeeded.

Test-and-Set

A classic atomic primitive is *test-and-set*: set a location to 1 and simultaneously return its old value. In pseudocode, a simple spin lock looks like:

```
while (test_and_set_acquire(lock) == 1) { }
```

Unlocking uses `store_release(lock, 0)`. These are abstract atomic operations, not ordinary C loads and stores: acquire/release ordering also protects the critical section's data. The operation is *atomic*: no competing PE may observe or interleave a partial update. Architecturally, that atomicity may be provided by a cache-coherent read-modify-write sequence or by another equivalent mechanism.

The weakness of this simple loop is coherence traffic. In the write-invalidate model used here, every failed `test_and_set()` still requests writable ownership of the lock variable. If many PEs spin on the same lock, they repeatedly invalidate one another's copies and keep forcing the block into a writable-owner state. The protocol spends a large fraction of its effort moving the lock variable rather than doing useful work.

Test-and-Test-and-Set

A better spin strategy is *test-and-test-and-set*:

```
while (1) {
  while (load_relaxed(lock) == 1) { }
  if (test_and_set_acquire(lock) == 0) break;
}
```

The inner read-only loop lets waiting PEs spin on a shared cached copy of the lock. The relaxed poll is atomic but does not acquire the lock; only the successful acquire operation admits the thread to the critical section. Only when the lock appears free do they attempt the atomic acquisition. This changes the coherence pattern substantially:

- while the lock is held, most waiting PEs perform only repeated local reads,
- when the owner releases the lock, the write invalidates the shared copies,
- only then do contenders attempt the more expensive atomic ownership request.

The result is still a spin lock, but one that generates far less coherence traffic than naive repeated test-and-set.

Barriers and Busy Waiting

Barriers are another important synchronization primitive. They ensure that all participating threads reach a common point before any are allowed to continue. They are often necessary, but they are not free: a barrier can force faster threads to idle while waiting for the slowest one. As with locks, the architectural cost appears as traffic, serialization, and delay rather than a violation of coherence correctness.

More generally, synchronization exposes the interaction between the programming model and the coherence system. A workload with frequent lock acquisitions, spinning, or barriers may obey all correctness rules and still perform poorly because shared synchronization variables become hot spots in the coherence fabric.

8.14 Scaling and Architectural Limits

Amdahl's Law still governs the big picture. If the useful work splits into independent pieces with negligible communication, the workload is often called *embarrassingly parallel*. Having no serial work in an Amdahl model gives ideal linear scaling, but does not by itself eliminate communication or resource contention. Real machines fall short of that ideal once a shared resource saturates. A single shared cache saturates quickly. A shared interconnect can also become a bottleneck. Even a well-designed coherence protocol can lose efficiency if the workload repeatedly bounces writable blocks between PEs. On a bus, the shared medium itself eventually limits scaling. In larger systems, the directory's storage cost and message traffic become the next design pressure point. Synchronization hot spots and false sharing are additional practical limits because they inject communication even when the algorithm's useful work is highly parallel.

Another difficulty is verification. Coherence protocols contain stable states, transient states, race windows, and multiple simultaneously active agents. The full system state is the cross-product of one block's state in every relevant cache and directory. That makes formal verification difficult but essential: the protocol must be correct not only in clean textbook traces but also under overlapping requests and rare timing corners.

This is why the interconnect, the memory model, and the programming style cannot be separated cleanly. A coherence protocol is not only a correctness mechanism; it also determines how much communication traffic the machine must generate in order to preserve the illusion of one shared memory.

8.15 What to Carry Forward

For each block, ask three separate questions: who may read, who may write, and who is responsible for the current value? Those questions explain the extra states better than memorizing their letters. An S copy can be current even when backing memory is stale, provided an O owner carries the dirty responsibility; an M writer cannot coexist with another readable copy in this write-invalidate model.

Track state and traffic separately, and state the implementation choices before counting messages or memory accesses. NUMA placement, directory layout, false sharing, and contention can change

cost without changing the program's intended result. The next chapter asks what else the program must request: indivisible updates and ordering between different locations.

8.16 Practice Questions

The following questions reinforce the main ideas of Chapter 8 and are intended for self-checking after the summary.

Concept Checks

1. Explain the difference between SIMD and MIMD. Then explain why a machine with multiple independent cores and a SIMD unit inside each core is still classified overall as MIMD.
2. Compare a shared-memory multiprocessor with a message-passing multiprocessor. What does the programmer gain and lose in each model?
3. Why is a traditional shared bus a natural fit for snooping coherence, while a point-to-point link such as PCIe is not naturally a broadcast snooping medium?
4. In the incoherent-cache example with initial value $x = 1$, explain why the sequence "P_0 writes 22; P_1 writes 44; P_2 executes `x++`" can produce 2 instead of 45.
5. State the three standard coherence properties in words and explain why one of them must be phrased in terms of "sufficient time" rather than instant visibility.
6. For one cache block in an MSI protocol, explain what it means for that block to be in states `M`, `S`, and `I`. In particular, explain what memory does or does not know in each case.

Trace and Protocol Exercises

7. Starting from all caches invalid for one block, use MSI to trace the access sequence `(P1,R)`, `(P1,W)`, `(P2,R)`, `(P3,R)`, `(P2,W)`, `(P1,R)`, `(P3,W)`. Use this traffic model: clean misses fetch from memory; a dirty owner supplies the requester directly and also updates memory on a shared downgrade; a writer that already holds S needs only invalidation permission. Complete each transaction before starting the next, and assume no replacements. For each step, record the state of the block in each cache and whether the access causes a main-memory read or a main-memory write.
8. Repeat the previous exercise for MESI, but focus only on the steps whose outcomes differ from MSI. Which steps use `E`, and how does that change later behavior?
9. Starting from both caches in state `S` for the same block, show the state changes when one cache performs a write hit. Which bus transaction occurs, and what must the other cache do?
10. What does the transient state `IM` mean? Give a short scenario in which a cache would enter `IM` rather than moving immediately from `I` to `M`.
11. Explain the practical value of the `E` state in MESI. Why does the protocol need some way to detect whether another cache already has the block?
12. A cache in `M` snoops `GetS`. Compare the MSI outcome with the MOESI outcome. What extra benefit does the `O` state provide, and when must memory be updated in each case?

Directory and Simulator Exercises

13. In a directory-based MSI protocol, distinguish the responsibilities of the cache agent from those of the directory controller. Which side handles processor-facing `LOAD`/`STORE` requests, and which side is responsible for global ordering and invalidation fan-out?

14. In a directory-based MSI protocol, compare what the directory must do for a `GetM` when the current other copies are all shared versus when one other cache has the dirty copy.

15. Why does a directory serve as both a sharer tracker and a write-ordering point in a scalable coherence system?

16. In a controller-oriented MESI implementation, why might a cache in `E` send `GETX` and enter a transient `EM` state rather than simply changing locally to `M`? What information is the directory learning through that exchange?

17. A system has 16 PEs, supports 128 GiB of memory, and uses 64-byte coherence blocks. Estimate the number of directory entries. Then estimate the metadata cost of a full MSI-style bit-vector directory if each entry stores one sharer bit per PE plus two control bits.

18. Using the same system, suppose a limited directory stores at most K sharer IDs, each $\lceil \log_2 16 \rceil$ bits wide, plus one overflow bit. What is the largest K for which this scheme still uses fewer sharer-tracking bits than the full 16-bit sharer vector? Count only the specified ID and overflow bits, ignoring occupancy, validity, and other control fields. In what kinds of sharing patterns does this scheme perform poorly?

19. Consider the trace

    ```
    (P0,W,Y), (P1,W,Y), (P0,R,X), (P1,R,Y),
    (P0,W,Y), (P1,R,X), (P0,W,Y)
    ```

    in a cold MOESI write-back, write-invalidate system with no replacements. A dirty owner supplies another cache directly; clean read misses use memory, and upgrades from a current S or O copy require permission but no new data transfer. Compare two cases: (a) X and Y are in the same cache block, and (b) X and Y are in different blocks. Which case causes more cache-to-cache transfers and why? Which case better avoids false sharing?

20. A controller-oriented simulator tracks memory reads, memory writes, cache-to-cache transfers, and silent-upgrade-style events. Assume cold caches, a lone clean reader receives E, a dirty owner supplies shared reads and retains O, and an evicted O owner writes back rather than passing ownership to a remaining sharer. For the sequence

    ```
    (P0,R), (P0,W), (P1,R), (P0,evict)
    ```

    on one block under a MOESI-like protocol with intervention, which of those counters must increase, and at which steps?

21. Which coherence-control and traffic questions can a simulator answer using only states and messages, and what stronger data-correctness claim would still require additional checking?

Scaling and Synchronization

22. Why does one shared lower-level cache fail to scale cleanly even if it is coherent? Explain the role of bandwidth and port contention.

23. Explain why a sliced lower-level cache plus sliced directory reduces bottlenecks compared with one centralized directory and one shared lower-level cache.

24. In the array-sum examples, explain why the shared-memory version needs a lock around the update of `globalSum`, while the message-passing version does not require the same kind of critical section.

25. Compare repeated test-and-set with test-and-test-and-set for a spin lock. Which one causes more coherence traffic, and why?

26. Why do barriers and locks remain necessary even in a perfectly coherent shared-memory system?

Chapter 9

Synchronization and Memory Consistency

One thread prepares a result and sets a flag to announce that it is ready. Another sees the flag and reads the result. The story sounds complete, yet a coherent machine can still expose the flag before the data unless the program requests the required ordering. Shared-memory correctness therefore needs three separate ideas: coherent copies, indivisible updates, and rules for ordering communication.

9.1 Protocol States Versus Protocol Behavior

Chapter 8 introduced stable coherence states such as `M`, `S`, `I`, and later `O`. The next important refinement is that a list of stable states does *not* completely define the protocol. Two systems may expose the same end states yet differ in how they move data, when they write back to memory, and how much traffic they generate while ownership changes are in flight.

Suppose one PE holds a block in `M` and another PE requests that block. Table 9.1 summarizes several common responses.

In MSI and MESI, a completed transition to shared copies requires the backing store to be current. Do not extend that rule to an S holder in MOESI: it may share current data with an O owner while memory is stale. There, "clean" means the S holder has no dirty writeback obligation, not that its bytes must match backing memory. If the protocol instead supports `O`, then memory may remain stale temporarily because one cache is explicitly responsible for the dirty shared copy. Thus, the presence or absence of `O` changes *when* write-back is required. The distinction matters most when memory is far away and cache-to-cache transfer is much cheaper than forcing every ownership change to go through DRAM.

This is why it is safer to think of protocol states as *permissions plus responsibilities* rather than as the full protocol. The state says what a cache is allowed to do and what cleanup it owes the rest of the system. The implementation still decides whether ownership changes happen through retry, direct intervention, deferred write-back, or some combination of those techniques.

9.2 Atomic Primitives for Shared-Memory Synchronization

Once multiple threads share writable state, the machine needs atomic operations that let software coordinate safely. Chapter 8 already introduced simple spin locks based on *test-and-set* and *test-and-test-and-set*. Those remain important because they reveal a core architectural truth: a synchronization primitive is not just a correctness device, but also a traffic pattern imposed on the coherence system.

A naive repeated `test_and_set()` forces the lock variable into a writable-owner state over and over again. That makes the modified copy bounce from PE to PE and fills the interconnect with invalidation traffic. *Test-and-test-and-set* reduces that cost by letting waiters spin mostly on shared cached copies and attempt the expensive atomic step only when the lock appears free. Both

Mechanism	Behavior when another PE requests a block currently held in `M`
Abort and retry	The current owner stalls the requester, completes the required ownership change or write-back, and then forces the requester to try again later.
Intervene	The current owner supplies the data directly, effectively pretending to be memory for that transfer and avoiding an unnecessary trip all the way to main memory.
Owned-state optimization	If the protocol supports `O`, the old owner may keep dirty responsibility while another cache receives a readable copy, so the memory write-back can be delayed until a later eviction or ownership loss.

Table 9.1: The stable state set does not by itself determine the transfer behavior. Different protocol implementations can move the same block in different ways and incur different write-back costs.

strategies can implement a correct lock with atomic polling/acquisition and acquire/release ordering; they differ in the communication they generate.

Load-Link / Store-Conditional

Another classic primitive pair is *load-link / store-conditional* (LL/SC). A load-link reads a memory value and establishes a reservation on a region containing that location. A later store-conditional attempts the write and succeeds only if the reservation and the ISA's other conditions permit it. A conflicting write can invalidate the reservation; an implementation may also clear it on conservative coherence events or for other permitted reasons. The reservation granule need not equal one variable or one cache block.

One way to express a lock acquisition using LL/SC is:

```
do {
  old = load_link(lock);
  if (old == 0)
    success = store_conditional(lock, 1);
  else
    success = 0;
} while (!success);
```

If `store_conditional()` succeeds, the thread acquired the lock. If it fails, the loop retries. As with the previous lock, successful acquisition needs acquire ordering and unlock needs release ordering; the abstract LL/SC loop does not specify an ISA encoding for those orders. A conflicting write can clear the reservation, but failure need not prove that another thread changed the value: an ISA can permit reservation loss or spurious failure for other implementation reasons.[1]

Architecturally, LL/SC works well when the machine can cheaply observe invalidation events affecting the reservation. That is natural in a broadcast snooping setting. In a larger directory-based system, however, the reservation must survive longer-latency ownership transfers and more complex

[1] For a concrete example, see the RISC-V atomic-extension specification in Appendix F.

message paths, so the mechanism becomes harder to scale cleanly. The primitive remains elegant, but its implementation cost depends strongly on the interconnect and coherence organization.

Fetch-and-Op and Fetch-and-Add

A more general atomic primitive is *fetch-and-op*: return the old value of a location while atomically applying some operation to it. In abstract form,

$$\texttt{fetch_and_op}(x, c): \qquad old = x, \qquad x \leftarrow x \ op \ c, \qquad \text{return } old. \tag{9.1}$$

The most common special case is *fetch-and-add*, where the operation is integer addition. Implementations may provide that primitive directly in hardware, build it from LL/SC, or realize it through another atomic read-modify-write path.

Fetch-and-add is extremely useful for dynamic work distribution. Instead of statically assigning iterations to threads, the runtime can let each worker claim the next available iteration number as it becomes free:

```
while ((i = fetch_and_add(next, 1)) < N) {
  body(i);
}
```

This is *dynamic work claiming* with one iteration per claim. Guided self-scheduling is a related policy that starts with larger chunks and reduces their size as less work remains; it is not the same as repeatedly claiming one iteration.[2] It handles uneven iteration costs naturally: a faster core simply claims more iterations over time, while a slower core claims fewer.

High-level interfaces often hide these mechanics. *OpenMP* is the standard directive-based shared-memory API. A compiler may translate a parallel loop into a runtime structure that internally resembles repeated work claiming. *MPI*, by contrast, is the standard message-passing interface; it exposes explicit communication rather than a shared atomic counter. The architectural lesson is that even high-level APIs eventually bottom out in a small set of atomic ownership and ordering operations.

9.3 Reducing Synchronization Contention

Correct synchronization is necessary, but naive synchronization can turn one hot variable into the dominant source of coherence traffic. Two common responses are to reduce the number of contenders hitting the same lock at once and to structure barrier release so consecutive barrier instances do not interfere with one another.

Tree and Tournament Locks

A *tree lock* or *tournament lock* replaces one heavily contended global lock with a hierarchy of smaller locks. For example, with 16 PEs one design might let: P_0 through P_3 compete first on one local lock, P_4 through P_7 compete on another, P_8 through P_{11} and P_{12} through P_{15} do the same, and then only the local winners advance to higher-level locks. That structure reduces the invalidation flood at the top of the tree because not all 16 contenders compete for the same variable at the same instant. Instead, most of the traffic stays localized within smaller competing groups.

[2] See the OpenMP worksharing-loop specification listed in Appendix F.

The price is latency. Acquiring the final global lock now requires winning several stages in sequence. Tree locks therefore trade reduced contention for longer acquisition paths. They are attractive when raw contention is the dominant problem and the extra hierarchy depth is cheaper than one enormous burst of invalidations on a single lock variable.

Barrier Construction with Fetch-and-Add

Fetch-and-add also provides a simple software barrier. The main idea is to count arrivals and release all threads when the last one arrives. The subtle part is making one barrier instance distinct from the next, especially if the barrier sits inside a loop. A common solution is a *sense-reversing barrier*, in which each thread keeps its own local phase bit.

A compact form is:

```
Shared atomics: count = 0, phase = 0
Per thread: localPhase = 0

barrier(N):
  localPhase = 1 - localPhase
  if (fetch_add_acq_rel(count, 1) == N-1) {
    store_relaxed(count, 0)
    store_release(phase, localPhase)
  } else {
    while (load_acquire(phase) != localPhase) { }
  }
```

The last arriving thread is the one for which

$$\texttt{fetch_and_add}(count, 1) = N - 1. \tag{9.2}$$

That thread resets the counter and flips the shared phase. All other threads spin until they observe the new phase value. The phase bit plays the role of the "toggle" idea often used in informal explanations: it prevents one completed barrier instance from being confused with the next one. Assume a fixed group of $N \geq 1$ threads, each arriving exactly once per phase, with no cancellation. The counter's acquire/release read-modify-writes form an arrival chain, collecting each thread's pre-barrier writes at the last arriver. That thread publishes the phase with release ordering; each waiter's successful acquire load then makes all pre-barrier work happen before its post-barrier work. The relaxed counter reset is safe here because it precedes that release and the next generation follows the completed phase. A phase bit alone, with ordinary racing C/C++ counter and flag accesses, would not provide this guarantee.

Hardware can implement a similar idea more directly. On a bus-based machine, one simple design gives each PE a dedicated arrival line. A PE asserts its line when it reaches the barrier, and everyone leaves once all lines are asserted. This is conceptually simple and resembles the one-hot signaling style used for other shared bus signals, but it does not scale well because it consumes dedicated hardware resources and assumes a shared observable medium. A reusable hardware barrier also needs a completion/reset or phase handshake; simply dropping arrival lines can let a fast participant erase the event before a slower one observes it.

Review Emphasis: Barrier Versus Lock A lock protects a critical section: at most one thread owns the protected region at a time. A barrier is a phase boundary: all participating threads wait until the last one arrives, then all proceed. The two are easy to confuse because barriers can be implemented using atomic operations and locks can be used inside synchronization libraries, but their purpose is different. Locks provide mutual exclusion. Barriers regroup threads that are allowed to run independently during the phase before the barrier. This is why barriers are common in bulk-synchronous programming models, including GPU-style kernel phases, but become expensive if inserted after every small step.

9.4 Transactional Memory

Locks are not the only way to protect a critical section. Another idea is *transactional memory*: execute the critical section optimistically, then commit only if its required isolation can be validated. The programmer's conceptual model is close to a database transaction:

1. begin a transaction,
2. read and write the shared data optimistically,
3. commit if no conflict occurred,
4. otherwise abort and retry.

In hardware transactional memory, the processor typically tracks the transaction's read and write sets using the cache hierarchy. The tracked footprint must remain within the implementation's tracking capacity; eviction and conflict behavior depend on the design. Read/read sharing is not a conflict, whereas an overlapping write can conflict with another transaction's read or write set. Cache-line tracking can also cause false conflicts between different variables in the same line. One lock-elision strategy is:

```
Try a bounded number of transactions:
  begin with an abort target
  read the fallback lock; abort if it is held
  execute the critical section
  commit and finish
On repeated abort:
  acquire the fallback lock
  execute the critical section
  release the fallback lock
```

The transactional path must subscribe to the same lock used by the fallback *before* accessing protected data. Reading that lock inside the transaction makes a real lock acquisition conflict with speculative execution. Without that connection, "fall back to a lock" does not ensure exclusion between the two paths. Some hardware transactions provide no forward-progress guarantee, so retrying forever is not a correct general fallback policy.

The attraction is clear. Nonconflicting transactions can execute together even when the lock-based form of those critical sections would serialize them. If conflicts are rare, many threads may complete without ever spinning on a shared lock.

The limitations are equally important. Transactions can fail because of true data conflicts, because the read or write set is too large for the available cache tracking, or because interrupts,

exceptions, or implementation-specific hazards force an abort. Security and speculation concerns also make this design space delicate: mechanisms that delay architectural visibility or suppress exceptional behavior must be engineered very carefully to avoid side channels. As a result, practical support is vendor-specific and often paired with conservative fallbacks.

An alternative is *software transactional memory*. Here the runtime records metadata in software, checks for conflicts explicitly, and retries when necessary. It adds software instrumentation and metadata costs while retaining the optimistic-concurrency programming model; the performance tradeoff depends on the workload and implementation.

9.5 Memory Consistency Beyond Coherence

Coherence and consistency are related, but they are not the same guarantee. *Cache coherence* constrains the order in which all processors observe writes to *one memory location*. *Memory consistency* defines the allowed outcomes and ordering of memory operations across threads, including operations to *different locations*. A machine can therefore be perfectly coherent and still violate a programmer's intended synchronization if it allows aggressive reordering across different addresses.

Data Races

For the C/C++ load/store examples here, a *data race* involves potentially concurrent accesses to the same memory location, at least one write, at least one non-atomic access, and no happens-before order between them. Such a race is undefined behavior, not merely a choice among hardware interleavings. Two relaxed atomic accesses do not by themselves create a language-level data race, but a protocol built from them can still be logically wrong. That broader timing-dependent failure is often called a *race condition*.

The subtle point is that the race may involve more than one variable. An incorrectly constructed barrier, for example, can race on both the arrival counter and the phase or completion flag. Even if both variables are accessed atomically and coherently, the combined protocol can fail if its cross-variable ordering is insufficient. Coherence therefore removes only one class of failure; it does not automatically make parallel code race-free.

Strict Consistency

The strongest imaginable model is *strict consistency*. In the idealized strict-consistency model used here, a read returns the most recent write in a single global real-time order. If an exercise supplies that complete order, it determines what each read sees. The model does not make thread scheduling deterministic: different execution timings can still produce different real-time orders. The cost is extreme. Strict consistency is effectively saying that the machine must behave as though there were only one instantly authoritative memory, no hidden reorderings, and no cache behavior that could let processors observe temporarily different views of the world. That makes it useful as a thought experiment and almost useless as a high-performance implementation target. In worked problems, a supplied actual-time order or global-clock order must therefore be respected exactly under strict consistency.

Sequential Consistency

The best-known practical teaching model is *sequential consistency*. It weakens strict consistency slightly by discarding the requirement that the global order match physical time exactly. Instead, it requires only two things:

1. each processor's memory operations appear in the global order in the same order as that processor's program,
2. all processors observe one common total interleaving of those per-processor streams.

This is equivalent to imagining that the program ran on a uniprocessor that interleaved whole-thread memory operations in some unpredictable but legal order. Thus, in an interleaving exercise, any total order is legal as long as each thread's local program order is preserved, even if the chosen interleaving does not match the originally supplied real-time order.

A classic example uses two shared variables initially set to 0:

```
Processor 1:          Processor 2:
  A = 1                 B = 1
  BB = B                AA = A
```

This is an abstract memory-model litmus test, with `AA` and `BB` as thread-local results. Use atomic `A` and `B` when implementing it in C/C++; ordinary unsynchronized accesses would instead invoke undefined behavior. Coherence alone only says that all processors will agree on the order of writes to `A`, and separately on the order of writes to `B`. It says nothing about the cross-address ordering between `A = 1` and `B = 1`. If the machine allows the later loads to become visible before the earlier stores, then outcomes such as `AA = 0` and `BB = 0` become possible even though each individual location remains coherent.

Under sequential consistency, however, the result `AA = 0` and `BB = 0` is impossible. To get `AA = 0`, Processor 2's load of `A` would have to appear before Processor 1's earlier store of `A`. To get `BB = 0`, Processor 1's load of `B` would have to appear before Processor 2's earlier store of `B`. Those two requirements create a cyclic ordering contradiction once both processors' program orders must be preserved. Sequential consistency is therefore much stronger than plain coherence.

It is still expensive to implement. Modern processors want to issue loads and stores out of order, buffer writes, speculate past unresolved misses, and overlap communication aggressively. Sequential consistency constrains the observable result, not the literal issue order of every instruction. A processor may overlap or speculate internally if validation, replay, or other mechanisms preserve a legal sequentially consistent outcome.

Weak and Relaxed Consistency

Real multiprocessors therefore use weaker or more selective models. There is no single weak-consistency design; there are many variants. The common idea is that not every memory access deserves the same ordering guarantee.

One useful mental model is to divide shared accesses into two broad categories:

- *ordinary data* accesses, which may be reordered subject to the language and machine model,
- *synchronization* accesses, whose specified ordering establishes the required communication between threads.

Contention does not automatically select stronger semantics; the program must use the right operations and memory orders even when a synchronization variable happens to be uncontended. Atomic operations such as `test_and_set()`, `fetch_and_add()`, and LL/SC-based updates can support synchronization, but atomicity alone does not imply acquire or release ordering. The instruction or API must also request the ordering needed by the protocol. For example, an atomic counter used only to allocate distinct work indices can need weaker ordering than an atomic flag that publishes data.

Memory Fences and Acquire/Release Ordering

One way to request stronger ordering is an explicit *memory fence*; some loads, stores, and atomic instructions carry the required acquire/release ordering directly. Common fence styles include:

- a *store-ordering fence*, which at least orders earlier stores before later stores in the model being discussed,
- a *load-ordering fence*, which at least orders earlier loads before later loads,
- and a *full memory fence*, which orders the specified earlier loads and stores before the specified later loads and stores.

Fence names are ISA-specific: do not infer load-to-store or store-to-load ordering from the word "fence" alone. Check which predecessor and successor access classes the actual instruction orders. Restricting that overlap can cost performance, especially when earlier memory operations are slow.

This is why locks are more than just atomic variables. An *acquire* operation must prevent later accesses inside the critical section from floating before the lock is obtained. A *release* operation must prevent earlier critical-section updates from remaining hidden after the lock is released. Put differently, a correct lock implementation behaves like an ordering device as well as a mutual-exclusion device.

The practical synchronization bug usually appears in a message-passing pattern such as:

```
Producer:                  Consumer:
  store_relaxed(x, 1)        while (load_relaxed(flag) == 0) { }
  store_relaxed(flag, 1)     z = load_relaxed(x)
```

Here both variables are atomic and initially 0, so this fragment illustrates insufficient ordering without introducing a C/C++ data race. If the consumer observes `flag == 1`, the programmer expects `z` to become 1 as well. Coherence by itself does not guarantee that expectation, because the machine might make the write to `flag` visible before the earlier write to `x` becomes visible to the consumer. The two variables are each coherent on their own, but the cross-variable ordering required by the synchronization protocol is missing.

One clear repair is to publish and consume the flag with release and acquire operations; in this version `x` may be ordinary non-atomic data:

```
Producer:                  Consumer:
  x = 1                      while (load_acquire(flag) == 0) { }
  store_release(flag, 1)     z = x
```

Assume `flag` is atomic, initially 0, the consumer's successful load reads the producer's release store, and no other thread modifies `x`. Then the release/acquire handoff makes the earlier write of `x`

available to the consumer's later read. Figure 9.1 traces that relationship. In C or C++, use the language's atomic operations or synchronization library so both the compiler and hardware honor it; ordinary shared variables plus invented fence function names are not a complete implementation.

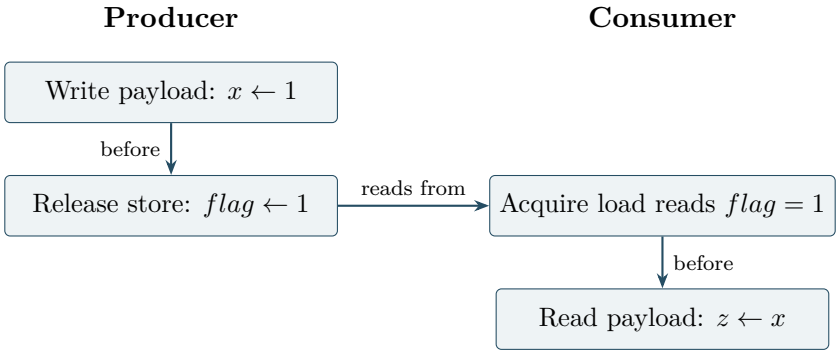


Figure 9.1: A release/acquire handoff orders the payload across two threads. The acquire must read the value published by the release. With the stated assumptions, the final read obtains $x = 1$. Coherence of each location alone does not establish this chain.

One brute-force solution would be to forbid essentially all load/store reordering and march memory operations through the system in a globally rigid order. That is simple to explain and expensive to implement. Real machines instead treat synchronization operations specially. An acquire prevents later memory actions from moving before the synchronization event, while a release prevents earlier actions from moving after it. Locks, unlocks, atomic synchronization primitives, and explicit memory fences rely on this idea. The result is a compromise: ordinary data accesses may still be reordered for performance, but synchronization points restore enough order for correctly written parallel programs to communicate.

Review Emphasis: The Consistency Failure Mode The recurring mistake is assuming that coherence makes the flag protocol correct. Coherence makes all processors agree on the order of writes to `x`, and separately on the order of writes to `flag`. It does not, by itself, force the write to `x` to become visible before the write to `flag`. That cross-address guarantee is exactly the job of consistency rules, fences, and acquire/release semantics.

This is the larger architectural lesson of the chapter pair. Chapter 8 showed how the machine preserves one shared-memory illusion for a single block at a time. This chapter adds the second half of the story: once software uses that shared memory for synchronization, the machine must also provide atomic updates and cross-address ordering guarantees. Without those additional guarantees, coherence alone is not enough.

9.6 What to Carry Forward

Coherence, atomicity, and ordering solve different problems. A lock needs an indivisible acquisition and the ordering that makes protected data visible to its next owner. A reusable barrier needs both generation tracking and a happens-before path from every arrival to every departure. Transactions need a correct abort and fallback strategy, not just optimistic execution.

When a shared-memory example fails, name the missing guarantee before adding a fence or a new state. Then check the language model as well as the processor model: ordinary racing C/C++

accesses cannot be justified by listing possible coherent hardware outcomes. The next chapter follows the messages themselves through the network that carries all this coordination.

9.7 Practice Questions

The following questions reinforce the main ideas of Chapter 9.

1. Explain why "the states are not the protocol." Use a block currently held in `M` by one cache and requested by another cache to illustrate your answer.
2. Compare abort/retry, intervene, and the `O` state as ways of handling a request to a block currently held dirty in another cache. Which choice can delay write-back the longest, and why?
3. Why does repeated `test_and_set()` cause more coherence traffic than test-and-test-and-set?
4. Describe the roles of *load-link* and *store-conditional*. What coherence event causes the store-conditional to fail?
5. What is the difference between *fetch-and-op* and *fetch-and-add*? Why is fetch-and-add useful for dynamic loop scheduling?
6. A workload uses one global lock and shows a severe invalidation storm when many threads enter the critical section together. Why might a tree lock help, and what latency cost does it introduce?
7. In a fetch-and-add barrier, why is a phase or toggle bit needed in addition to the arrival counter?
8. Compare a software barrier based on an atomic counter with a hardware barrier built from one dedicated signal line per PE. What assumptions does the hardware version make about the interconnect?
9. When should a programmer use a lock, and when should a programmer use a barrier? Give one example where each is the natural primitive.
10. What problem is transactional memory trying to avoid compared with conventional lock-based critical sections? Give two reasons a transaction may still abort.
11. Distinguish clearly between cache coherence and memory consistency.
12. In the producer/consumer example with `x` and `flag`, explain how a consumer can legally observe `flag == 1` yet still read an old value of `x` if the machine provides coherence but not sufficient consistency ordering.
13. Consider two threads:

```
Thread A:  A1: tA = M[0x3008]
           A2: tA = tA + 4
           A3: M[0x3004] = tA

Thread B:  B1: tB = M[0x3004]
           B2: tB = tB + 9
           B3: M[0x3008] = tB
```

Assume the initial values are $M[0x3008] = 20$ and $M[0x3004] = 10$. If the actual-time order is $A1, B1, A2, B2, A3, B3$, give the final values under strict consistency. Then give two different legal interleavings under sequential consistency that do *not* follow that actual-time order, and state the final memory values in each case.

14. What does sequential consistency guarantee, and why is it usually more expensive than weaker models with acquire/release ordering?
15. Define a data race precisely. Why can a program still contain a data race even when every individual location is cache coherent?
16. Compare strict consistency with sequential consistency. Which one is tied directly to physical time, and which one only requires preservation of each processor's program order?
17. In the two-processor example with `A`, `B`, `AA`, and `BB`, explain why the outcome `AA = 0, BB = 0` is forbidden under sequential consistency.
18. What do store fences, load fences, and full memory fences each prevent? Why are they expensive?
19. Why is read-only shared data usually less problematic for weak consistency than data protected by locks or atomics?

Chapter 10

Parallel Interconnection Networks: Topology, Routing, and Flow Control

An ownership request, a cache block, and an acknowledgment all have to travel somewhere. On a small machine a shared bus may carry them; on a larger one, several messages can compete for links and buffers along different routes. A useful network must answer three questions: where can a message go, which path will it take, and what happens when the next step is busy? Topology, routing, and flow control answer those questions in turn.

10.1 Why Parallel Interconnects Are Their Own Topic

Shared-memory multiprocessors ultimately depend on an interconnection fabric that moves requests, data, invalidations, acknowledgments, and coherence messages among processors, caches, directories, and memories. That fabric is not a minor implementation detail. Its structure determines whether the machine can scale, how quickly requests reach their destinations, and how badly latency grows when many transfers contend for the same resources.

Parallel interconnects and wide-area networks share many ideas, including packets, routing, congestion control, and error handling. The emphasis here is tightly coupled processor and memory traffic, where a few router cycles or blocked buffers can directly stall computation. A *flit* (flow-control digit) is the unit for which the network manages buffering and forwarding permission. In the course's simple on-chip model, one link transfers one flit per cycle. That is a modeling choice, not a definition: a physical link can serialize a flit over several transfers, and off-chip interconnects can use serial lanes too. Table 10.1 summarizes the contrast.

10.2 Direct and Indirect Topologies

A *topology* is the graph describing which nodes are connected directly to which other nodes. Two broad categories are common.

Design aspect	Parallel interconnection network	Conventional communication network
Transfer granularity	Flits and physical transfer width are modeled explicitly	Packet and frame formats are prominent; physical links may also use serial lanes
Reliability focus	Low latency matters, but link errors and retries still need handling	Heterogeneous paths and failures make end-to-end recovery important
Central design questions	Topology, routing, flow control, buffering, router design	Addressing, congestion management, reliability, heterogeneous paths

Table 10.1: Parallel interconnection networks optimize for tightly coupled communication inside one machine rather than for general-purpose wide-area connectivity.

Step	3-bit Gray code
0	000
1	001
2	011
3	010
4	110
5	111
6	101
7	100

Table 10.2: One Gray-code ordering for the eight nodes of a 3-dimensional cube. Consecutive entries differ in exactly one bit.

In a *direct topology*, each processor or node has explicit links to selected neighboring nodes. The processors themselves are vertices in the communication graph. Examples include meshes, tori, and the historically important *hypercube*.

In an *indirect topology*, processors connect to a separate network of switching elements. The communication graph is therefore built from routers or switches rather than from processor-to-processor links alone. A *crossbar* is the cleanest example: any input may be connected to any output, so the switch is functionally all-to-all. That flexibility is attractive, but full crossbars become expensive very quickly as the processor count grows.

Indirect organizations were sometimes called *dance hall* machines in older architecture discussions because the processors all had to meet in the central switching fabric rather than walking directly to one another. The metaphor is informal, but the architectural point is real: the switch fabric becomes the place where path selection, contention, and buffering are resolved.

10.3 Direct Networks: Hypercubes, Meshes, and Tori

10.3.1 Hypercubes and Gray-Code Intuition

A *d-dimensional hypercube* connects nodes so that each node has one link per dimension and neighboring nodes differ in exactly one coordinate. The standard counting facts are

$$N_{nodes} = 2^d, \qquad degree = d. \tag{10.1}$$

Thus a 3-dimensional hypercube has 8 nodes, each with degree 3. A 10-dimensional hypercube would have 1024 nodes, each with degree 10.

Binary coordinates define the cube's adjacency: labels at Hamming distance one are neighbors. *Gray code* gives one traversal in which consecutive labels differ in one bit. It is a path through the cube, not a replacement for the adjacency rule; the cube also has edges not used by that traversal. Table 10.2 shows one 3-bit Gray-code sequence.

Hypercubes were historically attractive because their diameter grows only logarithmically with node count. The physical difficulty is packaging. As the conceptual dimension count grows, the machine still has to be laid out in ordinary physical space, and routing many long links cleanly becomes increasingly awkward. That is one reason hypercubes became less popular in commercial systems even though the graph-theoretic properties are elegant.

10.3.2 K-ary N-Dimensional Networks

A broader family of direct networks is the *k-ary n-dimensional* topology. It has n dimensions and k nodes in each dimension, so the total node count is

$$N_{nodes} = k^n. \tag{10.2}$$

A 4-ary 2-dimensional network therefore has $4^2 = 16$ nodes.

The two most common members of this family are the *mesh* and the *torus*. A mesh connects each node to its neighbors along each dimension but does not wrap at the edges. A torus adds wraparound links so that each dimension becomes a ring. Thus a *k-ary n-torus* is the direct generalization of a 1-dimensional ring to multiple dimensions. For $k \geq 3$, an ordinary k-ary n-torus has degree $2n$, with two distinct neighbors per dimension. At $k = 2$, those two directions reach the same neighbor; distinguish simple-graph neighbor count from a design with parallel physical links.

A 4-ary 2-torus can be visualized as a 4-by-4 grid in which the left and right edges wrap together and the top and bottom edges wrap together. That wraparound is what gives the torus its name: the graph can be drawn on the surface of a donut without edge crossings. Modern many-core and server-class designs have often favored mesh- or torus-like organizations because they map naturally onto on-chip planar layouts.

10.3.3 Rings as a Pedagogical Baseline

Before moving to richer meshes and tori, many teaching simulators start with a simple deterministic ring. A ring is just the 1-dimensional torus case, so it is still a direct network. Its value is clarity: message latency is easy to interpret as hop count times link latency, and congestion is easy to see because all traffic shares one simple path structure.

For example, a single-flit message traversing five 1-cycle links takes five cycles if routing, injection, and ejection add no delay. A longer packet also pays serialization delay, and a real router can add processing and queueing time. If each node can consume at most one message per cycle, then additional delay appears as queueing rather than as route ambiguity. Many teaching models intentionally use this simplified structure, sometimes together with very deep input queues, so that the main architectural questions are message ordering, hop count, and contention rather than finite-buffer loss.

10.3.4 Physical Layout and Wire-Length Balance

Logical dimensionality is only part of the story. The network must still be fabricated in a planar physical layout. Longer wires have higher capacitance, and higher capacitance increases delay. An otherwise elegant logical network can therefore perform badly if its physical layout creates a few unusually long wires that become timing bottlenecks.

A simple ring illustrates the problem. If an 8-node 1-dimensional torus is drawn naively as a straight row with one long wraparound connection from the last node back to the first, that one link becomes much longer than the others. The result is an *unbalanced* layout: one path suffers distinctly higher delay than the rest.

A folded placement avoids that long return span without changing the ring's logical neighbors. For eight equally spaced physical slots, place successive logical nodes in slots $0, 2, 4, 6, 7, 5, 3, 1$. Every

connection then spans at most two slot spacings rather than seven. Figure 10.1 makes the layout change explicit.[1]

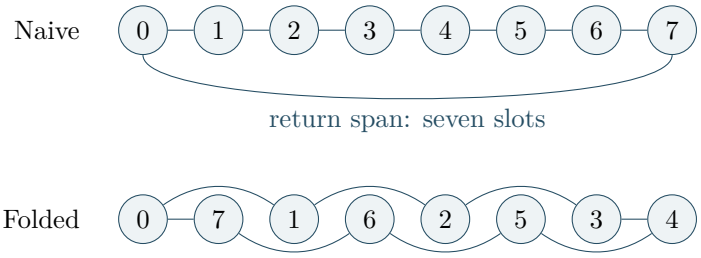


Figure 10.1: The same ring, placed differently. Labels are logical node IDs; horizontal positions are equally spaced physical slots. Both connect 0 through 7 and back to 0, but the folded placement's maximum horizontal link span is two slots.

This example uses horizontal span as a simple wire-cost proxy; bends, metal layers, repeaters, and physical routing still matter. Equal wire lengths are not mathematically impossible for a planar ring: a regular polygon already provides a counterexample. The engineering question is which placement fits the available floorplan and timing budget.

10.3.5 Bisectional Bandwidth

A standard figure of merit for direct networks is *bisectional bandwidth*. Conceptually, split the network into two equal halves and sum the bandwidth of the links crossing that cut. The *bisection bandwidth* is the minimum such value over all equal-halves cuts. If every link has the same bandwidth W_{link}, then the value reduces to

$$B_{bisect} = N_{cut} \times W_{link}, \tag{10.3}$$

where N_{cut} is the number of links crossing the minimum bisection. Here each undirected edge represents a full-duplex connection with bandwidth W_{link} *per direction*, and the quoted cut capacity is one-way. Counting simultaneous traffic in both directions would double that number.

Bisection bandwidth bounds the aggregate traffic that must cross a bottleneck balanced cut. It does not predict the exact runtime of a real workload, because real traffic creates conflicts and hot spots, but it does reveal a hard structural limit on how much data can move between large halves of the machine. The minimum is over *all* equal-halves partitions, not just visually straight cuts. For the regular square mesh and torus considered here, a dimension-aligned cut attains the minimum and makes the count easy to see. A diagonal or zig-zag partition is still a candidate if it divides the nodes equally; it does not change the definition, and counting extra crossings cannot increase the minimum.

A 4-ary 2-mesh with 1-unit links has a bisection bandwidth of 4 link-units because the natural equal cut crosses four links. The corresponding 4-ary 2-torus has a bisection bandwidth of 8 link-units because the wraparound links create a second independent set of crossings. In this 4×4 comparison, the torus therefore has twice the mesh's one-way bisection bandwidth. That does not make a torus automatically superior in every design, but it does explain why wraparound links are architecturally attractive.

[1]The April 21, 2026 class notes, *Day 27*, p. 1, contrast an unbalanced ring with a folded placement.



Figure 10.2: Equal halves in a 4×4 mesh and torus. Brown links cross the dashed partition. The torus's horizontal wraparound links add four crossings; vertical wraparound links are omitted because they stay within each half. Count each bidirectional link once using the same bandwidth convention in both panels.

Stage	Destination bit consumed	Example switch choice
0	1	choose the output corresponding to bit 1
1	0	choose the output corresponding to bit 0
2	1	choose the output corresponding to bit 1

Table 10.3: One bit of the destination is consumed at each stage in an omega network. Exact straight/cross labeling depends on the wiring convention, but the routing decision is driven directly by the destination bits.

10.4 Indirect Networks and Multi-Stage Fabrics

A full crossbar is not the only indirect topology. A cheaper alternative uses several stages of small switches. The standard building block is a 2-by-2 switch element. Each such switch can either pass its two inputs straight through or cross them. By composing many of these elements across multiple stages, the network can connect many inputs to many outputs without paying full crossbar cost.

One of the classic examples is the *omega network*. Between successive switch stages, the wires are connected in a *perfect shuffle* pattern. For an $N = 2^s$-input omega network built from 2-by-2 switches, the number of stages is

$$stages = \log_2 N. \tag{10.4}$$

An 8-input omega network therefore has 3 stages, which fits naturally with a 3-bit destination address.

The attractive property is routing simplicity. At each stage, one bit of the destination address determines whether the local switch should send the flit along one output or the other. Table 10.3 shows a simple route example for destination `101` in an 8-endpoint omega network.

The weakness of this omega network is internal *blocking*. Two transfers may need the same internal link or the same switch output at the same time. Unlike a nonblocking crossbar, the omega network does not generally offer an independent path for every pair of simultaneous requests. Thus the network may be fast when contention is low and much slower when several requests collide in the same internal region.

10.5 Routing Algorithms

Routing answers the question: given a source and a destination, which path should the message follow? The answer is not purely geometric. The chosen algorithm affects congestion balance, hardware complexity, deadlock risk, and average latency.

10.5.1 Deterministic Routing

A *deterministic* algorithm always chooses the same path for a given source-destination pair. The standard example is *dimension-order routing* (also called dimensional routing). In a 2-dimensional mesh or torus, the message may always traverse the x direction first and then the y direction. The method is simple, but deadlock freedom still depends on the channel dependencies. XY routing in an ordinary mesh can avoid cyclic channel dependencies; wrapping a dimension into a torus introduces cycles and requires additional precautions.

The weakness is load concentration. If many source-destination pairs are forced to use similar middle links, those links become hot spots even when alternate paths exist elsewhere in the network. Thus deterministic routing is attractive for simplicity but often poor at balancing congestion.

10.5.2 Adaptive Routing

An *adaptive* router uses local or global information about congestion to choose among multiple legal next hops. If one output queue is full or heavily loaded, the router may choose a different direction and still attempt to reach the same destination. This can reduce hot spots substantially.

The difficulty is correctness. If routers are allowed to keep redirecting traffic in response to temporary congestion, they can create cycles in which packets keep moving but never actually reach the destination. That is *livelock*. Adaptive routing therefore needs carefully designed restrictions or escape paths to guarantee forward progress.

10.5.3 Oblivious Routing and Valiant's Idea

Oblivious routing chooses paths without consulting current congestion. It includes deterministic traffic-independent routing as well as randomized choices; these labels are not three mutually exclusive categories. The classic randomized example is *Valiant's algorithm*. To route from source X to destination Y:

1. choose a random intermediate node Z, commonly uniformly over the nodes,
2. route from X to Z using a simple deterministic method such as dimension-order routing,
3. then route from Z to Y the same way.

If the selection permits $Z = X$ or $Z = Y$, one leg simply has zero length.

The point is load balancing. Deterministic routing tends to punish the same central links repeatedly, while Valiant-style oblivious routing spreads traffic more evenly across the network. The cost is that many messages travel farther than the geometric shortest path. That is often a good trade: equalizing congestion can improve overall throughput even when some individual routes become longer. Shortcutting a route changes the policy and can change its load balance and deadlock argument; it is not automatically a free optimization.

10.6 Flow Control and Congestion Management

Routing chooses a path. *Flow control* decides whether the next step along that path is currently safe. If the desired buffer or output link is full, the network must decide what to do with the blocked traffic.

10.6.1 Circuit Switching and Packet Switching

The oldest answer is *circuit switching*. A source first sends a setup request that reserves the entire route. Once the reservation succeeds, the source transmits along that dedicated path, and finally sends a teardown message to release the resources. Circuit switching can be very predictable and is attractive for real-time traffic, but it ties up resources even when the path is only intermittently active. Under load, this can create fairness problems and long waits.

The more common answer is *packet switching*. A message is divided into packets, each of which carries its own routing information. A packet typically contains a header, a payload, and a tail marker or trailer. Different packets may take different routes to the same destination, and no path must be reserved for the entire duration of a conversation. That flexibility is why packet switching dominates general-purpose interconnects.

10.6.2 Packets and Flits

A *packet* is the logical message unit that contains routing information. A *flit* is the flow-control unit used to move the packet through the network. The packet may therefore be broken into a header flit, one or more body flits, and a tail flit. The header carries the routing information; the body flits carry most of the data; the tail marks the end of the packet.

This distinction matters because flits do not behave like fully independent packets. In the wormhole model used here, a packet's flits retain their order within a virtual channel. The full route need not be repeated in every body flit, although local control information such as a virtual-channel ID is still needed. That saves bandwidth, but it also means the network cannot freely reorder flits belonging to the same packet.

10.6.3 What Happens at Congestion?

Suppose a router wants to forward a packet but the desired downstream resource is full. Several broad responses are possible.

One option is a *negative acknowledgment* (NACK): tell the source the packet did not arrive so the source can retry later. That is workable in some best-effort systems but poor for real-time streams, where repeated drops create visible or audible artifacts.

A second option is *misrouting*: take some alternate path instead of waiting. This can preserve motion, but unless the routing rules are designed carefully it can also create livelock.

A third option is *wait with back pressure*. The router buffers the packet locally, retries later, and uses *back pressure* to prevent upstream senders from overflowing the waiting space. Back pressure can be implemented with a simple on-off scheme (stop when full, resume when space opens) or with a *credit-based* scheme in which the sender tracks how many downstream buffer slots are currently available. An on-off stop signal needs enough spare space for flits already in flight while the signal travels upstream. With credits, the sender spends a credit when it sends a flit and regains it only

when a returned credit reports a freed downstream slot. For example, four initial credits and three sends leave one available credit until returns arrive, even if the link is momentarily idle.

10.7 From Store-and-Forward to Virtual Channels

10.7.1 Store-and-Forward

In pure *store-and-forward* switching, a router waits until it has received the entire packet before sending that packet to the next router. The method is simple but expensive in latency because each hop adds a full-packet reception delay. Large packets therefore move slowly across long paths.

10.7.2 Virtual Cut-Through

The packet-buffered *virtual cut-through* model reduces that latency. As soon as the header arrives and the next hop is known to be available, the router begins forwarding the packet before the entire packet has arrived locally. This pipelines transmission across links and can reduce end-to-end delay substantially.

Its tradeoff is whole-packet buffer reservation, which distinguishes this model from wormhole's flit-level buffering. If the packet is allowed to begin moving through the router, the downstream side must be able to absorb the entire packet without losing ordering. That often means buffering or reserving enough space for the whole packet, which can be wasteful.

10.7.3 Wormhole Flow Control

Wormhole flow control pushes the idea further by moving the packet as a chain of flits. Only a small amount of buffering is needed at each hop, often effectively one flit per virtualized buffer. This makes the network much more storage-efficient than packet-sized buffering.

The weakness is *head-of-line blocking*. If the header flit is blocked waiting for a resource, the rest of the packet is stretched across several links and routers behind it. Those occupied resources may then prevent unrelated traffic from advancing even though the conflict originated only at the blocked head. Wormhole is therefore fast and storage-efficient but vulnerable to interference between unrelated flows.

Head arrival is not packet arrival Assume a path of H links, each moving one flit per cycle with one-cycle traversal, no extra router delay, and no contention. Measure from the first flit's transmission start to the last flit's arrival. For a packet of F flits, store-and-forward (SF) and cut-through (CT) give

$$T_{\mathrm{SF}} = HF, \qquad T_{\mathrm{CT}} = H + F - 1. \tag{10.5}$$

Unblocked wormhole forwarding has the same pipelined timing in this model. With four links and six flits, the whole packet takes 24 cycles with store-and-forward and 9 with cut-through, even though the cut-through header arrives after only 4 cycles. Different link rates, pipeline stages, or stalls require a different timing calculation.

10.7.4 Virtual Channels

The standard modern response is the *virtual channel*. A physical link is divided logically into several independent channels, each with its own buffering and state. Each packet is assigned a

virtual-channel identifier, and the identifier travels with the flits so the receiver knows which logical stream each flit belongs to.

Conceptually, each virtual channel has states such as *idle*, *waiting for resources*, and *active*. If one packet is blocked because its next-hop destination is unavailable, another packet using a different virtual channel on the same physical link can still move. This breaks much of the head-of-line blocking that would occur in a single-channel wormhole design.

Virtual channels let flits from different packets interleave on a physical link and can reduce head-of-line blocking when the packets need different downstream resources. They do not multiply that link's bandwidth: a one-flit-per-cycle link still sends only one flit per cycle in total. Nor does adding queues by itself guarantee progress or prevent deadlock. The price is complexity: the router must manage multiple buffers, virtual-channel allocation, and additional arbitration logic.

10.7.5 Deadlock, Livelock, and Starvation

These are different failure modes. In *deadlock*, packets hold resources while waiting in a cycle, so none of them can advance. For example, four packets can each hold one channel and wait for the next channel around a ring. In *livelock*, a packet keeps moving or being redirected but never reaches its destination. In *starvation*, other traffic keeps winning arbitration while one packet is postponed indefinitely.

A sufficient deadlock-avoidance strategy in the usual channel-dependency model is to impose an acyclic resource order. Virtual-channel classes or an escape routing subnetwork can help implement such an order, but their allocation rules matter. Fair arbitration addresses starvation, while bounded detours or another progress argument address livelock. Request/response protocols can introduce additional dependencies at endpoints, so deadlock-free routing alone does not prove a deadlock-free coherent memory system.

10.8 Switch Architecture

A modern switch or router is not just a crossbar. It is a small pipeline of control and data-movement stages. A typical design includes:

- input buffers, often organized per virtual channel,
- routing logic that determines the requested output direction,
- a virtual-channel allocator,
- a switch allocator or arbiter that decides which requests win a contested output,
- and the crossbar traversal itself.

This structure is analogous in spirit to a processor pipeline: several flits can be in different internal stages of the router at the same time. That pipelining is essential for throughput. At the same time, the arbitration policy matters. The lecture handout asks the reader to compare ring injection with traffic on a road.[2] A useful interpretation is a roundabout: transit flits are already circulating, while locally injected flits wait to merge. Prioritizing transit traffic can keep the ring moving, but unlimited priority could starve a core trying to inject. The analogy therefore motivates both congestion control and a fairness rule, not one universal arbitration policy.

[2] `n8_interconn.pdf`, slide 28, "Packet Insertion: Ring Stop."

The large lesson is that topology alone does not determine performance. Real performance emerges from the interaction of topology, routing, buffering, virtual-channel structure, and router arbitration. A network with an impressive theoretical bisection bandwidth can still perform poorly if its switches create avoidable blocking or if its routing policy concentrates load onto a few hot links.

10.9 What to Carry Forward

A topology gives possible paths, not a complete performance or correctness story. Count hops, distinguish the first flit from the whole packet, and state the bandwidth direction before using a bisection number. Then ask what happens when a buffer fills: who waits, which resources remain held, and whether the routing and arbitration rules guarantee progress.

The same reasoning connects the whole book. A core overlaps work, a cache shortens data travel, coherence coordinates copies, and the network carries the resulting communication. Each optimization moves a limit rather than making all waiting disappear. Appendix D carries those tradeoffs into devices whose physical rules differ from conventional CMOS.

10.10 Practice Questions

The following questions reinforce the main ideas of Chapter 10.

1. What distinguishes a parallel interconnection network from a general communication network? Why does flit width matter in the former?
2. Compare direct and indirect topologies. In which one are the processors themselves the vertices of the communication graph?
3. A hypercube has 64 nodes. What is its dimensionality, and what is the degree of each node?
4. What does the phrase *k-ary n-dimensional* mean? How many nodes are in a 4-ary 3-dimensional direct network?
5. What is the difference between a mesh and a torus? Why does the torus usually have higher bisection bandwidth?
6. An 8-node unidirectional ring has 1-cycle links and no contention. Treat each message as one flit and ignore router, injection, and ejection delays. How many cycles does it take a message to travel from node 0 to node 5? If node 5 can consume at most one message per cycle and already has one older message queued on arrival, what extra service delay is introduced?
7. Why can a logically simple ring become physically problematic when laid out on a planar chip or board?
8. Define bisection bandwidth. A 4-ary 2-mesh and a 4-ary 2-torus use the same per-link bandwidth. Why does the torus have twice the bisection bandwidth?
9. For a regular square mesh or torus, why is a dimension-aligned equal cut a convenient way to find bisection bandwidth? Does the definition exclude other equal-halves partitions?
10. Compare a full crossbar with a multi-stage indirect network. Why might a designer choose the latter even though it can block?
11. What is a perfect shuffle, and why is it associated with omega networks?

12. An omega network has 16 endpoints. How many stages are required, and how many destination bits are consumed along one route?
13. Compare deterministic, adaptive, and oblivious routing. What problem is Valiant-style routing trying to solve?
14. What is the difference between a packet and a flit? For a six-flit packet crossing four one-flit-per-cycle, one-cycle links with no other delays, compute whole-packet latency with store-and-forward and with cut-through.
15. What are three possible responses when the desired downstream buffer or output is full?
16. Compare store-and-forward, cut-through, wormhole, and virtual-channel flow control. Which one most directly addresses head-of-line blocking?
17. Why does virtual-channel flow control make the router more complicated? Does adding virtual channels alone prove deadlock freedom or increase physical link bandwidth?

Appendix A

Pipeline Timing-Diagram Interpretation Checklist

A five-stage timing diagram is best read as a checklist rather than as raw symbols. Move cycle by cycle (column by column), then apply the prompts below.

A.1 How to Use This Checklist

Quick Reading Order

1. Use the legend to distinguish repeated stall labels from normal multi-cycle execution.
2. Mark killed instructions separately from idle cells; a dash can mean either in different diagram conventions.
3. Identify the event that caused each stall/squash (dependence, branch, resource conflict, or memory latency).
4. Compute measured CPI from the traced window.

1. Structural: Resource Organization

1.1 IF and MEM overlap

- **Ask:** Can instruction fetch occur in the same cycle as a load/store data access?
- **Cue:** One cycle shows IF for one instruction while another instruction is in MEM with a real data access.
- **Meaning:** Sustained overlap indicates concurrent instruction/data paths. Repeated IF stalls when load/store reaches MEM suggests a shared bottleneck.

1.2 Single-issue vs superscalar

- **Ask:** Is the machine *single-issue* or *superscalar* (*multi-issue*)?
- **Cue:** In simple single-issue diagrams, each stage has at most one instruction per cycle.
- **Meaning:** Two instructions starting in the same defined issue stage in one cycle demonstrate multi-issue. Two instructions merely occupying EX at once do not: a single-issue machine can have several operations in flight across pipelined or separate functional units.

1.3 Multi-cycle execute units

- **Ask:** Do some operations occupy EX for multiple cycles?
- **Cue:** One instruction appears in EX across consecutive cycles (EX, EX, EX, . . .).

- **Meaning:** First distinguish normal multi-cycle execution from an EX stall. A multi-cycle unit can accept a new operation every cycle if pipelined; latency alone does not determine how often younger operations may start.

2. Data: Dependence and Forwarding

2.1 RAW hazard location and cost

- **Ask:** Where are *RAW* hazards, and how many bubbles do they add?
- **Cue:** Consumer instruction repeats a stage (usually ID, sometimes IF) while waiting for producer timing.
- **Meaning:** Extra residence beyond the normal stage duration reveals waiting. Check whether the cause is the dependence itself or backpressure from another blocked instruction.

2.2 Forwarding strength

- **Ask:** Is forwarding present, and from where?
- **Cue:** Back-to-back ALU dependencies proceed without stalling even though producer WB occurs later.
- **Meaning:** Some result path supplies the consumer before ordinary register-file write-back. Use the producer/consumer separation to identify whether EX/MEM→EX, MEM/WB→EX, or another path is needed; one pair does not establish every bypass route.

2.3 Load-use behavior

- **Ask:** What is the load-use penalty?
- **Cue:** Load immediately followed by consumer, where consumer stalls one cycle.
- **Meaning:** Classic 5-stage with forwarding still needs a one-cycle load-use stall because load data appears at end of MEM.

2.4 WB/ID same-cycle timing

- **Ask:** Can ID read a register in the same cycle an older instruction writes it during the WB stage?
- **Cue:** Consumer ID aligns with producer WB, with no extra bubble.
- **Meaning:** Implies write-before-read register timing and/or WB→ID bypass behavior.

3. Control: Branch and Redirection Policy

3.1 Resolution stage and penalty

- **Ask:** Where is branch outcome resolved, and what is the visible penalty?
- **Cue:** Find where target/direction becomes known; then count wrong-path squashes or inserted bubbles before correct-path flow stabilizes.
- **Meaning:** Later resolution generally increases misfetch cost.

3.2 Delay slot vs squash slot

- **Ask:** Does the post-branch slot always execute, or can it be killed?
- **Cue:** Slot instruction either always completes (*delay slot*) or is conditionally killed (*squash slot*/branch-likely style).
- **Meaning:** A slot guaranteed to execute on both outcomes has ordinary delay-slot semantics. A single trace in which it executes cannot by itself exclude conditional-annul semantics; check both outcomes or the ISA definition.

3.3 Taken/not-taken inference (branch-likely style)

- **Ask:** Which branch outcome occurred?
- **Cue:** Whether the slot instruction executes or is squashed.
- **Meaning:** If the specified instruction annuls its slot on a not-taken outcome, a squashed slot indicates not taken. Apply the instruction's actual annul rule rather than inferring one from the name "likely."

3.4 Prediction vs no prediction

- **Ask:** Does fetch speculate, and is speculation often correct?
- **Cue:** Continued fetching past unresolved branches followed by occasional squashes (speculation), versus front-end waiting for resolution (no speculation).
- **Meaning:** Wrong-path fetches later killed imply predictive/fixed speculation; repeated front-end pauses imply non-speculative handling.

4. Memory: Miss Effects in the Timeline

4.1 Miss penalty visibility

- **Ask:** Are long memory latencies visible?
- **Cue:** Load/store occupies MEM for multiple cycles (M, M, M, ...), with younger instructions backing up.
- **Meaning:** Indicates cache-miss (or similar long-latency memory) behavior.

4.2 Blocking vs non-blocking cache behavior

- **Ask:** Is cache behavior *blocking*, or can it do *hit-under-miss* (*non-blocking*)?
- **Cue:** Look for a later cache access serviced while an earlier miss is outstanding.
- **Meaning:** A later hit completing under that miss establishes hit-under-miss support. Independent ALU progress alone does not prove the cache is non-blocking. Conversely, a frozen core may be limited by in-order scheduling or a dependence even when its cache could accept another access.

5. Performance from the Diagram

5.1 Measured CPI of a trace window If a window spans C cycles and retires N_{ret} non-squashed instructions (Eq. (A.1)):

$$\mathrm{CPI}_{measured} = \frac{C}{N_{ret}}. \tag{A.1}$$

Squashed instructions consume cycles but do not count as retired instructions.

5.2 Stall-cycle attribution by cause Assign repeated-stage cells and squashes to causes:

- dependence waiting (RAW/load-use),
- control redirection (branch squashes/bubbles),
- structural conflicts (resource contention),
- memory-latency extension (repeated MEM).

This gives a practical CPI breakdown from one timing chart. Do not add every repeated cell in every instruction's row: one blocked instruction can make several younger instructions wait during the same cycle. Count lost progress once under the stated issue model, separate normal fill/drain from stalls, and state how overlapping causes are attributed.

6. Common Gotchas

- ALU instructions may still appear in MEM in pedagogical 5-stage diagrams; this alone does not prove a data-memory access.
- IF stalls can be secondary backpressure from a downstream stall (often ID), not a fetch-unit limitation by itself.
- Squash and stall are different: squashed instructions are killed; stalled instructions are preserved in place with repeated stage labels.

Appendix B

Formula Reference

This appendix gathers the core formulas used across Chapters 1–10. Keep the stated model with each equation: symbols are reused across topics, latency units must agree, and overlapping costs must not be added twice.

Chapter 1 Formula Set

Chapter 1-F1 (Vector Serial vs Pipelined Cycles)

$$\text{serial cycles} = nk, \qquad \text{pipeline cycles} \approx k + n - 1. \tag{B.1}$$

Here n is the number of elements and k is the per-element latency in cycles; the pipeline accepts one element per cycle without stalls.

Chapter 2 Formula Set

Chapter 2-F1 (CPU Time Decomposition)

$$\text{CPU time} = \text{IC} \times \text{CPI} \times \text{CT}. \tag{B.2}$$

Chapter 2-F2 (Frequency Form)

$$\text{CPU time} = \frac{\text{IC} \times \text{CPI}}{f}. \tag{B.3}$$

Here f is clock frequency; it is not the accelerated time fraction used in Amdahl's law below.

Chapter 2-F3 (Instruction-Mix Total Cycles)

$$\text{Total cycles} = \sum_i \left(\text{IC}_i \times \text{CPI}_i\right). \tag{B.4}$$

Chapter 2-F4 (Average CPI)

$$\text{CPI}_{avg} = \frac{\text{Total cycles}}{\text{Total instructions}}. \tag{B.5}$$

Chapter 2-F5 (Speedup Definition)

$$S = \frac{T_{old}}{T_{new}}. \tag{B.6}$$

Chapter 2-F6 (Amdahl's Law)

$$S_{Amdahl} = \frac{1}{(1-f) + \frac{f}{s}}. \tag{B.7}$$

f is the fraction of *baseline execution time* accelerated by factor s; the fixed-work model ignores new overhead and overlap changes.

Chapter 2-F7 (Amdahl Upper Bound)

$$S_{\text{upper}} = \frac{1}{1-f}. \tag{B.8}$$

For $f < 1$ as $s \to \infty$.

Chapter 2-F8 (Gustafson's Law)

$$S_{Gustafson} = \alpha + (1-\alpha)N. \tag{B.9}$$

α is the serial fraction measured in the scaled N-processor run, not the baseline fraction from Amdahl's law.

Chapter 2-F9 (Equal-Weight Mean of Times)

$$\bar{T}_{arith} = \frac{1}{n}\sum_{i=1}^{n} T_i. \tag{B.10}$$

Chapter 2-F10 (Combined Rate for Equal-Work Segments)

$$\bar{R}_{harm} = \frac{n}{\sum_{i=1}^{n} \frac{1}{R_i}}. \tag{B.11}$$

For positive rates on consecutive equal-work segments. Equal-time segments instead use an arithmetic rate mean; in general, divide total work by total time.

Chapter 3 Formula Set

Chapter 3-F1 ((C, B, S) Definitions)

$$\text{capacity} = 2^C \text{ bytes}, \qquad \text{block size} = 2^B \text{ bytes}, \qquad \text{associativity} = 2^S \text{ ways}. \tag{B.12}$$

Capacity here means data bytes, excluding tags and control metadata. The simple byte-addressed geometry requires $C \geq B + S$.

Chapter 3-F2 (Total Blocks and Sets)

$$\text{blocks} = 2^{C-B}, \qquad \text{sets} = \frac{2^C}{2^B \cdot 2^S} = 2^{C-B-S}. \tag{B.13}$$

$$\text{lines} = \text{sets} \cdot \text{ways} = 2^{C-B-S} \cdot 2^S = 2^{C-B}. \tag{B.14}$$

Direct-mapped corollary: if $S = 0$, then ways $= 1$ and lines $=$ sets. Set-associative corollary: if $S > 0$, then lines $>$ sets.

Chapter 3-F3 (Address-Field Widths, General AW)

$$|\text{Offset}| = B, \qquad |\text{Index}| = C - B - S, \qquad |\text{Tag}| = AW - (C - S). \tag{B.15}$$

$$\text{Address} = [\text{Tag} \mid \text{Index} \mid \text{Offset}]. \tag{B.16}$$

AW is the address width used by the modeled lookup, not automatically the ISA register width or the logarithm of installed RAM size.

Chapter 3-F4 (Address-Field Widths, 32-bit Address)

$$|\text{Offset}| = B, \qquad |\text{Index}| = C - B - S, \qquad |\text{Tag}| = 32 - (C - S) = 32 - C + S. \tag{B.17}$$

Chapter 3-F5 (Way Selection Relationship)

$$\text{ways} = 2^S, \qquad \log_2(\text{ways}) = S. \tag{B.18}$$

Chapter 3-F6 (Single-Level Average Access Time)

$$AAT = HT + MR \cdot MP. \tag{B.19}$$

MP is the additional service time after the current level's hit lookup in this additive model.

Chapter 3-F7 (Bit Overhead from Valid/Dirty)

$$\text{overhead fraction} = \frac{n_v + n_d}{2^{B+3}}, \qquad n_v = 1, \quad n_d \in \{0, 1\}. \tag{B.20}$$

Valid/dirty bits per block divided by data bits per block; this does not include tag or replacement-state overhead.

Chapter 3-F8 (Full-Block Miss Penalty Model)

$$MP_{full} = T_1 + (N_{sb} - 1)T_f, \qquad N_{sb} = \left\lceil 2^B / W \right\rceil. \tag{B.21}$$

W is bytes per transfer, T_1 is time to the first completed sub-block, and T_f is the interval between later sub-blocks.

Chapter 3-F9 (Early-Restart Approximation)

$$MP_{ER} \approx T_1 + \frac{N_{sb} - 1}{2} T_f. \tag{B.22}$$

Assumes an in-order fill and a uniform distribution over requested sub-block positions.

Chapter 3-F10 (Recursive Two-Level Form)

$$AAT_{L1} = HT_{L1} + MR_{L1} \cdot AAT_{L2}. \tag{B.23}$$

The L2 term is complete service time per L1 miss; L2 miss ratios are conditional on accesses reaching L2 in this serial-lookup model.

Chapter 3-F11 (VIPT Constraint)

$$C - S \leq P. \tag{B.24}$$

2^P bytes per page; this is the conventional VIPT index-within-page-offset constraint without extra synonym-handling machinery.

Chapter 4 Formula Set

For the pipeline timing formulas, N is the number of balanced stages, M is the instruction count, T_s is stage logic delay, and T_{reg} is register overhead; assume one new instruction per cycle and no stalls unless stated.

Chapter 4-F1 (Pipelined Clock Approximation)

$$T_{clk,pipe} \approx T_s + T_{reg}. \tag{B.25}$$

Chapter 4-F2 (Ideal Long-Run Pipeline Speedup)

$$S_{ideal} \approx \frac{NT_s}{T_s + T_{reg}}. \tag{B.26}$$

Chapter 4-F3 (Finite-Length Pipelined Time)

$$T_{pipe} \approx (N + M - 1)T_{clk,pipe}. \tag{B.27}$$

Chapter 4-F4 (Unpipelined Time for M Instructions)

$$T_{unpiped} \approx MNT_s. \tag{B.28}$$

Chapter 4-F5 (Practical CPI Model)

$$CPI = 1 + \text{stalls}_{avg}. \tag{B.29}$$

For the single-issue model with ideal CPI 1, counting non-overlapping added stall cycles per retired instruction.

Chapter 4-F6 (Branch-Stall CPI Contribution)

$$CPI \approx 1 + f_{branch} \cdot \text{stall}_{branch,avg}. \tag{B.30}$$

Chapter 4-F7 (History-Table Indexing)

$$index_{GShare} = PC_{selected} \oplus GHR, \tag{B.31}$$

$$index_{GSelect} = \text{concat}(PC_p, GHR_h), \qquad N_{counters} = 2^{p+h}. \tag{B.32}$$

For GShare, the selected PC and history fields have equal width. Exclude alignment bits when the predictor's indexing convention requires it.

Chapter 4-F8 (Perceptron Prediction and Training)

$$y = w_0 + \sum_{i=1}^{n} w_i x_i, \qquad x_i \in \{-1, +1\}. \tag{B.33}$$

Predict taken when $y \geq 0$. If wrong or $|y| \leq \theta$, update $w_0 \leftarrow w_0 + t$ and $w_i \leftarrow w_i + t x_i$, where t encodes the actual outcome. The exercises use $\theta = \lfloor 1.93n + 14 \rfloor$ and assume no weight overflow.

Chapter 5 and Simulation Formula Set

Chapter 5-F1 (Two-Level AAT with Explicit L1 Penalty)

$$AAT_{L1} = HT_{L1} + MR_{L1} \cdot MP_{L1}, \qquad MP_{L1} = AAT_{L2}. \tag{B.34}$$

Chapter 5-F2 (L2 AAT Model)

$$AAT_{L2} = HT_{L2} + MR_{L2,read} \cdot DRAM_TIME. \tag{B.35}$$

Chapter 5-F3 (DRAM Block Service Model)

$$DRAM_TIME = DRAM_AT + DRAM_AT_PER_WORD \cdot WORDS_PER_BLOCK. \tag{B.36}$$

Here $DRAM_AT$ excludes the per-word transfer time; Chapter 3's T_1 includes the first completed sub-block transfer.

Chapter 6 Formula Set (Dynamic ILP, Superscalar, and Out-of-Order Execution)

Chapter 6-F1 (IPC and CPI Relationship)

$$IPC = \frac{1}{CPI}. \tag{B.37}$$

Use the same instruction count and cycle interval for both quantities, normally retired instructions for architectural IPC.

Chapter 6-F2 (CPI/IPC Threshold Equivalence)

$$CPI < 1 \iff IPC > 1. \tag{B.38}$$

Chapter 6-F3 (Retire-Throughput Bound)

$$IPC_{retire} \leq \min(IPC_{fetch},\ IPC_{dispatch},\ IPC_{execute},\ IPC_{state\ update})\,. \tag{B.39}$$

A sustained-rate bound, not a claim that every short window has matching fetch and retirement counts.

Chapter 6-F4 (Simplified Scoreboard Fire Condition)

$$\text{Fire if } \neg busy(src_1) \wedge \neg busy(src_2) \wedge \neg busy(dest) \wedge \neg busy(FU_{needed}). \tag{B.40}$$

The simple FICO model captures sources before younger writers pass. A busy FU means unable to accept another operation, not merely occupied by an older pipelined operation.

Chapter 6-F5 (Result-Bus Completion Bound)

$$IPC_{complete} \leq N_{result\ buses}. \tag{B.41}$$

Applies to result producers that each require one broadcast, not necessarily to stores or branches using other completion paths.

Chapter 6-F6 (Physical Register File Port Estimate)

$$N_{data\ read\ ports} \approx 2W_{issue}, \qquad N_{write\ ports} \approx N_{result\ buses}. \tag{B.42}$$

For a simple full-bandwidth PRF with two source values per issued operation. Polling two ready bits for every scheduling entry instead costs up to $2N_{sched}$ one-bit checks per cycle.

Chapter 6-F7 (Precise Restartable-Fault Property)

$$\begin{aligned} &\text{restartable fault at } i \Rightarrow \text{all older instructions are committed} \\ &\qquad\qquad \wedge\ i \text{ and all younger instructions have no committed effects} \end{aligned} \tag{B.43}$$

Chapter 6-F8 (RAT Rename Mapping)

$$P_{src1} = RAT[A_{src1}], \qquad P_{src2} = RAT[A_{src2}], \qquad RAT[A_{dest}] \leftarrow P_{free}. \tag{B.44}$$

Chapter 6-F9 (Ready-Based Fire Condition)

$$fire_i \Leftarrow ready(P_{src1}) \wedge ready(P_{src2}) \wedge free(FU_i). \tag{B.45}$$

Chapter 6-F10 (ROB Retirement Condition)

$$retire \Leftarrow ready(ROB_{head}) \wedge \neg exception(ROB_{head}). \tag{B.46}$$

Chapter 6-F11 (ROB Exception Rule)

$$\begin{aligned} &exception(ROB_{head}) \Rightarrow \text{flush younger instructions} \\ &\qquad\qquad \wedge\ \text{transfer control to the retry or handler PC} \end{aligned} \tag{B.47}$$

Chapter 7 Formula Set (VLIW and Software Pipelining)

Chapter 7-F1 (Dependence Height)

$$height(i) = \begin{cases} lat(i), & succ(i) = \emptyset \\ lat(i) + \max\limits_{j \in succ(i)} height(j), & \text{otherwise} \end{cases} \tag{B.48}$$

For the acyclic graph and latency convention used in Chapter 7; compact cyclic loop graphs instead need recurrence analysis.

Chapter 7-F2 (Earliest Legal List-Scheduling Cycle)

$$cycle(i) = \min\{c \mid \forall j \in pred(i),\ c \geq cycle(j) + \delta_{ji},\ \text{resources are legal in } c\}\,. \tag{B.49}$$

δ_{ji} is the required edge separation; for a RAW edge in the simple fixed-latency model, it is $lat(j)$.

Chapter 7-F3 (Resource-Constrained Initiation Interval)

$$ResII = \max_t \left\lceil \frac{N_t}{U_t} \right\rceil \tag{B.50}$$

Assumes disjoint classes of units that each accept one operation per cycle; multi-cycle resource occupancy requires a richer reservation model.

Chapter 7-F4 (Recurrence-Constrained Initiation Interval)

$$RecII = \max_c \left\lceil \frac{Lat(c)}{Dist(c)} \right\rceil \tag{B.51}$$

For dependence cycles with positive total iteration distance, using the sum of required edge latencies.

Chapter 7-F5 (Minimum Initiation Interval Bound)

$$MII = \max(ResII, RecII), \qquad II \geq MII. \tag{B.52}$$

The lower bound need not be achievable by a legal schedule or by a bounded heuristic search.

Chapter 8 Formula Set (Multiprocessors and Coherence)

Chapter 8-F1 (Ideal Embarrassingly Parallel Scaling)

$$S(N) = N \qquad \text{for ideally balanced, independent work with no added overhead.} \tag{B.53}$$

This idealization excludes communication, synchronization, serial work, and capacity or bandwidth bottlenecks.

Chapter 8-F2 (Directory Entry Count)

$$N_{entries} = \frac{2^m}{2^v} = 2^{m-v}, \tag{B.54}$$

For a dense directory covering all 2^m bytes of memory with 2^v-byte blocks.

Chapter 8-F3 (Full Bit-Vector Directory Storage Estimate)

$$bits_{dir} \approx 2^{m-v}\,(N + n_{ctrl})\,, \tag{B.55}$$

where N is the number of PEs and n_{ctrl} represents the small set of non-sharer control bits such as dirty, valid, or overflow indicators.

Chapter 8-F4 (Limited K-Pointer Directory Storage Estimate)

$$bits_{dir} \approx 2^{m-v} \left(K \left\lceil \log_2 N \right\rceil + n_{ctrl}\right). \tag{B.56}$$

Chapter 9 Formula Set (Synchronization and Memory Consistency)

Chapter 9-F1 (Fetch-and-Op Semantics)

$$\texttt{fetch_and_op}(x, c): \qquad old = x, \qquad x \leftarrow x \ op \ c, \qquad \text{return } old. \tag{B.57}$$

Chapter 9-F2 (Dynamic Work Claim via Fetch-and-Add)

$$i = \texttt{fetch_and_add}(next, 1), \qquad \text{execute iteration } i \text{ iff } i < N. \tag{B.58}$$

Chapter 9-F3 (Last Arrival in a Counter-Based Barrier)

$$\text{last arriver if } \texttt{fetch_and_add}(count, 1) = N - 1. \tag{B.59}$$

This identifies the last arrival; a reusable barrier also requires phase tracking and memory-ordering semantics, as in Chapter 9.

Chapter 10 Formula Set (Parallel Interconnection Networks)

Chapter 10-F1 (K-ary N-Dimensional Node Count)

$$N_{nodes} = k^n. \tag{B.60}$$

Chapter 10-F2 (Hypercube Size and Degree)

$$N_{nodes} = 2^d, \qquad degree = d. \tag{B.61}$$

Chapter 10-F3 (Bisection Bandwidth for Uniform Links)

$$B_{bisect} = N_{cut} \times W_{link}. \tag{B.62}$$

Uses the minimum equal-halves cut. Chapter 10 reports one-way capacity with W_{link} specified per direction of each full-duplex edge.

Chapter 10-F4 (Omega Network Stage Count)

$$stages = \log_2 N. \tag{B.63}$$

For N a power of two and 2-by-2 switching elements.

Chapter 10-F5 (Whole-Packet Latency in the Unit-Link Model)

$$T_{\mathrm{SF}} = HF, \qquad T_{\mathrm{CT}} = H + F - 1. \tag{B.64}$$

H one-cycle links, one flit per link per cycle, F flits per packet, no extra router delay, and no contention; measure from first transmission start to final-flit arrival. SF denotes store-and-forward; CT denotes cut-through.

Appendix C

Glossary of Core Definitions

This appendix collects first-pass definitions used throughout Chapters 1–10 and the advanced-topics appendix.

Parallelism Doing multiple useful operations concurrently to increase throughput.

Speculation Executing work before all required conditions are confirmed, with recovery if assumptions are wrong.

Prediction Estimating uncertain future behavior to guide speculative execution.

Instruction-Level Parallelism (ILP) Overlap of independent instructions within one thread.

Very Long Instruction Word (VLIW) Instruction encoding style that packs several compiler-scheduled ops into one instruction word.

Polycyclic Scheduling Early name for overlapping multiple loop iterations in one compact repeating schedule.

Software Pipelining Loop-scheduling technique that overlaps operations from different iterations and emits a prologue, steady-state kernel, and epilogue.

Thread-Level Parallelism (TLP) Concurrent execution of multiple threads on one or more cores.

Processing Element (PE) One independently executing processor, core, or other execution agent in a multiprocessor.

Digital Signal Processor (DSP) Processor class optimized for regular numeric workloads such as filtering, radio, and image-processing kernels.

SIMD One instruction stream operating on multiple data elements in lockstep.

MIMD Multiple instruction streams operating on multiple data streams independently.

SPMD Single program, multiple data: workers run the same program on different data but may branch and advance independently.

Shared Memory Organization in which participants can access common memory locations; their virtual addresses need not be identical across separate processes.

Message Passing Programming model in which participants communicate explicitly through messages, whether or not the underlying hardware shares memory.

Partitioned Global Address Space (PGAS) Programming model with globally addressable data partitioned by affinity to particular participants.

Uniform Memory Access (UMA) Organization with approximately uniform underlying main-memory access cost across processors and locations, ignoring cache and queueing effects.

Non-Uniform Memory Access (NUMA) Organization in which physical memory placement affects access cost; coherent NUMA can still provide shared memory.

Accelerator Specialized processing engine, often attached to a host CPU through an external link and using distinct memory.

Bus Shared observable broadcast medium across which all attached agents can see the same transactions.

Point-to-Point Interconnect Link organization in which packets move between specific endpoints rather than across one shared observable medium.

Cache Coherence Property that keeps multiple cached copies of the same location mutually consistent enough to preserve shared-memory semantics.

Snooping Coherence Coherence method in which each cache monitors shared interconnect traffic and reacts when another processor requests a block it holds.

Directory-Based Coherence Coherence method in which a directory tracks which caches currently hold a block and coordinates invalidations or forwarding.

MSI Coherence protocol whose stable states are modified, shared, and invalid.

MESI Coherence protocol whose stable states are modified, exclusive, shared, and invalid.

MESIF Coherence family adding a designated clean forwarder to the MESI states.

MOESI Coherence protocol whose stable states are modified, owned, exclusive, shared, and invalid.

MOESIF Coherence protocol whose stable states are modified, owned, exclusive, shared, invalid, and forward.

GetS Coherence request for a shared readable copy of a block.

GetM Coherence request for exclusive writable ownership of a block.

WriteBack Propagation of a dirty cache block to lower memory.

Exclusive (E) Clean state in which exactly one cache holds the block and may later write it without first broadcasting an upgrade request.

Owned (O) Read-only ownership state carrying dirty-data responsibility; other current S copies may exist while backing memory remains stale.

Forward (F) Clean shared state designating one cache as the responder for later shared requests.

Hit Line Shared indication line used to tell a requester that some other cache already holds the block, thereby distinguishing `E` from `S` in a snooping protocol.

Intervene Snooper action in which a cache supplies data directly in response to another processor's request.

Abort Snooper response instructing a requester to wait or retry because ownership transfer is not yet complete.

Transient State Intermediate coherence state used while a block is waiting for a request or ownership transfer to complete.

Presence Bit Vector Directory field identifying caches or processors that may hold a block; conservative extra bits can cause unnecessary probes without missing a real holder.

Limited Directory Directory organization that records only a bounded number of sharers explicitly and uses an overflow policy when that bound is exceeded.

Recall Directory command requiring a cache to return the current block value and surrender ownership.

Invalidate Directory command requiring a cache to discard its current copy of a block.

Ack Directory-style acknowledgment indicating that a required coherence action completed.

Nack Directory-style negative acknowledgment indicating that a request cannot yet be completed and should be retried later.

Sharing Pattern Characteristic way in which a block is accessed across processors, such as private, read-only, producer-consumer, migratory, or read-write shared.

False Sharing Performance problem in which unrelated variables placed in the same cache block force unnecessary invalidations or transfers because one of them is actively shared.

Coherence Miss Miss caused by coherence actions such as invalidation or ownership transfer rather than by compulsory, capacity, or conflict effects alone.

Inclusive Cache Hierarchy Hierarchy in which data present in an upper level is guaranteed to also be present in the next lower level.

Exclusive Cache Hierarchy Hierarchy in which adjacent levels are intended not to hold duplicate copies of the same block.

Non-Exclusive Cache Hierarchy Hierarchy that allows overlap between adjacent levels without requiring it for every block.

Critical Section Region of code that must not be executed concurrently by multiple threads if shared data is to remain correct.

Lock Synchronization primitive that grants exclusive access to a critical section.

Mutex Mutual-exclusion lock allowing at most one thread into a protected critical section at a time.

Barrier Synchronization point that forces all participating threads to wait until the last one arrives.

Spin Lock Lock implementation in which a waiting thread repeatedly polls a shared variable instead of blocking.

Atomic Read-Modify-Write Indivisible memory operation that reads a location, transforms it, and writes the new value with no visible intermediate state.

Test-and-Set Atomic read-modify-write primitive that writes a 1 and returns the old value.

Test-and-Test-and-Set Spin-lock strategy that polls with atomic read-only loads and attempts atomic acquisition when the lock appears free.

Load-Link / Store-Conditional (LL/SC) Pair in which a load establishes a reservation and a later store attempts an atomic update; invalid reservations force failure, and the ISA may permit additional failure causes.

Fetch-and-Op Atomic primitive that returns the old value of a location while applying some operation to it.

Fetch-and-Add Fetch-and-op specialization in which the atomic update is addition.

Tree Lock Lock organization that reduces contention by arranging locks hierarchically so contenders meet in stages.

Tournament Lock Another name for a hierarchical tree lock in which contenders advance through multiple levels.

Sense-Reversing Barrier Barrier that uses a phase bit so one barrier instance is not confused with the next.

OpenMP Standard shared-memory parallel-programming interface based largely on compiler directives and runtime support.

MPI Message Passing Interface; standard API for distributed-memory or message-passing parallel programming.

Transactional Memory Model that executes a region optimistically and commits only when its required atomicity and isolation can be validated.

Hardware Transactional Memory Transactional-memory implementation that uses hardware structures such as caches to track speculative read and write sets.

Software Transactional Memory Transactional-memory implementation that tracks conflicts and retries through software metadata rather than specialized hardware.

Memory Consistency Rules defining allowed cross-thread memory outcomes, including ordering across different addresses.

Strict Consistency Memory model in which all processors observe memory operations in one global order that matches actual time.

Sequential Consistency Strong memory model in which execution appears as one total interleaving that preserves each processor's program order.

Weak Consistency Family of relaxed models that provide strong ordering mainly at explicit synchronization operations rather than at every ordinary load and store.

Acquire Synchronization action that prevents later memory accesses from moving before it.

Release Synchronization action that prevents earlier memory accesses from moving after it.

Store Fence ISA-specific ordering operation; the teaching model at least orders earlier stores before later stores, not necessarily every cross-type pair.

Load Fence ISA-specific ordering operation; the teaching model at least orders earlier loads before later loads, not necessarily every cross-type pair.

Memory Fence Explicit ordering operation that constrains permitted memory reordering around a program point.

Flit Flow-control digit; the unit of transfer controlled by the network's flow-control mechanism.

K-ary N-Dimensional Topology Direct-network family with n dimensions and k nodes in each dimension, for a total of k^n nodes.

Direct Topology Network topology in which processors or nodes are connected directly to selected neighboring nodes.

Indirect Topology Network topology in which processors communicate through a separate switching fabric.

Hypercube Direct topology with 2^d nodes in which each node has degree d and neighbors differ in one coordinate.

Mesh Direct topology in which nodes connect to neighbors along one or more dimensions without wraparound edge links.

Torus Direct topology in which each dimension wraps around so the network forms one ring per

dimension.

Gray Code Binary numbering scheme in which adjacent code words differ in exactly one bit.

Bisection Bandwidth Minimum aggregate link bandwidth crossing any equal-halves cut of a network.

Crossbar Indirect switching fabric that can connect any input to any output.

Multi-Stage Interconnect Indirect network built from several stages of small switches rather than one full crossbar.

Perfect Shuffle Fixed wiring permutation commonly used between stages of an omega network.

Omega Network Multi-stage interconnect in which one destination bit guides the routing choice at each stage.

Routing Process of choosing the path a flit or packet takes through the interconnect.

Deterministic Routing Routing policy that always chooses the same path for a given source and destination.

Dimension-Order Routing Deterministic routing policy that traverses dimensions in a fixed order such as x before y.

Adaptive Routing Routing policy that chooses among multiple next hops based on current congestion information.

Oblivious Routing Non-adaptive routing policy that ignores current load but may still use randomized path selection for balance.

Valiant's Algorithm Oblivious routing method that sends traffic first to a randomly chosen intermediate node and then to the final destination.

Flow Control Rules determining when data may advance through the network and what happens when downstream resources are full.

Circuit Switching Communication method that reserves an entire route before data transmission begins.

Packet Switching Communication method in which independently routed packets carry their own destination information.

Packet Message unit that carries routing information and may be divided into multiple flits for transport.

Back Pressure Upstream throttling mechanism used to prevent downstream buffers from overflowing.

Credit-Based Flow Control Back-pressure scheme in which a sender tracks how many downstream buffer slots remain available.

Store-and-Forward Flow-control method that buffers the entire packet at each hop before transmitting it onward.

Cut-Through Routing Packet-forwarding method that begins sending to the next hop as soon as enough of the header has arrived to determine the route.

Virtual Cut-Through Packet-buffered cut-through model that reserves enough downstream space for an entire packet, unlike wormhole's flit-level buffering.

Wormhole Flow Control Flit-level forwarding method in which one packet may stretch across

several routers and links at once.

Head-of-Line Blocking Throughput loss caused when one blocked packet prevents unrelated traffic behind it from advancing.

Virtual Channel Logical subdivision of one physical link into multiple independently buffered and arbitrated channels.

Deadlock Cyclic resource waiting that prevents the involved packets from advancing.

Livelock Continued movement or redirection without eventual delivery.

Starvation Indefinite postponement of one contender while others continue to receive service.

Router Interconnect component that examines incoming traffic, chooses outputs, and manages buffering and arbitration.

Data Race For the C/C++ load/store examples, potentially concurrent conflicting accesses without happens-before ordering, with at least one write and one non-atomic access; such a race is undefined behavior.

Embarrassingly Parallel Workload divisible into independent pieces with negligible communication, allowing near-ideal scaling when resources permit.

Pipelining Temporal parallelism where different stages process different instructions simultaneously.

Superscalar (Multi-Issue) Pipeline organization that can issue more than one instruction per cycle.

Register Renaming Mapping reused architectural names to distinct producer identities, such as physical registers or ROB tags, to remove WAR/WAW name conflicts without removing RAW dependences.

Op One encoded operation within a VLIW instruction stream.

Multi-Op Group of ops intended to execute in the same cycle in a VLIW design.

Tail Bit Per-op field marking whether the current op ends the current multi-op.

Pause Field Encoded delay field that inserts a specified number of idle cycles before the next multi-op begins.

Iterative Modulo Scheduling (IMS) Practical software-pipelining algorithm that searches for a legal modulo schedule by trying increasing initiation intervals.

Initiation Interval (II) Number of cycles between the start of successive loop iterations in a software-pipelined schedule.

Minimum Initiation Interval (MII) Lower bound on II, equal to $\max(ResII, RecII)$.

Resource-Constrained Initiation Interval (ResII) Lower bound on II imposed by recurring resource demand and capacity; counting units alone assumes one-cycle initiation and disjoint resource classes.

Recurrence-Constrained Initiation Interval (RecII) Lower bound on II imposed by loop-carried dependence cycles.

Prologue Non-repeating prefix of a software-pipelined loop that fills the pipeline before steady state.

Kernel Repeating steady-state portion of a software-pipelined loop.

Epilogue Non-repeating suffix of a software-pipelined loop that drains remaining work after loop

exit.

Point Latency Model Scheduling model that treats an op as occupying one issue point and becoming ready after a fixed latency.

Tomasulo's Algorithm Out-of-order scheduling method that combines renaming, reservation stations, and result broadcast to wake dependent instructions when operands become ready.

Reservation Station Scheduling entry that holds an instruction, its operand state, and dependency tags until it can execute.

Common Data Bus (CDB) Shared result-broadcast path used to distribute a completed value and its producer tag to waiting instructions.

Result Bus Generalized form of the CDB; modern designs often use more than one such broadcast path.

Architected Register Register name defined by the ISA and visible to software.

Architected Register File Register file containing the committed architectural state visible to software.

Physical Register Internal hardware register used to hold renamed values beyond the architected register set.

Register Alias Table (RAT) Speculative mapping from architectural names to their latest renamed physical producers, whether their values are ready or not.

Ready Bit Status bit indicating that a register or operand value has already been produced and may be consumed.

Reorder Buffer (ROB) Structure that tracks in-flight instructions for in-order retirement and precise exceptions; result values may live in the ROB or in a separate physical register file.

Future File Register organization that holds the most recent speculative values while a separate architected register file holds committed state.

Messy Register File Informal name for the current speculative register state that may contain out-of-order completions not yet reflected in a coherent sequential state.

Precise Exception Exception model with a defined sequential architectural boundary. For a restartable fault, all older instructions are committed and neither the faulting instruction nor younger instructions have committed effects.

Imprecise Exception Exception model that reports an error without guaranteeing an exact sequential recovery point.

Checkpoint/Repair Recovery method that periodically captures coherent register state and restores it when an exception or mis-speculation occurs.

Instruction Boundary Program-order prefix boundary used to define the state a checkpoint must represent for recovery.

Simultaneous Multithreading (SMT) Execution organization in which multiple hardware threads share one core's issue and execution resources.

Speculative Side Channel Information leak caused when squashed speculative work still changes microarchitectural state such as caches or predictors.

Runtime Time to complete one job (response/turnaround time).

Throughput Completed work per unit time.

Amdahl's Law Speedup limit for fixed-size workloads when only a fraction is accelerated.

Gustafson's Law Scaled-speedup model where workload size grows with processor count.

Average Access Time (AAT) Expected memory-access latency, typically modeled as hit time plus miss rate times miss penalty.

SRAM Fast, low-latency, higher-area memory used for on-chip caches.

DRAM Dense, lower-cost, higher-latency memory used for main memory capacity.

Hit Time (HT) Latency to serve an access from the current cache level.

Miss Ratio (MR) Fraction of accesses at a level that miss.

Miss Penalty (MP) Extra latency to service a miss from lower levels.

Compulsory Miss In a cold demand-only trace, the first reference to a block.

Capacity Miss Non-first miss that also occurs in the stated fully associative reference cache of the same capacity and block size.

Conflict Miss Non-first miss avoided by that fully associative reference cache under the stated replacement policy, revealing a placement restriction.

Write-Back Write policy that defers propagation of modified data until eviction, an explicit write-back, or a coherence transaction requires it.

Write-Through Write policy that also sends each write hit to the next level, possibly through a buffer.

Write-Allocate Write-miss policy that fetches and installs the block before write.

Write-No-Allocate Write-miss policy that bypasses install and writes lower level directly.

Qubit Two-level quantum system described by normalized complex amplitudes; measurement in a basis returns one basis outcome rather than all amplitudes.

Decoherence Loss of useful quantum coherence through uncontrolled interaction with the environment.

NISQ Noisy intermediate-scale quantum computing; a noisy-device regime with limited usable circuit depth, not a permanent calendar label.

Logical Qubit Error-corrected qubit represented by many physical qubits and maintained through repeated decoding.

Phase Oracle Reversible quantum operation that encodes a classical function by changing the phase of selected basis states.

Fault-Tolerant Quantum Computing Quantum-computing approach that uses encoded logical qubits and error correction to run longer computations reliably.

Superconducting Logic Classical digital logic using superconducting devices, often with pulse signaling; it is not automatically quantum computation.

Rapid Single Flux Quantum (RSFQ) Superconducting pulse-logic family that represents bits using quantized magnetic-flux pulses.

Reciprocal Quantum Logic (RQL) AC-powered superconducting logic family related to RSFQ and used as one candidate technology for energy-efficient processors.

Operand Queue Storage structure that holds produced values until compiler-scheduled consumer instructions need them.

Barrel Processor Multithreaded processor that issues from different resident threads in rotation to keep a deep pipeline occupied.

Cryostat Cooling system used to maintain cryogenic temperatures for superconducting or quantum hardware.

Energy-Delay Product (EDP) Efficiency metric combining energy and runtime; lower EDP indicates a better energy/performance tradeoff.

SUP Superconducting Unit of Performance; proposed organization combining latency-oriented and throughput-oriented superconducting resources.

SYNDRA Static Synchronous Dataflow RISC Architecture; proposed compiler-scheduled architecture targeting single-thread performance.

STRIDE Proposed many-thread superconducting architecture targeting aggregate throughput.

PIPT Physically indexed, physically tagged cache organization whose set index and tag are defined by the physical address.

VIPT Virtually indexed, physically tagged cache organization that overlaps indexing with translation.

TLB Translation lookaside buffer caching virtual-to-physical translations and associated access-permission information.

Structural Hazard Stall risk from insufficient hardware resources.

Data Hazard Stall risk from operand read/write ordering conflicts.

Control Hazard Stall risk from unresolved branch/jump control flow.

RAW Read-after-write dependence hazard.

WAR Write-after-read dependence hazard.

WAW Write-after-write dependence hazard.

Data Dependence Graph Graph representation of instruction ordering constraints induced by true and false dependences.

Dependence Height Priority heuristic measuring the remaining latency on the longest dependent path below an instruction.

List Scheduling Heuristic compiler scheduling method that repeatedly places the highest-priority ready instructions into the earliest legal cycle.

Memory Disambiguation Compile-time or runtime reasoning used to determine whether two memory references may access the same location.

Forwarding (Bypassing) Sending a produced value directly to a consumer before register write-back.

Branch Target Buffer (BTB) Predictor structure that supplies likely branch targets early in fetch.

Aliasing Multiple names or addresses referring to the same storage; in predictor tables, the term also describes unrelated branches sharing an entry.

Return-Address Stack (RAS) Small stack predictor for call/return target addresses.

Direction Prediction Branch predictor decision for taken versus not taken.

One-Bit Predictor Dynamic branch predictor that records only the last observed branch outcome.

Smith Predictor Saturating-counter branch predictor, commonly implemented with a two-bit counter per entry.

GShare Predictor indexing by XOR of selected PC bits and global history; the model specifies whether address-alignment bits are discarded.

GSelect Global-history predictor that forms its index by concatenating selected PC bits and global-history bits.

Yeh/Patt Predictor Two-level branch predictor that uses recorded branch history to index a table of direction counters.

Perceptron Predictor Branch predictor that computes a weighted sum over branch-history bits and predicts from the sign of that sum.

Hybrid Predictor Predictor that chooses among multiple underlying predictors, often through a tournament-style selector.

BTFNT Backward taken, forward not taken static branch heuristic.

Delay Slot ISA semantic in which the instruction after a branch executes before control transfer takes effect.

Squash Slot Post-branch instruction position that may be conditionally killed based on branch outcome/policy.

WBWA Write-back, write-allocate pairing.

WTWNA Write-through, write-no-allocate pairing.

Blocking Cache Cache behavior where an outstanding miss prevents servicing younger accesses.

Non-Blocking Cache Cache behavior allowing selected younger accesses to proceed during a miss.

Hit-Under-Miss Ability to serve cache hits while a miss is still outstanding.

Acronym List

AAT Average Access Time

ALU Arithmetic Logic Unit

AVX Advanced Vector Extensions

ACK Acknowledgment

AW Address Width

BTFNT Backward Taken, Forward Not Taken

BTB Branch Target Buffer

CDB Common Data Bus

CISC Complex Instruction Set Computer

CPI Cycles Per Instruction

CPU Central Processing Unit

CT Cycle Time
DDG Data Dependence Graph
DSP Digital Signal Processor
DISP Dispatch and Decode
DRAM Dynamic Random-Access Memory
EDP Energy-Delay Product
EX Execute
FICI Fire In Order, Complete In Order
FIFO First In, First Out
FICO Fire In Order, Complete Out of Order
FOCO Fire Out of Order, Complete Out of Order
FP Floating Point
FU Functional Unit
GPU Graphics Processing Unit
HT Hit Time
HTM Hardware Transactional Memory
IC Instruction Count
ID Instruction Decode
IF Instruction Fetch
II Initiation Interval
ILP Instruction-Level Parallelism
IMS Iterative Modulo Scheduling
IPC Instructions Per Cycle
ISA Instruction Set Architecture
LFU Least Frequently Used
LL/SC Load-Link / Store-Conditional
LIP Least Recently Used Insertion Policy
LLC Last-Level Cache
LRU Least Recently Used
MEM Memory
MII Minimum Initiation Interval
MIMD Multiple Instruction, Multiple Data
MIP Most Recently Used Insertion Policy
MISD Multiple Instruction, Single Data
MESI Modified, Exclusive, Shared, Invalid
MP Miss Penalty
MR Miss Ratio

MRU Most Recently Used

MSI Modified, Shared, Invalid

MOESI Modified, Owned, Exclusive, Shared, Invalid

MOESIF Modified, Owned, Exclusive, Shared, Invalid, Forward

MPI Message Passing Interface

NACK Negative Acknowledgment

NISQ Noisy Intermediate-Scale Quantum

NoC Network on Chip

OpenMP Open Multi-Processing

OS Operating System

PC Program Counter

PCIe Peripheral Component Interconnect Express

PE Processing Element

PFN Physical Frame Number

PGAS Partitioned Global Address Space

PIPT Physically Indexed, Physically Tagged

RAS Return-Address Stack

RAT Register Alias Table

RAW Read After Write

RecII Recurrence-Constrained Initiation Interval

ResII Resource-Constrained Initiation Interval

RISC Reduced Instruction Set Computer

ROB Reorder Buffer

RMW Read-Modify-Write

RQL Reciprocal Quantum Logic

RS Reservation Station

RSFQ Rapid Single Flux Quantum

SIMD Single Instruction, Multiple Data

SIMT Single Instruction, Multiple Threads

SISD Single Instruction, Single Data

SMT Simultaneous Multithreading

SRAM Static Random-Access Memory

STM Software Transactional Memory

SSE Streaming SIMD Extensions

SSD Solid-State Drive

SU State Update

SUP Superconducting Unit of Performance

TLP Thread-Level Parallelism

TLB Translation Lookaside Buffer

TM Transactional Memory

VIPT Virtually Indexed, Physically Tagged

VLIW Very Long Instruction Word

VPN Virtual Page Number

WAR Write After Read

WAW Write After Write

WB Write Back

WBWA Write-Back, Write-Allocate

WTWNA Write-Through, Write-No-Allocate

Appendix D

Advanced Topics: Quantum and Superconducting Architectures

The core chapters asked how to overlap work, move values, and manage limited resources. Those questions remain useful when the physical devices change. This appendix connects the advanced lecture topics to two different research directions: quantum accelerators and *classical* superconducting digital processors. Both can involve cryogenic equipment, but a superconducting logic bit is not thereby a qubit. The named superconducting proposals below are research architectures, not claims of deployed commercial performance.

D.1 Quantum Computing Architecture

Quantum computers are best viewed as accelerators rather than replacements for classical computers. A useful quantum system still needs a classical computer to prepare work, drive the quantum device, interpret measurements, and coordinate repeated trials. The expected advantages are problem-specific. Two motivating examples are simulation of physical and molecular systems and integer factoring. A sufficiently capable fault-tolerant quantum computer could threaten RSA-style cryptography; this is not a claim that a small noisy device can already factor practical keys.

A qubit has a state of the form

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{D.1}$$

The symbols $|0\rangle$ and $|1\rangle$ name basis states, and α, β are complex amplitudes. Measuring in that basis returns 0 or 1 with probabilities $|\alpha|^2$ and $|\beta|^2$. It does not print every value present in a superposition; useful algorithms must arrange interference before measurement.

D.1.1 Why the Hardware Is Difficult

Classical logic restores signals toward separated digital levels. Quantum computation also depends on amplitudes and relative phase, so an uncontrolled rotation or environmental interaction can corrupt information without looking like an ordinary classical bit flip. Isolation, calibration, and precise control are therefore central architectural concerns.

A useful first distinction separates unwanted environmental coupling from imperfect operations. *Decoherence* is loss of the useful coherence and correlations of the intended quantum state through interaction with the environment. A stray photon, thermal effect, vibration, or other coupling can destroy the state that the algorithm depends on. *Imperfect control* means that the hardware operation itself is inaccurate: a laser pulse may be mistimed, mis-aimed, too wide, too weak, too strong, or the measurement apparatus may fail to collect the intended signal.

Noise exposure depends on both elapsed time and the operations performed. Reducing costly gates or circuit depth can help, but instruction count alone is not a reliability metric: different gates

have different error rates, and connectivity can require extra movement operations. A compiler must consider gate type, duration, parallelism, and mapping together rather than treating every removed instruction as an equal gain.

D.1.2 NISQ and Fault Tolerance

NISQ, noisy intermediate-scale quantum computing, names the noisy-device regime discussed by John Preskill in 2018.[1] Useful computation in that regime is constrained by qubit count, errors, and usable circuit depth. Whether an application beats its best classical alternative depends on the problem, implementation, and comparison, not on the presence of superposition alone.

The contrasting goal is *fault-tolerant quantum computing*. Instead of treating one noisy physical qubit as one reliable algorithmic qubit, many noisy physical qubits are entangled and managed together to form a more reliable *logical qubit*. There is no universal physical-to-logical qubit ratio; the code, physical error rates, and target reliability determine the overhead. This creates a new architecture problem: the machine must repeatedly extract error information and run a decoder quickly enough to keep the logical qubits usable. Syndrome measurements reveal error information without directly measuring the encoded logical value; the classical decoder uses those results to guide recovery. Decoder design and topology-aware error correction are therefore architectural topics, not just physics topics.

D.1.3 Quantum Programs and Reversibility

Quantum gates describing the evolution of an isolated system must be reversible (unitary). Measurement and reset are different kinds of operation and need not be reversible on the measured system. The gate requirement is a major difference from ordinary classical functions. For example, a classical function $f(x)$ that maps many inputs to one output may not have an inverse, so it cannot be used directly as a quantum operation.

A Boolean function $f(x) \in \{0, 1\}$ can instead be represented by a reversible phase operation:

$$U_f|x\rangle = (-1)^{f(x)}|x\rangle. \tag{D.2}$$

If $f(x) = 0$, the state is unchanged. If $f(x) = 1$, the state receives a negative phase. Applying the same operation twice cancels the phase change, so this operation is reversible. It does not make a large function free to compute: the oracle still needs an implementation and its cost belongs in the algorithm's resource accounting.

The algorithmic benefit comes from superposition and interference. A quantum operation can act on a superposition of possible inputs, and later operations can make the desired answers reinforce while undesired answers cancel. This is not generic parallelism: the algorithm must match the mathematical structure of the problem. That is why quantum speedups exist for some problems but not for arbitrary computation.

A phase change needs interference to become visible For the one-bit function $f(x) = x$, the phase oracle is the Z gate: it leaves $|0\rangle$ unchanged and negates $|1\rangle$. Starting from $|0\rangle$, a Hadamard gate H prepares $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. The oracle changes this to $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, and a second H maps it to $|1\rangle$. Figure D.1 follows that change.

[1] See the paper listed in Appendix F; NISQ is a device regime, not a permanent calendar label or a claim about every current platform.

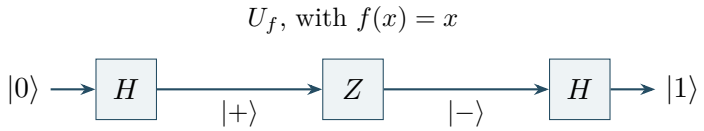


Figure D.1: A relative phase becomes a readable outcome through interference. Direct computational-basis measurement before the final H would still give 0 or 1 equally often. This small example explains phase processing, not a general quantum speedup.

Measuring either $|+\rangle$ or $|-\rangle$ directly in the computational basis gives equal 0/1 probabilities. The final H turns their *relative* phase difference into different measurement outcomes. A common phase multiplying the entire state, such as replacing every amplitude by its negative, is instead unobservable by itself.

D.2 Superconducting Processor Architecture

The superconducting architecture topic asks whether classical processors can be built differently from conventional CMOS designs. The motivation is energy. Modern data centers and supercomputers consume large amounts of power, and simply scaling familiar CMOS microarchitecture is increasingly expensive. Superconducting circuits offer extremely low switching energy, but they impose constraints that break many familiar processor structures.

D.2.1 Why Ordinary Processor Designs Do Not Map Cleanly

A superconductor can have zero DC resistance below its operating limits, but that does not make switching, clocking, biasing, or refrigeration free. Single-flux-quantum-style logic represents information using short quantized pulses rather than ordinary long-lived CMOS voltage levels. That can be very energy efficient, especially on wires, but it creates a practical problem: a tiny pulse cannot be freely fanned out to many destinations. Every split or copy of a signal may require explicit hardware.

This makes common high-performance structures expensive:

- multi-ported register files need many read and write paths,
- result buses and bypass networks need broad fanout,
- out-of-order schedulers need associative wakeup and tag broadcast,
- branch prediction and recovery need fast tables and cheap mis-speculation recovery,
- and multi-ported caches require duplicated indexing, tags, comparators, and wiring.

These are exactly the structures used heavily in conventional high-IPC CMOS cores. The architectural response is to move more work to the compiler and design the hardware around predictable data movement.

D.2.2 SYNDRA: Compiler-Scheduled Single-Thread Performance

The research proposal names its single-thread-oriented architecture *SYNDRA*, Static Synchronous Dataflow RISC Architecture. Its Statically Assigned DataGraph Execution (SADGE) model moves

scheduling work to the compiler.[2] The important connection to the core chapters is static dataflow: make producer-consumer relationships explicit instead of rediscovering them with a large dynamic scheduler.

An illustrative implementation can route a result toward its scheduled consumer and hold it in an *operand queue* until needed, rather than returning every value through a centralized multi-ported register file. This describes the architectural idea, not a complete specification of the proposed core's pipeline or ISA.

This resembles VLIW and software scheduling from Chapter 7, but with a different hardware motivation. When timing is predictable, the compiler can arrange delivery distances and interconnect controls ahead of time. The goal is to keep useful instruction-level parallelism while avoiding the most expensive fanout-heavy structures of out-of-order execution.

Control flow can likewise be separated into three conceptual jobs:

- *prepare* supplies a possible branch target early,
- *compare* computes the condition early,
- *decide* selects control flow when the condition is known.

This separates where the program might go from whether it will go there. This can create time to prefetch or configure a path, but a static plan still needs a correct response to variable memory latency and unresolved conditions.

Banking illustrates the same resource tradeoff without depending on a particular prototype. Instead of one fully multi-ported cache, divide storage into independently accessed banks. For example, odd and even indices can map to different half-sized banks. This gives multiple accesses per cycle when the accesses land in different banks, while avoiding the area cost of a monolithic multi-ported cache. Two accesses to the same single-ported bank still conflict.

D.2.3 STRIDE: Thread-Level Parallelism Through Barrel Processing

The proposed STRIDE core attacks the problem from the other direction: many-thread throughput rather than only single-thread ILP. It is a barrel processor. Issue opportunities rotate among resident hardware threads rather than always advancing one thread. If the pipeline is very deep, the core needs many resident threads so that a thread's long-latency operation can make progress before that thread is scheduled again.

The tradeoff is clear. For a simple one-wide barrel model with $N_{threads}$ runnable threads and no lost slots, aggregate IPC can be 1 while each thread averages only $1/N_{threads}$. That is an illustrative model, not the proposed STRIDE core's numerical performance specification. The whole core can still sustain high system throughput because many threads are in flight at once. This style is attractive for abundant independent work, but can be a poor fit when the priority is low single-thread completion time.

D.2.4 SUP: Combining the Two Modes

The proposed larger unit combines SYNDRA and STRIDE into a *superconducting unit of performance (SUP)*. The reason is that high-performance computing programs often alternate between phases:

[2] Pulkit Gupta's May 5, 2026 Georgia Tech proposal abstract identifies SYNDRA, STRIDE, and SUP; see Appendix F.

- single-threaded or latency-sensitive setup,
- coarse-grained parallel work with some synchronization,
- and massively parallel phases such as matrix or signal-processing kernels.

SYNDRA targets single-thread-oriented phases, while STRIDE targets highly threaded work. Any hybrid also needs a software interface for choosing execution resources, transferring state, and accounting for transition overhead. Sharing data or memory resources could reduce duplication, but the benefit must be weighed against contention and control cost rather than assumed from the word "hybrid."

D.2.5 Cryogenic System Organization

Cooling cost cannot be ignored. For an ideal refrigerator moving heat from T_{cold} to T_{hot}, the minimum input work per unit of removed heat is

$$\frac{W_{input,min}}{Q_{cold}} = \frac{T_{hot}}{T_{cold}} - 1. \tag{D.3}$$

Temperatures are absolute, in kelvin. Between 4 K and 300 K, removing 1 W at the cold stage requires at least 74 W of input work even in this ideal limit; a real refrigerator requires more. The actual multiplier depends on the equipment, load, and staging, not one universal "watts per cold watt" constant. Compare end-to-end energy or throughput per wall-plug watt, including cooling and support electronics, rather than quoting only the logic's switching energy.

A full system can use temperature staging. Superconducting logic sits near liquid-helium temperatures. Some support electronics may sit at a warmer stage, but their device behavior and interface cost must be characterized at that temperature. Links to other modules or warm network equipment also add interface power and heat load.

D.3 Final Review Checkpoints

For final-review purposes, focus on the following claims.

1. In the hybrid model discussed here, a quantum accelerator is coordinated by a classical host.
2. Quantum speedups are problem-specific; physics simulation and integer factoring are canonical examples.
3. Decoherence and imperfect control are the two broad error categories used here.
4. Quantum resource cost includes gate type, error rate, duration, depth, and mapping, not instruction count alone.
5. NISQ machines are limited by noise, qubit count, and short useful program lengths.
6. Fault-tolerant quantum computing uses many physical qubits plus decoding to create more reliable logical qubits.
7. Closed-system quantum gates are unitary; measurement and reset need not be reversible. A phase oracle changes relative phases, which interference can convert into measurement information.
8. Superconducting logic can be much more energy efficient than CMOS, but cooling overhead and architecture constraints still matter.

9. Near-zero resistance does not make conventional fanout, register files, bypass networks, and branch predictors free.
10. SYNDRA proposes compiler-scheduled dataflow; operand queues illustrate an alternative to routing every value through a centralized multi-ported register file.
11. STRIDE targets many-thread throughput; barrel processing explains the trade between aggregate progress and individual-thread issue rate.
12. A combined SUP-style design uses the single-thread core for latency-sensitive phases and the threaded core for highly parallel phases.

D.4 Practice True/False Prompts

Mark each statement true or false, then correct the false statements.

1. A quantum computer is expected to replace the classical computer entirely for all workloads.
2. Reducing quantum instruction count can improve reliability because both decoherence and gate/control errors accumulate during execution.
3. NISQ machines are already assumed to be good enough for long, reliable, general-purpose quantum programs.
4. Fault tolerance in quantum computing usually means encoding one logical qubit across many unreliable physical qubits.
5. A classical function embedded as a closed-system quantum gate must be implemented reversibly.
6. Superconducting processors can use all conventional out-of-order CPU structures cheaply because wire resistance is near zero.
7. Compiler scheduling becomes more important when hardware fanout, multi-ported storage, and associative wakeup are expensive.
8. A barrel processor improves the latency of one thread by issuing that same thread every cycle.
9. Banking a cache can provide multiple accesses per cycle when the accesses map to different banks.
10. A SUP-style design combines single-thread-oriented and throughput-oriented execution resources because real programs often move through different parallel phases.

Appendix E

Comprehensive Final Review Checklist

This appendix is a compact review map for the full course. It is not a substitute for the chapter practice questions; it is a last-pass checklist of the concepts that repeatedly connect worked problems, implementation exercises, and review questions.

E.1 Core Topic Checklist

Architecture and Performance

- Know the grain sizes of parallelism: bit-level, instruction-level, thread-level, and inter-program parallelism.
- Distinguish SIMD, MIMD, packed-word SIMD, and pipelined vector execution.
- Use the CPU-time equation time $= IC \times CPI \times CT$ and know which term an optimization changes.
- Apply Amdahl's law for fixed-size speedup and Gustafson's law for scaled workloads.
- Choose rate averages from the workload: harmonic for equal-work segments, arithmetic for equal-time segments, with appropriate weights otherwise.

Caches and Memory Hierarchy

- Split addresses into tag, index, and offset fields from (C, B, S) cache notation.
- Classify misses as compulsory, capacity, conflict, or coherence misses when multiprocessors are involved.
- Connect each cache optimization to the term it improves: hit time, miss ratio, or miss penalty.
- Compare write-back/write-through and write-allocate/write-no-allocate behavior.
- Explain why VIPT caches must keep index plus offset bits within the page offset unless extra synonym-handling machinery exists.
- Treat tiling as a working-set transformation: choose tiles so the active data footprint fits in cache and does not create severe set conflicts.

Pipelines and Branch Prediction

- Interpret IF/ID/EX/MEM/WB timing diagrams cycle by cycle.
- Identify structural, data, and control hazards and connect them to inserted stalls or squashes.
- Explain why forwarding fixes many RAW hazards but does not eliminate the classic load-use stall.

- Compare delay slots, squash slots, branch target buffers, return-address stacks, and direction predictors.
- Work through one-bit, two-bit, GSelect, GShare, Yeh/Patt, perceptron, and hybrid predictor lookup/update behavior.

Dynamic and Compiler-Scheduled ILP

- Distinguish FICI, FICO, and FOCO execution models.
- Explain how Tomasulo-style scheduling uses reservation stations, tags, result broadcast, and register renaming.
- Explain precise recovery at a program-order boundary, including no committed effects from a restartable faulting instruction; compare reorder-buffer and checkpoint/repair mechanisms.
- Draw data dependence graphs and classify pure, anti-, and output dependences.
- Use dependence height as a list-scheduling priority.
- Explain VLIW as a trade: simpler dynamic hardware in exchange for stronger compiler scheduling responsibility.
- For software pipelining, compute $ResII$, $RecII$, and $MII = \max(ResII, RecII)$, then reason about prologue, kernel, and epilogue code.

Multiprocessors, Synchronization, and Consistency

- Distinguish shared-memory and message-passing systems.
- Separate the communication model from UMA/NUMA physical placement and from whether accelerators share memory.
- Explain coherence as ordering and propagation for one memory location, not as full synchronization correctness.
- Trace MSI/MESI/MOESI-style state transitions and know what `E`, `O`, and `F` add.
- Distinguish abort/retry from intervene behavior when dirty data is requested by another cache.
- Explain false sharing as unnecessary coherence traffic caused by unrelated variables sharing one cache block.
- Compare snooping and directory-based coherence, including why limited-pointer directories trade space against overflow behavior.
- Explain why test-and-test-and-set reduces coherence traffic compared with repeated test-and-set.
- Use locks for mutual exclusion around critical sections and barriers for regrouping a set of threads at a phase boundary.
- Define a data race and explain why coherence alone does not remove data races.
- Distinguish C/C++ data-race undefined behavior from an ordering bug using relaxed atomic operations.
- Compare strict, sequential, and weak/relaxed consistency models.

- Know what store fences, load fences, full fences, acquire operations, and release operations prevent.

Interconnection Networks

- Distinguish direct topologies, such as meshes and tori, from indirect topologies, such as crossbars and multi-stage networks.
- For k-ary n-dimensional networks, compute node count as k^n.
- Define bisectional bandwidth as the aggregate bandwidth crossing the minimum equal-halves cut.
- Explain why a torus has more bisectional bandwidth than the corresponding mesh.
- Compare routing choices without treating all labels as exclusive: traffic-independent deterministic routing is also oblivious.
- Explain Valiant's algorithm as randomized load balancing through an intermediate node.
- Distinguish packet-level and flit-level flow control.
- Compare store-and-forward, cut-through, wormhole, and virtual-channel flow control.
- Distinguish header latency from whole-packet latency, and deadlock from livelock and starvation; virtual channels do not automatically prove progress or add physical bandwidth.

E.2 Review Clarifications

Tiling

Tiling is useful when the program has reuse but the untiled working set is too large for the cache. The transformation changes loop order so the program operates on cache-sized pieces before moving on. For review problems, the first check is usually the footprint inequality: active arrays, tile dimensions, and bytes per element must fit within the effective cache capacity. The second check is placement: a tile that fits by byte count can still perform poorly if the active blocks conflict in the same sets.

Iterative Modulo Scheduling

Modulo scheduling is easiest to understand as loop unrolling followed by rerolling. Unrolling exposes overlap among different iterations, but unrolling too far makes code large. Software pipelining keeps the repeating overlapped pattern as a compact kernel and adds a prologue and epilogue to handle startup and drain.

Iterative modulo scheduling begins at the lower bound $MII = \max(ResII, RecII)$. $ResII$ comes from resource pressure. $RecII$ comes from loop-carried dependence cycles. The compiler tries to schedule the loop at that initiation interval; if it cannot, it increments II and tries again. An operation that would appear in cycle x of the unrolled schedule maps to row $x \bmod II$ of the modulo reservation table. Operations that conceptually wrap around the compact kernel explain why prologue and epilogue code are needed.

Memory Consistency

The core consistency trap is that coherence is per-location, while synchronization usually depends on ordering between locations. In the producer/consumer pattern, a coherent machine can still let the consumer observe a flag update before it observes the data update that the flag was meant to publish. The fix is not merely better coherence; the fix is an ordering rule around synchronization, implemented through acquire/release semantics or explicit fences.

Strict consistency ties the visible memory order to actual time. Sequential consistency allows a single legal interleaving that preserves each thread's program order, even if that interleaving is not the real-time order. Weak or relaxed models allow ordinary accesses to move more freely but require synchronization accesses to carry stronger ordering guarantees. In C/C++, use the language's atomic operations or synchronization library; undefined behavior from ordinary racing accesses is not repaired merely by a coherent cache or an invented fence call.

Barriers and Locks

A lock protects a critical section: only one thread should execute that protected code at a time. A barrier is a phase boundary: every participating thread waits until all have arrived, and then all may proceed. Barriers are natural in bulk-synchronous programs such as GPU-style kernel phases. They are expensive when overused because fast threads idle while waiting for slow ones.

Bisectional Bandwidth

Bisectional bandwidth is a topology figure of merit. For a legal equal-halves cut, sum the bandwidths of the links crossing the cut. The bisectional bandwidth is the minimum such aggregate over the allowed equal splits. For the regular square meshes and tori used here, a dimension-aligned equal cut attains the minimum. Other equal-halves partitions are still legal candidates; the definition always takes the minimum over all of them. This value is a potential maximum, not a guarantee: routing, buffering, and traffic conflicts can keep real throughput below it.

E.3 Fast Self-Check Prompts

1. Given a cache geometry, can you compute tag/index/offset bits and explain whether a miss is compulsory, capacity, conflict, or coherence?
2. Given a branch predictor table and branch history, can you show both the prediction and the update?
3. Given a dependence graph, can you identify false dependences and explain how renaming removes them?
4. Given a loop body and functional-unit counts, can you compute $ResII$, $RecII$, and MII?
5. Given a coherence trace, can you count memory reads, memory writes, invalidations, and cache-to-cache transfers?
6. Given a lock implementation, can you explain which part provides atomicity and which part provides ordering?
7. Given two threads accessing two variables, can you decide which outcomes are legal under strict consistency, sequential consistency, and relaxed ordering?

8. Given a network topology, can you compute bisectional bandwidth and explain why routing may still perform worse than that theoretical limit?
9. Given a packet/flit scenario, can you explain whether store-and-forward, cut-through, wormhole, or virtual channels best describes the behavior?
10. Given the advanced-topics appendix, can you explain why quantum accelerators need classical control and why classical superconducting cores impose different architectural tradeoffs from CMOS?

Appendix F

Lecture Sources and Historical Notes

The course materials supplied with this project are the primary reference for its scope and terminology. Slide numbers refer to the printed identifiers inside the handouts, which may differ from PDF page numbers because a PDF page can contain two slides.

How Attribution Works

Lecture examples are paraphrased, not reconstructed quotations. They preserve the teaching point while making modeling assumptions explicit. Examples added to connect ideas, rather than recovered from a lecture, are presented as the book's own illustrations.

Examples Used in the Narrative

Independent additions, Chapter 1. `n1_intro.pdf`, INTRO-6, contrasts three dependent additions with three independent additions on three adders. The book states the equal-latency, available-resource assumptions in its diagram.

Scheduling the next crew, Chapter 1. January 13, 2026 lecture, Notability note *1. Syllabus*, *Recording 2*, approximately 14:41–15:25. The construction example separates estimating a completion time from acting on that estimate.

The transistor-budget question, Chapter 1. January 15, 2026 lecture, Notability note *2. ECE 6100*, *Recording 3*, approximately 06:47–08:05, and slides INTRO-18–19. The book retains the design question without treating the power wall as the end of single-core improvement.

The Cray vector example, Chapter 1. `n1_intro.pdf`, INTRO-13–14, contrasts streamed vector execution with packed-word arithmetic. The book separates the lecture's simplified pipeline from historical hardware timings.

Buying a jet-simulation machine, Chapter 2. `n2_perf.pdf`, PERF-18–20, uses an extravagant many-processor machine to expose Amdahl's limit, then changes the question to a larger simulation for Gustafson's law. The book uses 100 hours as a convenient baseline rather than the handout's rounded week-to-seconds conversion.

Passenger time and airline throughput, Chapter 2. January 20, 2026 lecture, Notability note *3. ECE 6100*, *Recording 1*, approximately 02:00–05:15. The book keeps the qualitative Concorde-versus-larger-aircraft comparison and states its simplified capacity assumptions.

Buying the slower machine, Chapter 2. `n2_perf.pdf`, PERF-9–11, compares rates of 4 and 4 with rates of 2 and 7. The workload must contain equal work of each type for the harmonic-mean comparison.

The supercomputer joke, Chapter 2. January 15, 2026 lecture, Notability note *2. ECE 6100*, *Recording 2*, approximately 06:03–06:32.

Choosing a victim, Chapter 3. `n3_cache_all.pdf`, CACHE-20, asks which of four blocks should be replaced after a short access history. The book follows that example to distinguish knowing the past from knowing the next use.

A full set, Chapter 3. January 29, 2026 lecture, Notability note *6. ECE 6100*, *Recording 3*, approximately 03:59–05:58. Five blocks competing for a four-way set motivate the distinction between set pressure and a capacity miss in the fully associative reference cache.

Deep recursion and a small predictor, Chapter 4. February 19, 2026 lecture, Notability note *12. ECE 6100*, return-address-prediction recording segment, approximately 13:22–18:39. The factorial example motivates a repeated-bottom-entry pop policy. The book supplies its own three-entry, five-return trace rather than treating those sizes as a recovered hardware specification.

Why renaming mattered, Chapter 6. `n5_ilp_part1.pdf`, ILP-14, connects the System/360 Model 91's long floating-point operations and small architectural register set to dynamic scheduling and renaming.

One result bus, Chapter 6. `n5_ilp_part1.pdf`, ILP-23–24, follows an independent cache hit past an older miss. The book makes response eligibility and next-cycle wakeup explicit and separates D's cycle-6 result from its cycle-7 broadcast.

Forwarding a value or a promise, Chapter 6. March 12, 2026 lecture, Notability note *18. 6100*, ROB-bypass recording segment, approximately 28:54–31:26. The comparison with the earlier pipeline's forwarding paths explains why dispatch can need either a ready value or the tag of an unfinished older producer.

Overlapping loop iterations, Chapter 7. The software-pipelining handout, ILP-86–87, supplies the pointer-load/store loop. The book states alias assumptions, uses an aligned address step and a decrementing trip counter, and shows the final load-to-store delay explicitly. Its filename is `n5_vliw_spipe_handout_s26-1.pdf`.

Folding a ring, Chapter 10. The April 21, 2026 handwritten class notes, *Day 27*, p. 1, contrast a long wraparound wire with a folded placement. The book's numbered diagram makes the physical slot mapping explicit and treats horizontal span as a wire-cost proxy.

Merging into the ring, Chapter 10. `n8_interconn.pdf`, slide 28, asks the reader to connect packet injection with a car-road situation. The roundabout interpretation in the book is an explanatory analogy, not a reconstructed quotation or a universal traffic-priority rule.

Primary Research and Historical References

Multicore and energy efficiency. Intel's March 7, 2006 Core microarchitecture introduction documents the contemporary emphasis on multicore performance and energy efficiency. It provides historical context for the transistor-budget discussion in Chapter 1.

Vector registers and chaining. Cray Research, *CRAY-1 Computer Systems Hardware Reference Manual*, HR-0004F, May 1982, sections on vector registers and chaining. The manual describes eight vector registers with up to 64 elements each and explains the conditions for forwarding an element stream between functional units. Available from the Computer History Museum's Bitsavers archive.

Dynamic scheduling. R. M. Tomasulo, *An Efficient Algorithm for Exploiting Multiple Arithmetic*

Units, IBM Journal of Research and Development, 11(1), 1967, pp. 25–33. This is the original description of the Model 91 floating-point scheduling method. doi:10.1147/rd.111.0025.

Two-level predictor organizations. Tse-Yu Yeh and Yale N. Patt, *Alternative Implementations of Two-Level Adaptive Branch Prediction*, ISCA, 1992. The paper distinguishes global and per-address histories and shared versus separate pattern tables. Chapter 4 identifies its local-history example within that larger family.

Perceptron predictors. Daniel A. Jiménez and Calvin Lin, *Dynamic Branch Prediction with Perceptrons*, HPCA, 2001. The authors' paper, Sections 3 and 5, supplies the training rule, linear-separability limitation, and empirically tuned history-length threshold used in Chapter 4.

Modulo scheduling. B. Ramakrishna Rau, *Iterative Modulo Scheduling: An Algorithm for Software Pipelining Loops*, MICRO, 1994, pp. 63–74. The paper distinguishes the initiation-interval lower bound from a constructed legal schedule and describes bounded scheduling attempts. Texas Instruments' C7000 optimization guide illustrates pipeline startup, steady state, draining, and register-allocation constraints in a production compiler.

Network resource dependencies. William J. Dally and Charles L. Seitz, *Deadlock Free Message Routing in Multiprocessor Interconnection Networks*, Caltech technical report, 1986, available through CaltechAUTHORS. Chapter 10 uses its channel-dependency perspective rather than treating the presence of virtual channels as a proof of deadlock freedom.

Noisy quantum devices. John Preskill, *Quantum Computing in the NISQ Era and Beyond*, Quantum 2, 79, 2018; author manuscript. This supplies the historical device-regime terminology, not a fixed prediction about every later platform.

Quantum query models. John Watrous's IBM Quantum course provides the query-model context. The book's one-qubit HZH example is a direct matrix identity illustrating interference, not an assertion that oracle-query advantages automatically become whole-program speedups.

SYNDRA, STRIDE, and SUP. Pulkit Gupta, *Architectures and Primitives for Composable Superconducting Supercomputers*, Georgia Tech Ph.D. proposal announcement, May 5, 2026. The abstract verifies the names and high-level architectural goals. Appendix D labels its operand-queue, control-flow, banking, and one-wide barrel examples as illustrations rather than a full implementation specification or a verified hardware benchmark.

Refrigeration cost. Ray Radebaugh, *Refrigeration for Superconductors*, NIST author copy, Section II. The Carnot coefficient of performance gives the ideal cooling-work bound used in Appendix D; actual equipment has additional losses.

Reading the Remaining Course Handouts

The chapter sequence follows `n1_intro.pdf`, `n2_perf.pdf`, `n3_cache_all.pdf`, `n4_pipe_part1.pdf`, `n5_ilp_part1.pdf`, the VLIW/software-pipelining handouts, `n6_multiprocv1.1.pdf`, `n7_synch.pdf`, and `n8_interconn.pdf`. These filenames identify the local course sources; the handouts are not redistributed with the book. Course-specific categories, including the broad use of speculation for locality-based caching, are identified in the text when a narrower definition is also common.

Specifications Behind Implementation Caveats

Memory placement. The Linux kernel's NUMA documentation explains local allocation and the cost of remote placement and migration. Chapter 8 distinguishes this physical organization from the shared-memory programming model.

Dirty shared data. AMD's *Software Optimization Guide for the AMD Family 10h and 12h Processors*, publication 40546, section on MOESI and cache-line ownership, describes an O owner with S sharers while memory can be stale. This is the distinction drawn in Chapters 8 and 9; state names alone are not a complete protocol specification.

Barriers and language-level ordering. Mellor-Crummey and Scott's scalable-synchronization pseudocode includes the sense-reversing centralized barrier. Chapter 9 spells out acquire/release orders rather than treating the original abstract pseudocode as ordinary C/C++ accesses. The C++ working draft defines data races and happens-before and atomic synchronization orders; its language rules are separate from a hardware-only litmus model.

Transactional fallback. Intel's lock-elision anti-patterns explain why the transactional path must read the fallback lock. Its fallback discussion also emphasizes that best-effort hardware transactions need a nontransactional path.

Renamed state and retirement. The RISC-V BOOM implementation documentation distinguishes explicit physical-register storage from data-in-ROB storage, explains operand-read ports and bypassing, and describes retirement and committed-store draining. Chapter 6 uses these concrete designs to avoid treating every renamed core as a future file with a separate committed data copy.

Atomic operations and reservations. The RISC-V atomic-extension specification provides a concrete example of selectable acquire/release ordering and permitted store-conditional failure. These details motivate the distinction between atomicity and ordering in Chapter 9; other ISAs have their own rules.

Dynamic and guided loop scheduling. The OpenMP worksharing-loop specification distinguishes fixed-size dynamic chunks from guided chunks that decrease as work is consumed. Chapter 9 uses this distinction when describing a fetch-and-add work counter.

Persistence and cached writes. Intel's persistent-memory programming discussion explains why writes can remain in volatile caches or buffers and need appropriate propagation to reach a persistence domain. Chapter 3 separates this requirement from write-through policy and cache coherence.

Translation faults. The RISC-V supervisor-level specification, under virtual-address translation, illustrates that page-table validity and permission checks can raise faults. Chapter 3 distinguishes a cached-translation miss from a page fault and from storage I/O.